

5G RAN

NG and Xn Self-Management Feature Parameter Description

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 06 (2025-11-07).

Technical Changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for NG and Xn self-management by the UMPThb board. For details, see: • 5.1.5 Xn-C Interface Specifications • 6.3.3 Hardware	None	Low-frequency TDD High-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added support for X2 self-management in SA networking, and revised descriptions of scenarios where the user planes of the X2 and Xn interfaces use the same IP address. For details, see 5.1.3 Xn Self-Removal .	None	Low-frequency TDD High-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Optimized control-plane dual-stack alarm reporting for the Xn interface. For details, see: • 5.1.1.2 IPv4/IPv6 Dual-Stack Transmission • 5.3.2.1 Prerequisite Functions • 5.4.1.1 Data Preparation • 5.4.1.2 Using MML Commands	Modified parameter: Added the XN_CP_DUALSTACK_CHK_SW option to the gNBInterfaceParam.InterfaceOmSw parameter.	Low-frequency TDD High-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Editorial Changes

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FBFD-021104	SA Option 2 Architecture	4 NG Self-Management 5 Xn Self-Management
FOFD-010080	IPsec	6 Direct IPsec for Xn Interface

2.3 Differences Between NSA and SA

Function Name	Difference	Chapter/Section
NG self-management	Supported only in SA networking	4 NG Self-Management
Xn self-management	Supported in both NSA and SA networking, with the following differences: AMF-based Xn self-setup with peers automatically configured: Supported only in SA networking. MAE-based Xn self-setup with peers automatically configured: Supported in both SA networking and NSA networking.	5 Xn Self-Management
Direct IPsec for Xn interface	None	6 Direct IPsec for Xn Interface

2.4 Differences Between High Frequency Bands and Low Frequency Bands

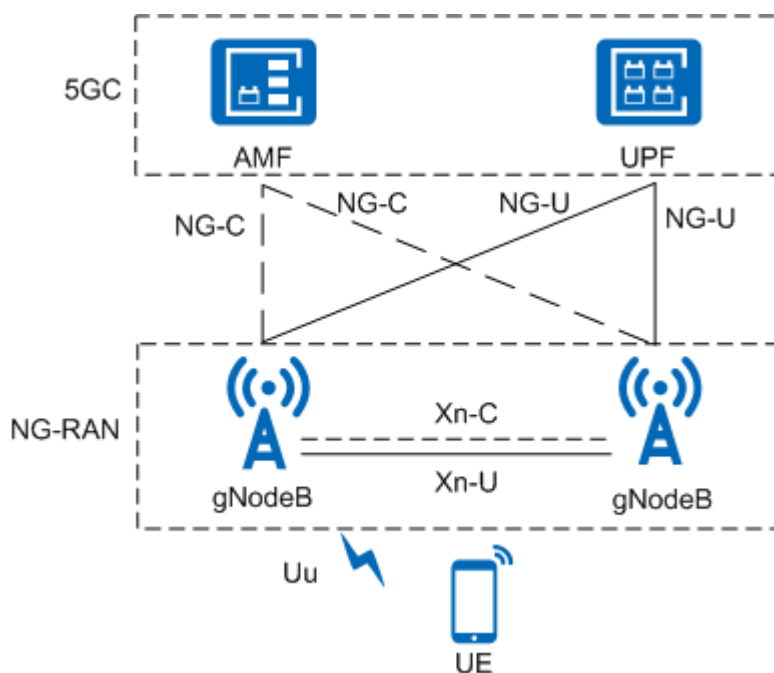
This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
NG self-management	None	4 NG Self-Management
Xn self-management	None	5 Xn Self-Management
Direct IPsec for Xn interface	None	6 Direct IPsec for Xn Interface

3 Overview

3GPP Release 15 is the first version which supports NR standalone (SA) networking. [Figure 3-1](#) illustrates the logical architecture with base stations in integrated deployment mode.

Figure 3-1 Logical structure of the NR network



The NR SA architecture includes the 5G Core Network (5GC) and NG Radio Access Network (NG-RAN). The 5GC mainly consists of the access and mobility management function (AMF) and user plane function (UPF), while the NG-RAN consists of gNodeBs. The following describes the interfaces between network elements (NEs):

- NG-C: The gNodeB is connected to the AMF through the NG-C interface to implement NG control plane functions.
- NG-U: The gNodeB is connected to the UPF through the NG-U interface to implement NG user plane functions.

- Xn-C: gNodeBs are connected through Xn-C interfaces to implement Xn control plane functions.
- Xn-U: gNodeBs are connected through Xn-U interfaces to implement Xn user plane functions.

To simplify configuration operations and reduce operators' operation costs, NG and Xn Self-Management is introduced. The following sub-functions are supported:

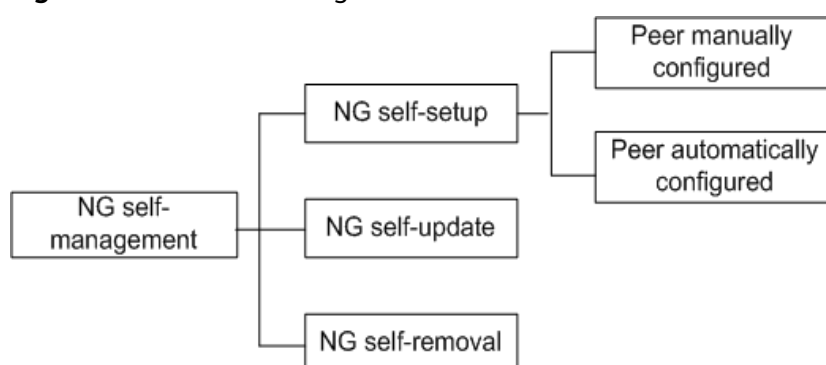
- NG self-management: self-setup, self-update, and self-removal
- Xn self-management: self-setup, self-update, and self-removal

4 NG Self-Management

4.1 Principles

Figure 4-1 shows NG self-management.

Figure 4-1 NG self-management



4.1.1 NG Self-Setup

NG self-setup is performed in either of the following modes:

- NG self-setup with peers manually configured
NG peers are manually configured in this mode. If a gNodeB is connected to multiple AMFs, the address information of each AMF must be manually added.
- NG self-setup with peers automatically configured
A domain name server (DNS) is required in this mode, and the mapping between the AMF domain name and AMF IP address must have been configured on the DNS. A gNodeB queries and automatically obtains a peer AMF IP address from the DNS. The gNodeB then automatically obtains the UPF IP address from the AMF during NG interface bearer setup.

NOTE

The **gNBCUNgOp** MO must be configured for the NG interface in multi-operator sharing scenarios. For details, see *Multi-Operator Sharing*.

If the NG-U and S1-U interfaces need to use the same local IP address and the same peer IP address (that is, the S-GW and UPF use the same devices and IP addresses):

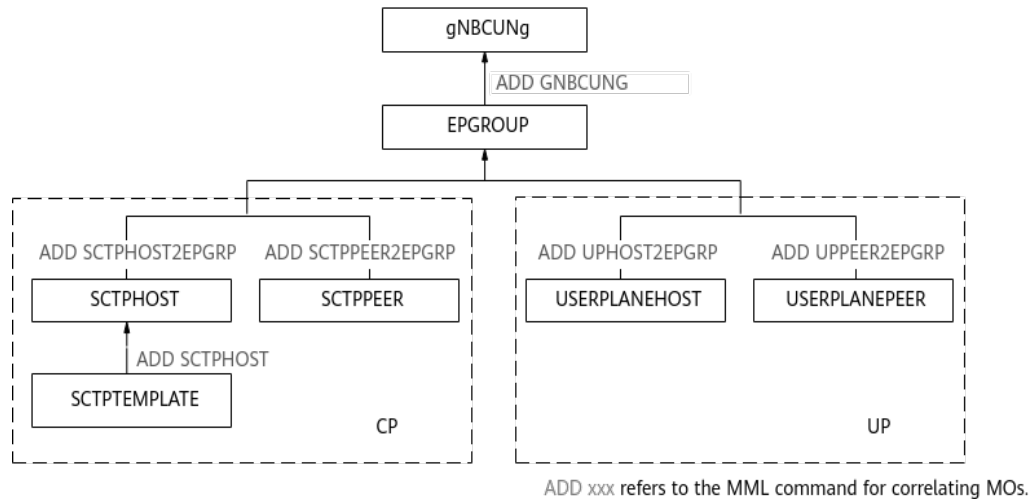
- The NG and S1 interfaces must be configured with the same user-plane **EPGROUP** MO. Otherwise, only the first user-plane path can be set up.
- A user-plane link cannot exist in two different user-plane endpoint groups (specified by the **EPGROUP** MO) at the same time. If you need to adjust a user-plane peer (specified by the **USERPLANEPEER** MO) to the user-plane endpoint group of a different interface, ensure that the user-plane peer has been removed from the user-plane endpoint group of the original interface.

4.1.1.1 IPv4/IPv6 Single-Stack Transmission

4.1.1.1.1 NG Self-Setup with Peers Manually Configured

An NG interface is classified into NG-C and NG-U interfaces. The NG-C interface transmits control-plane data, and the NG-U interface transmits user-plane data. [Figure 4-2](#) shows the relationships between MOs involved in NG self-setup with peers manually configured.

Figure 4-2 Relationships between MOs involved in NG self-setup with peers manually configured



The following describes how the MOs in [Figure 4-2](#) relate to each other.

- Control plane
 - **SCTPHOST** and **SCTPPEER** MOs are used to set parameters such as the IP addresses and port numbers of a control-plane link at the local and peer ends, respectively. Each **SCTPPEER** MO represents an AMF.
 - In IPv4 networking, the SCTP link uses the values of the **SCTPHOST.SIGIP1V4** and **SCTPPEER.SIGIP1V4** parameters as the local and peer IP addresses, respectively.

- In IPv6 networking, the SCTP link uses the values of the **SCTPHOST.SIGIPV6** and **SCTPPEER.SIGIPV6** parameters as the local and peer IP addresses, respectively.
- The **SCTPHOST.PN** and **SCTPPEER.PN** parameters must be set to the same value. According to 3GPP specifications, the recommended value is **38412**.

The gNodeB automatically sets up a multihoming SCTP link if second IP addresses are configured in the **SCTPHOST** and **SCTPPEER** MOs.

 NOTE

The **SCTPTEMPLATE** MO consists of optional parameters common for configuring SCTP links. The template is referenced by a control-plane link through the **SCTPHOST.SCTPTEMPLATEID** parameter, whose value is the same as the **SCTPTEMPLATE.SCTPTEMPLATEID** parameter value.

- User plane

USERPLANEHOST and **USERPLANEPEER** MOs are used to set parameters such as the local and peer IP addresses of a user-plane path, respectively.

- In IPv4 networking, the gNodeB uses a local NG user-plane IP address (specified by **USERPLANEHOST.LOCIPV4**) and a peer UPF IP address (specified by **USERPLANEPEER.PEERIPV4**).
- In IPv6 networking, the gNodeB uses a local NG user-plane IPv6 address (specified by **USERPLANEHOST.LOCIPV6**) and a peer IPv6 address (specified by **USERPLANEPEER.PEERIPV6**).

- Endpoint group

In the same endpoint group, links are set up between the local control plane and the peer control plane or set up between the local user plane and the peer user plane. When adding the local and peer control and user planes to endpoint groups by using the **EPGROUP** MO, note the following:

- The local and peer control planes of an NG interface must be added to the same endpoint group.
- The local and peer user planes of an NG interface must be added to the same endpoint group.

- NG interface

The **gNBCUNg** MO is used to add an NG interface object and bind it with the endpoint group.

After the preceding MOs are configured, the gNodeB and AMF automatically configure SCTP links (control plane) and NG-U links (user plane) of the NG interface and set up the NG interface and links.

4.1.1.1.2 NG Self-Setup with Peers Automatically Configured

NG self-setup with peers automatically configured is classified into NG-C self-setup and NG-U self-setup. For details about NG-C self-setup and NG-U self-setup, see [NG-C Self-Setup](#) and [NG-U Self-Setup](#), respectively.

NG-C Self-Setup

The **NGC_SON_SETUP_SW** option of the **gNBNgSonConfig.NgcSonSwitch** parameter determines whether to enable this function.

Figure 4-3 shows the relationships between MOs involved in NG-C self-setup.

Figure 4-3 Relationships between MOs involved in NG-C self-setup

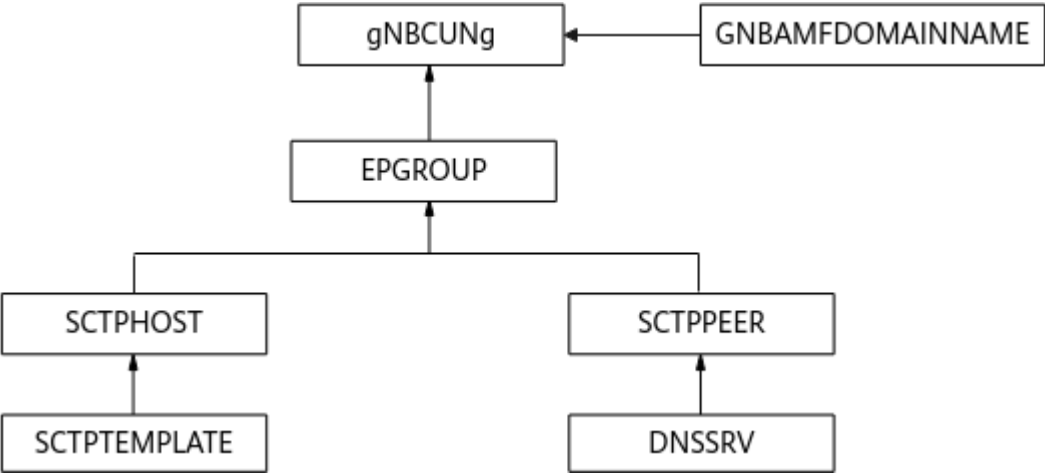
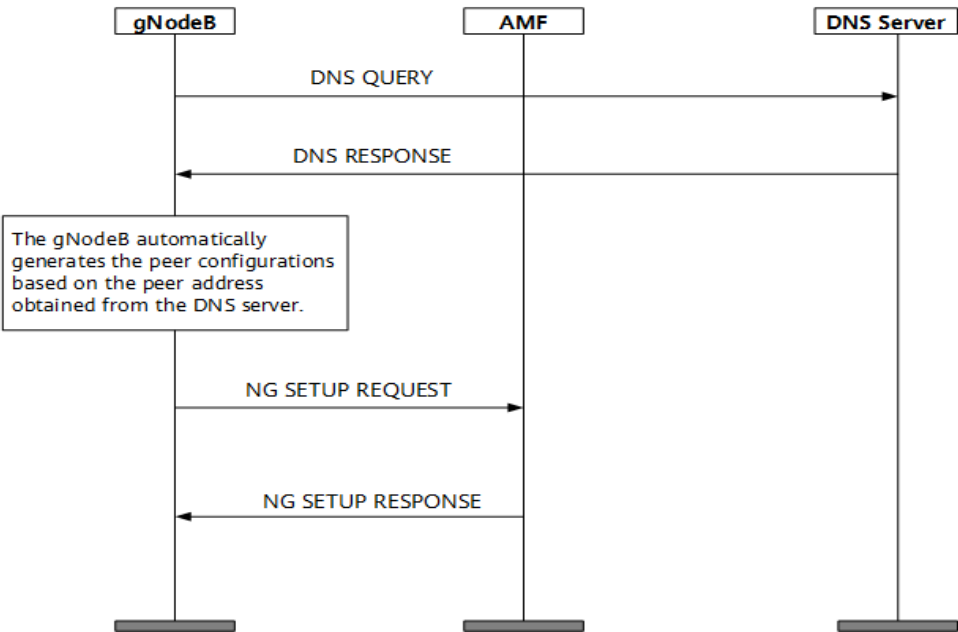


Figure 4-4 shows the NG-C self-setup procedure.

Figure 4-4 NG-C self-setup procedure



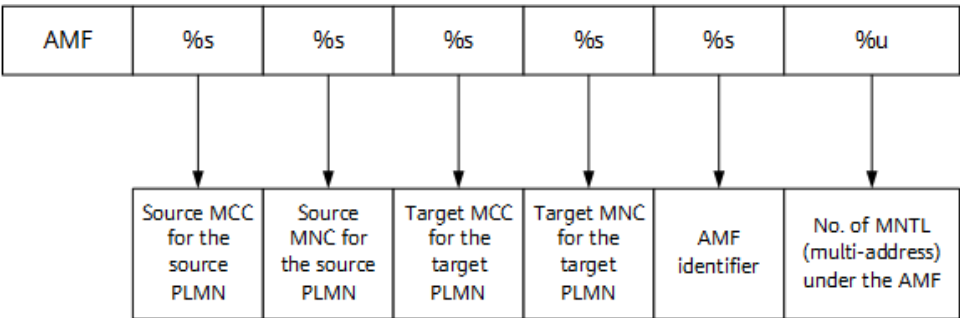
1. The gNodeB acting as a DNS client carries the AMF domain name. The gNodeB initiates a DNS query process to the DNS server and obtains the corresponding AMF IP address.
- Users can run the **ADD GNBAMFDOMAINNAME** command to configure an AMF domain name. Adding a new domain name will trigger DNS query. After a domain name is configured, the gNodeB queries the DNS server for the IP address corresponding to the domain name at a period specified by the **GTRANSPARA.DNSPRD** parameter. If this parameter is set to **0**, the gNodeB does not periodically initiate DNS queries. If an AMF domain name is not configured, the gNodeB cannot obtain the AMF IP address from the DNS server, meaning that the NG-C peer cannot be automatically configured.

During the preceding DNS query procedure, the gNodeB acquires only the IP address of the AMF and not the SCTP port number. The gNodeB automatically sets the SCTP peer port number to **38412** as recommended in 3GPP specifications. Therefore, the port number must be set to **38412** on both the gNodeB and AMF. Without such settings, the NG-C link becomes abnormal.

- 2. The gNodeB uses the obtained AMF IP address as the peer IP address. If the obtained peer IP address is not among the configured peer IP addresses (specified in the **SCTPPEER** MOs), the gNodeB automatically generates a new **SCTPPEER** MO and automatically adds it to the endpoint group.

The **SCTPPEER.USERLABEL** parameter value is automatically generated based on the information about the peer AMF. The format is "AMF%s%s%s%s%s %u", as shown in **Figure 4-5**.

Figure 4-5 Peer object user label value



- 3. The gNodeB automatically sets up an NG-C link based on the following MOs: (1) manually configured **SCTPHOST**, **EPGROUP**, and **gNBCUNg** MOs; (2) automatically generated **SCTPPEER** MO.

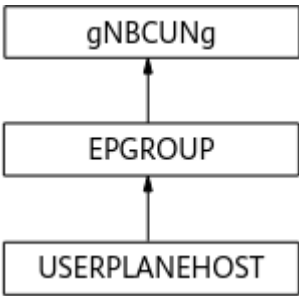
A gNodeB can connect to a maximum of four DNS servers, which are deployed in active/standby mode. If the DNS query on a DNS server fails, the gNodeB automatically queries other DNS servers. If the DNS query fails on all DNS servers, the gNodeB reports ALM-25958 DNS Resolution Failure. The DNS servers configured on the gNodeB are in active/standby mode. When there are multiple DNS servers, ensure that the configurations on these DNS servers are the same.

According to 3GPP and RFC requirements, the gNodeB functioning as a DNS client must support DNS server query for NAPTR, SRV, and A resource records. Base stations support two DNS query procedures: initial domain name > NAPTR > SRV > A and initial domain name > NAPTR > A. For details, see 3GPP TS 29.303 V16.2.0, RFC1034, and RFC1035. For the DNS server, the data mapping between the initial domain name and the full domain name, service level, and AMF IP address must be configured.

NG-U Self-Setup

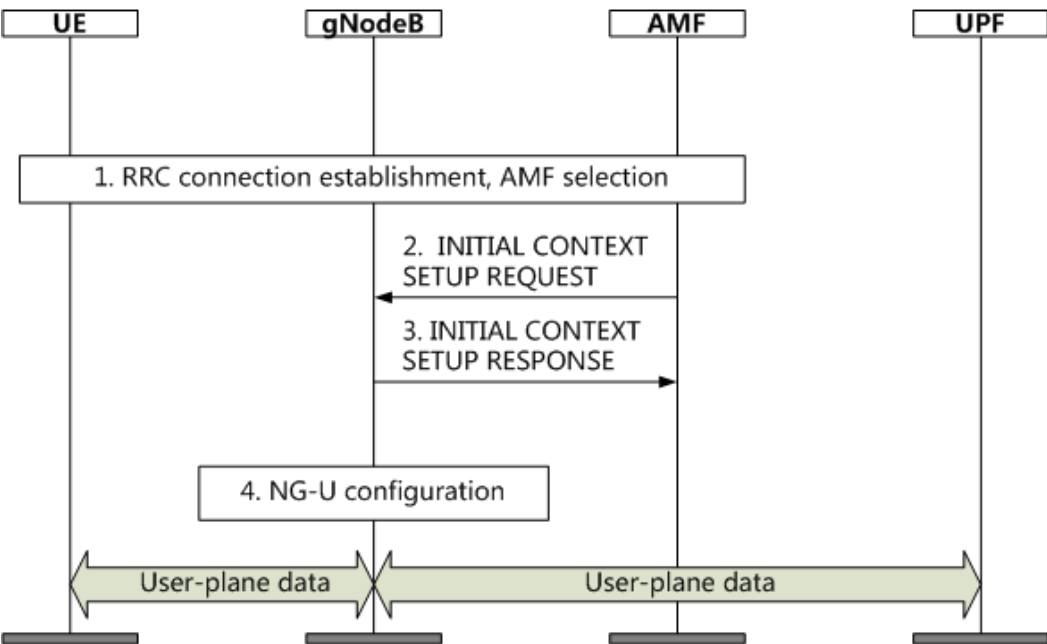
Figure 4-6 shows the relationships between MOs involved in NG-U self-setup.

Figure 4-6 Relationships between MOs involved in NG-U self-setup



During NG-U self-setup, the gNodeB automatically configures the **USERPLANEPEER** MO based on the peer transport layer address that is required for NG bearer setup and contained in the NG control-plane signaling messages. The address may be contained in multiple NG control-plane signaling messages, such as INITIAL CONTEXT SETUP REQUEST, PDU SESSION RESOURCE SETUP REQUEST, and HANDOVER COMMAND. The following uses the automatic configuration of the NG-U peer during initial context setup as an example. [Figure 4-7](#) shows the signaling procedure.

Figure 4-7 Signaling procedure for NG-U self-setup



1. The UE initiates a service.
2. The AMF sends the gNodeB an INITIAL CONTEXT SETUP REQUEST message containing the UPF IP address.
3. After receiving the UPF IP address, the gNodeB sends the AMF an INITIAL CONTEXT SETUP RESPONSE message.

If the UPF IP address is different from the existing configuration (specified by the **USERPLANEPEER** MO), the gNodeB generates a user-plane peer (**USERPLANEPEER** MO) according to the UPF IP address, and automatically adds the user-plane peer to the configured endpoint group (specified by the **EPGROUP** MO).

At the same time, the gNodeB automatically assigns a value to **USERPLANEPEER.USERLABEL** according to the NG interface's MCC and MNC, in the format of **UPF:MCC_xxx MNC_xxx**. **UPF:MCC_xxx MNC_xxx** specifies the operator for the user-plane peer.

4. The gNodeB automatically sets up an NG-U link based on the following MOs:
- (1) manually configured **USERPLANEHOST**, **EPGROUP**, and **gNBCUNg** MOs;
 - (2) automatically generated **USERPLANEPEER** MO.

4.1.1.2 IPv4/IPv6 Dual-Stack Transmission

The NG interface supports IPv4/IPv6 dual-stack transmission when the local gNodeB is configured with both IPv4 and IPv6 addresses. IPv4/IPv6 dual-stack transmission over the NG interface applies when a gNodeB uses different IP versions to connect to different AMFs/UPFs. For example, during a reconstruction from IPv4 to IPv6 transmission, some AMFs/UPFs use IPv6 and some use IPv4. When the NG interface supports IPv4/IPv6 dual-stack transmission, NG links can be set up using either of the following methods.

NG Self-Setup with Peers Manually Configured

During NG self-setup with peers manually configured, if the local gNodeB has been configured with both IPv4 and IPv6 addresses, the following requirement must be met:

Configuring both IPv4 and IPv6 addresses for a peer AMF/UPF should be avoided. If both the local and peer ends are configured with IPv4 and IPv6 addresses, both IPv4 and IPv6 links will be established, occupying twice the link resources, although one of the links will be unavailable. As a result, ALM-29815 gNodeB NG Interface Fault is generated. Additionally, the NG interface may fail to work properly because the control and user planes of the NG interface use different IP versions.

When the NG interface supports IPv4, IPv6, or IPv4/IPv6 dual-stack transmission, the IP version for an NG link is selected based on IP versions configured on the gNodeB, as listed in [Table 4-1](#).

Table 4-1 Rules for configuring IP versions for NG links

Local IP Version Configured on the gNodeB	AMF's IP Version Configured on the gNodeB	IP Version Configured on the AMF	IP Version for NG Link	Description
IPv4	IPv4	IPv4 and IPv6	IPv4	Common configuration
IPv6	IPv6	IPv4 and IPv6	IPv6	Common configuration

Local IP Version Configured on the gNodeB	AMF's IP Version Configured on the gNodeB	IP Version Configured on the AMF	IP Version for NG Link	Description
IPv4	IPv6	IPv4 and IPv6	Cannot be set up	None
IPv6	IPv4	IPv4 and IPv6	Cannot be set up	None
IPv4	IPv4 and IPv6	IPv4 and IPv6	IPv4	None
IPv6	IPv4 and IPv6	IPv4 and IPv6	IPv6	None
IPv4 and IPv6	IPv4	IPv4 and IPv6	IPv4	None
IPv4 and IPv6	IPv6	IPv4 and IPv6	IPv6	None
IPv4 and IPv6	IPv4 and IPv6	IPv4 and IPv6	IPv4 or IPv6	None

NOTE

If multiple TNLAs are set up between a gNodeB and an AMF, and the gNodeB is configured with both IPv4 and IPv6 addresses, the IP versions for NG links (as listed in [4.1.4 Multi-TNLA over the NG Interface](#)) are different from those in [Table 4-1](#).

NG Self-Setup with Peers Automatically Configured

During NG self-setup with peers automatically configured, the NG-U interface supports IPv4/IPv6 dual-stack transmission, while the NG-C interface does not. For details about NG-C self-setup and NG-U self-setup, see [NG-C Self-Setup](#) and [NG-U Self-Setup](#), respectively.

During NG-U self-setup, if the local gNodeB has been configured with both IPv4 and IPv6 addresses, the gNodeB automatically selects the IP address for the peer UPF based on the following rules:

- If the peer UPF is configured with only an IPv4 address, the IPv4 address is used for the peer.
- If the peer UPF is configured with only an IPv6 address, the IPv6 address is used for the peer.
- If the peer UPF is configured with both IPv4 and IPv6 addresses, the peer IP address is selected based on the value of the **EPGROUP.IPVERPREFERENCE** parameter.

4.1.1.3 Packet Filtering Protected NG Self-Setup

The procedures for packet filtering protected NG self-setup are similar to those described in [5.1.1.3 Packet Filtering Protected Xn Self-Setup](#), except for the differences in Xn and NG interfaces.

After ACL rule self-configuration in endpoint mode for packet filtering is enabled for the NG interface, the base station automatically generates ACL rules that

completely match the local and peer IP addresses of the user plane and control plane. Subsequent packet filtering is performed based on these new rules. Therefore, when this function is enabled, you are advised not to manually configure ACL rules for the NG interface. This prevents occupation of ACL rule specifications.

NOTE

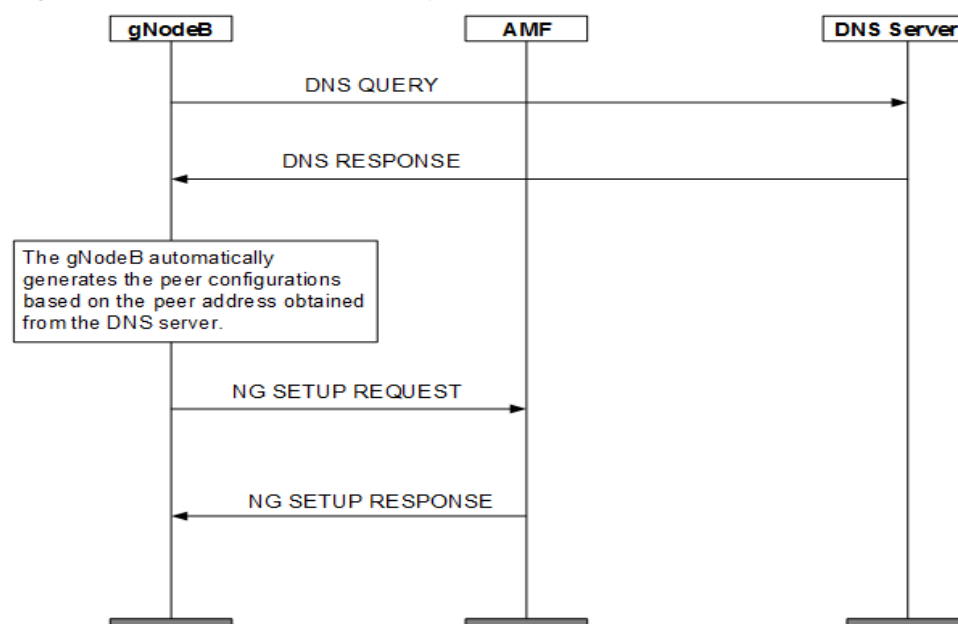
When manually configuring the **ACLRULE** MO for the S-GW on the gNodeB, ensure that the value of **ACLRULE.ACLID** for the ACL rule is the same as that of **PACKETFILTER.ACLID** (old model, with **GTRANSPARA.TRANS CFGMODE** set to **OLD**) or **PACKETFILTERING.ACLID** (new model, with **GTRANSPARA.TRANS CFGMODE** set to **NEW**) for packet filtering. Otherwise, transmission links will be disconnected.

4.1.2 NG Self-Update

The **NGC_SON_SETUP_SW** option of the **gNBNgSonConfig.NgcSonSwitch** parameter determines whether to enable this function. NG-C peer configurations are automatically updated when a DNS server has been deployed and AMF IP addresses have been updated on the DNS server.

An automatic update of the NG-C peer configurations is triggered when the AMF transport layer address obtained from the DNS query changes, as shown in [Figure 4-8](#).

Figure 4-8 DNS-based NG self-update



1. The gNodeB periodically performs DNS query or users manually perform DNS query to obtain the mapping between the AMF and the IP addresses.
2. Based on the new mapping between the AMF and the IP address, the gNodeB automatically updates the IP addresses and the remote ID specified by **SCTPPEER.REMOTEID** and re-establishes an NG-C link.

4.1.3 NG Self-Removal

NG-C Self-Removal

During network construction, if a redundant NG-C interface exists on the gNodeB due to AMF adjustment, NG-C self-removal is initiated if all the following conditions are met:

- The **NGC_SON_DELETE_SW** option of the **gNBNgSonConfig.NgcSonSwitch** parameter is selected.
- The removal of a gNodeB AMF domain name (**RMV GNBAMFDOMAINNAME**) or AMF adjustment results have been updated on the DNS server and the DNS query result indicates that the target AMF does not exist.

The NG-C self-removal procedure is as follows:

1. The gNodeB obtains the mapping between the domain name of the current AMF and the AMF IP address through DNS query. It can obtain the mapping only after the DNS queries of all the AMF domain names are successful.
2. The gNodeB compares the mapping between the old AMF domain name and the IP address with that between the new AMF domain name and IP address. Then, the gNodeB determines whether self-removal is required. If the DNS query result indicates that an AMF configured on the gNodeB does not exist, the gNodeB removes the NG-C interface of the AMF.

NOTE

During transmission network reconstruction, if NG-C links are redundant because NG-C self-removal conditions are not met, run the **RMV SCTPPEER** and **RMV SCTPPEER2EPGRP** commands to manually remove redundant NG-C links.

NG-U Self-Removal

The gNodeB can automatically remove the **USERPLANEPEER** MOs with **USERPLANEPEER.CTRLMODE** set to **AUTO_MODE** and the corresponding NG-U transmission links. However, the gNodeB does not automatically remove the **USERPLANEPEER** MOs with **USERPLANEPEER.CTRLMODE** set to **MANUAL_MODE** and the corresponding NG-U transmission links. This function is controlled by the **GEPMODELpara.UPAUTODELSW** parameter.

The gNodeB determines whether to automatically remove NG-U transmission links as follows:

- Self-removal based on link faults
If the GTP-U detection result indicates that all the transmission links corresponding to **USERPLANEPEER** MOs are faulty, the gNodeB periodically (every 60 minutes) attempts to remove those transmission links (up to 50 links at one time) that remain faulty for over one hour and the corresponding **USERPLANEPEER** MOs.
- Self-removal based on the aging mechanism
 - In case of an excess over the specifications of a base station
The gNodeB marks a transmission link as aged if the link is not used by any service.

- The gNodeB performs self-removal based on the aging mechanism once every hour and checks whether the number of established transmission links reaches or exceeds 90% of the maximum number of transmission links supported. If it does, the gNodeB removes aged transmission links (up to 50 links at one time) and the corresponding **USERPLANEPEER** MOs. If it does not, no aged transmission links are removed.
- When UEs need to establish new transmission links for initiating services, the gNodeB checks whether the number of established transmission links reaches or exceeds 90% of the maximum number of transmission links supported. If it does, the gNodeB removes aged transmission links (up to 50 links at one time) and the corresponding **USERPLANEPEER** MOs, and establishes new transmission links. If it does not, no aged transmission links are removed.
- If the number of established transmission links has reached the maximum number supported and there is no aged link when UEs need to establish new transmission links for initiating services, the gNodeB rejects new UEs' access requests.
- In case of an excess over the specifications of a single endpoint group "Self-removal in case of an excess over the specifications of a single endpoint group" takes effect for the gNodeB when the **SINGLE_EPUP_AUTO_DEL_SW** option of the **TRANSALGEXTPARA.TNLBEARERMNGALGSW** parameter is selected. The base station removes aged links when the number of peers in a single endpoint group exceeds 90% of the maximum number allowed. The condition for marking a link as aged and the self-removal mechanism are the same as those for "self-removal in case of an excess over the specifications of a base station".

 NOTE

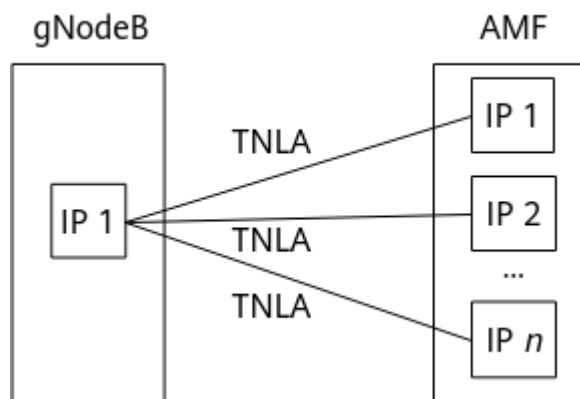
During transmission network reconstruction, if NG-U links are redundant because NG-U self-removal conditions are not met, run the **RMV USERPALNEPEER2EPGRP** and **RMV USERPALNEPEER** commands to manually remove redundant NG-U links.

4.1.4 Multi-TNLA over the NG Interface

Support for Self-Setup of Multiple TNLAs over the NG Interface

As defined in 3GPP specifications, multi-Transport Network Layer Association (multi-TNLA) is a connection mode that allows the self-setup of multiple NG-C connections between an AMF and a gNodeB, as shown in [Figure 4-9](#).

Figure 4-9 Self-setup of multiple TNLAs between an AMF and a gNodeB



One AMF can be configured with multiple SCTP IP addresses, and each SCTP IP address can be used for the self-setup of an NG-C interface with the same gNodeB. AMF's support for multiple SCTP IP addresses is configured on the AMF side. The configuration on the gNodeB side is the same as that for NG self-setup. For details, see [4.1.1.1 NG Self-Setup with Peers Manually Configured](#) and [NG-C Self-Setup](#). That is, after TNLA is introduced, no special configuration is required on the gNodeB side.

Primary link: The gNodeB establishes the first link with the AMF based on the manually configured base station and AMF addresses. This link is called the primary link.

Secondary link: The AMF sends an AMF CONFIGURATION UPDATE message carrying additional TNLA addresses of the AMF through the primary link. An NG interface established using any of these TNLA addresses is called a secondary link. That is, the primary link needs to be established before secondary links.

The **SCTPPEER** MOs manually configured on the gNodeB side are considered to be different AMFs. If multiple **SCTPPEER** MOs manually configured on the gNodeB side correspond to the same AMF, an NG interface between the base station and the AMF can be set up with only one of the **SCTPPEER** MOs in the method described in [4.1.1.1 NG Self-Setup with Peers Manually Configured](#). The NG interface setup fails for the remaining **SCTPPEER** MOs.

After the first NG interface is set up between the base station and the AMF, the AMF delivers the remaining core network addresses through the AMF CONFIGURATION UPDATE message to trigger NG self-setup. If the IP address and port number carried in the AMF CONFIGURATION UPDATE message (the port number is **38412** by default if it is not carried in the message) are the same as those in the **SCTPPEER** MO manually configured on the base station, self-setup using this IP address will fail.

For example, if the AMF supports three IP addresses (IP 1, IP 2, and IP 3), only one IP address needs to be configured in the **SCTPPEER** MO on the gNodeB. Assume that IP 1 is configured in the **SCTPPEER** MO on the gNodeB. After an NG interface is successfully set up between the gNodeB and AMF using IP 1, the AMF proactively sends an AMF CONFIGURATION UPDATE message containing IP 2 and IP 3. The self-setup of the remaining two NG interfaces can be completed through the TNLA setup procedure in the method described in [4.1.1.2 NG Self-Setup with Peers Automatically Configured](#).

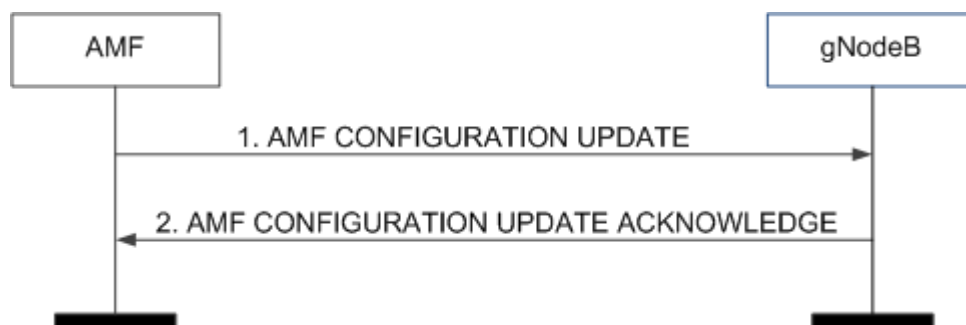
Self-setup of multiple TNLAs over the NG interface is controlled by the **MULTI_TNLA_SELF_SETUP_SW** option of the **gNBNgSonConfig.NgcSonSwitch** parameter.

 NOTE

Multi-TNLA is not equivalent to SCTP multi-homing. In SCTP multi-homing, only one link takes effect. In multi-TNLA, all links can take effect at the same time. For details about SCTP multi-homing, see *IPv4 Transmission* or *IPv6 Transmission*.

Figure 4-10 shows the self-setup procedure of a TNLA between a gNodeB and an AMF.

Figure 4-10 Self-setup procedure of a TNLA between a gNodeB and an AMF



1. The AMF sends an AMF CONFIGURATION UPDATE message to the gNodeB, instructing the gNodeB to add, remove, or update a TNLA.
 - Adding a TNLA
The AMF sends the gNodeB an AMF CONFIGURATION UPDATE message that carries the *AMF TNL Association to Add Item* IE.
 - Removing a TNLA
The AMF sends the gNodeB an AMF CONFIGURATION UPDATE message that carries the *AMF TNL Association to Remove Item* IE.
 - If the *AMF TNL Association to Remove Item* IE received by the gNodeB contains the *Port Number* IE, the gNodeB removes multiple TNLAs with the same *Endpoint IP Address* and *Port Number* IE values in the *AMF TNL Association to Remove Item* IE.
 - If the *AMF TNL Association to Remove Item* IE received by the gNodeB does not contain the *Port Number* IE, the gNodeB removes multiple TNLAs with the same *Endpoint IP Address* IE value in the *AMF TNL Association to Remove Item* IE.
 - Updating a TNLA
The AMF sends the gNodeB an AMF CONFIGURATION UPDATE message that carries the *AMF TNL Association to Update Item* IE.
2. After receiving a request from the AMF, the gNodeB adds, removes, or updates the TNLA, and replies to the AMF with an AMF CONFIGURATION UPDATE ACKNOWLEDGE message.

The AMF CONFIGURATION UPDATE ACKNOWLEDGE message from the gNodeB contains the addresses indicating successful and failed setups. For the failed

addresses, the AMF may continue the attempt by sending an AMF CONFIGURATION UPDATE message.

After a TNLA is set up, the gNodeB sends the first NGAP message RAN CONFIGURATION UPDATE through the TNLA, and the AMF associates the TNLA with the NG-C interface instance. For details, see chapter 7 "Transport layer" in 3GPP TS 38.412 V15.1.0.

TNLAs serving as additional links to the first established NG-C link will also be removed when the manually configured primary NG-C link is removed; and the gNodeB does not remove all links when the AMF delivers the AMF CONFIGURATION UPDATE message containing the *AMF TNL Association to Remove Item* IE and the IE contains the IP address of the primary NG-C link.

The **DSP GNBCUNGINTERFACE** command can be executed to query the information about an NG-C interface. If the value of **gNodeB CU NG Interface Attribute** is **CREATED_BY_GNODEB**, the NG link is the first transmission link proactively set up by the gNodeB. If the value is **CREATED_BY_AMF**, the NG link is a transmission link proactively set up by the AMF. After a TNLA established by the AMF is removed using the **RMV SCTPPEER2EPGRP** command, if the **ADD SCTPPEER2EPGRP** command is executed to add the **SCTPPEER** MO to the **EPGROUP** MO again, this **SCTPPEER** MO will be automatically removed.

In multi-TNLA scenarios, after the self-setup of an NG-C interface between the gNodeB and AMF, if the AMF is configured with both IPv4 and IPv6 addresses, the AMF delivers one of the addresses to the gNodeB. As a result, when the local gNodeB is configured with both IPv4 and IPv6 addresses, IP versions for NG links are different from those in [Table 4-1](#).

Table 4-2 Rules for configuring IP versions for NG links in multi-TNLA scenarios

Local IP Version Configured on the gNodeB	AMF's IP Version Configured on the gNodeB	IP Version Configured on the AMF	IP Version for NG Link	Description
IPv4	IPv4	IPv4 and IPv6	IPv4	Common configuration
IPv6	IPv6	IPv4 and IPv6	IPv6	Common configuration
IPv4	IPv6	IPv4 and IPv6	Cannot be set up	None
IPv6	IPv4	IPv4 and IPv6	Cannot be set up	None
IPv4	IPv4 and IPv6	IPv4 and IPv6	IPv4	None
IPv6	IPv4 and IPv6	IPv4 and IPv6	IPv6	None

Local IP Version Configured on the gNodeB	AMF's IP Version Configured on the gNodeB	IP Version Configured on the AMF	IP Version for NG Link	Description
IPv4 and IPv6	IPv4	IPv4 and IPv6	IPv4 and IPv6	None
IPv4 and IPv6	IPv6	IPv4 and IPv6	IPv4 and IPv6	None
IPv4 and IPv6	IPv4 and IPv6	IPv4 and IPv6	IPv4 or IPv6	None

Support for Self-Removal of Faulty TNLAs in the Context of Multiple TNLAs over the NG Interface

Self-removal of faulty TNLAs in the context of multiple TNLAs over the NG interface is controlled by the **FAULT_MULTI_TNLA_SELF_RMV_SW** option of the **gNBNgSonConfig.NgcSonSwitch** parameter.

- If this option is selected, self-removal of faulty TNLAs in the context of multiple TNLAs over the NG interface takes effect. If a TNLA is faulty for seven consecutive days, the gNodeB proactively removes the TNLA.
- If this option is deselected, self-removal of faulty TNLAs in the context of multiple TNLAs over the NG interface does not take effect.

4.1.5 NG-C Interface Specifications

The NG-C is the interface between a gNodeB and an AMF for implementing control plane functions. The specifications are as follows:

- When only one NG-C connection is set up between a gNodeB and an AMF, a maximum of 36 NG-C interfaces (36 AMF addresses) can be configured on the gNodeB side.
- When multiple NG-C connections are set up between a gNodeB and an AMF, a maximum of 32 NG-C interfaces can be configured between them. However, a maximum of 96 NG-C interfaces can be configured between a gNodeB and all AMFs (36 AMFs at most).

NOTE

- Only one IP address can be configured for each AMF on the gNodeB side. If an AMF has multiple addresses, other addresses are included in the AMF CONFIGURATION UPDATE message sent from the AMF to the gNodeB and do not need to be configured on the gNodeB side.
- When NG-C setup is based on DNS, the DNS servers corresponding to all MME domain names configured on the gNodeB must meet the following requirements:
 - A maximum of 32 NAPTR records are supported.
 - A maximum of 4 SRV records are supported.
 - A maximum of 2 A records are supported.

If the preceding restrictions are exceeded, domain name resolution will fail, resulting in NG-C setup failures.

4.1.6 Application Restrictions

- The gNodeB supports the self-setup of two **USERPLANEPEER** MOs at the same time. The self-setup request for additional **USERPLANEPEER** MOs will be rejected, which results in UE access failures.
- If different AMFs are configured with the same AMF name or GUAMI, the gNodeB can set up an NG interface with only one of the AMFs.

4.2 Network Analysis

4.2.1 Benefits

NG self-management simplifies configuration operations, reduces the operating expense (OPEX) of operators, and improves gNodeB resource usage.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

This function has no impact on the network.

4.3 Requirements

4.3.1 Licenses

None

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Automatic user plane removal	GEPMODEL PARA. UPAUTO DELSW	<i>NG and Xn Self-Management</i>	Ensure that the GEPMODEL PARA. UPAUTO DELSW parameter is set to ENABLE before enabling "self-removal in case of an excess over the specifications of a single endpoint group" (by selecting the SINGLE_EPUP_AUTO_DEL_SW option of the TRANSALGEXTPARA. TNLBEARERMN GALGSW parameter).
Low-frequency TDD High-frequency TDD	Self-removal in case of an excess over the specifications of a single endpoint group	SINGLE_EPUP_AUTO_DEL_SW option of the TRANSALGEXTPARA. TNLBEARERMN GALGSW parameter	<i>NG and Xn Self-Management</i>	Ensure that "self-removal in case of an excess over the specifications of a single endpoint group" is disabled before disabling automatic user plane removal (by setting the GEPMODEL PARA. UPAUTO DELSW parameter to DISABLE).

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Endpoint group IDs for the Xn interface	gNBCUXn.CpEpGroupId and gNBCUXn.UpEpGroupId	<i>NG and Xn Self-Management</i>	• The value of the gNBCUNg.CpEpGroupId parameter configured for the NG interface must be different from that of the gNBCUXn.CpEpGroupId parameter configured for the Xn interface. • The value of the gNBCUNg.UpEpGroupId parameter configured for the NG interface must be different from that of the gNBCUXn.UpEpGroupId parameter configured for the Xn interface.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Endpoint group IDs for the X2 interface	gNBCUX2.CpEpGroupId and gNBCUX2.UpEpGroupId	<i>X2 and S1 Self-Management in NSA Networking</i>	- The value of the gNBCUNg.CpEpGroupId parameter configured for the NG interface must be different from that of the gNBCUX2.CpEpGroupId parameter configured for the X2 interface. - The value of the gNBCUNg.UpEpGroupId parameter configured for the NG interface must be different from that of the gNBCUX2.UpEpGroupId parameter configured for the X2 interface.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards support this function. To learn which main control boards are NR-capable, see the related BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

4.3.4 Networking

For the quality of service (QoS) requirements of NG interface for transmission networking, see *Transmission Resource Management*.

4.3.5 Others

If the NG-C self-setup with peer information automatically obtained needs to be deployed, the following requirements must be met:

- A DNS server is configured on the core network.
- Connection between the gNodeB and DNS server is available.
- The mapping between the AMF domain name and AMF IP address has been configured on the DNS server.

NOTE

Ensure that there is a one-to-one mapping between the domain names and IP addresses.
The AMF IP addresses obtained from the DNS server must be IPv4 addresses.

4.4 Operation and Maintenance

4.4.1 When to Use

It is recommended that NG self-management be enabled when a large number of AMFs and UPFs exist in a network. This spares operators from configuring a large number of NG-C and NG-U interfaces.

4.4.2 Data Configuration

4.4.2.1 Data Preparation

The following table describes the parameters that must be set in an **SCTPTEMPLATE** MO to configure negotiation parameters for the NG-C transmission link.

Parameter Name	Parameter ID	Setting Notes
SCTP Parameters Template ID	SCTPTEMPLATE.SCTPTEMPLATEID	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Switch Back Flag	SCTPTEMPLATE.SWITC HBACKFLAG	Set this parameter based on the network plan.

The following table describes the parameters that must be set in an **SCTPHOST** MO to configure the NG-C host information.

Parameter Name	Parameter ID	Setting Notes
SCTP Host ID	SCTPHOST.SCTPHOSTID	Set this parameter based on the network plan.
IP Version	SCTPHOST.IPVERSION	Set the IP version of the local end to the same as the peer end.
First Local IP Address	SCTPHOST.SIGIP1V4	If SCTPHOST.IPVERSION is set to IPv4 , configure a local control-plane IPv4 address.
First Local IPv6 Address	SCTPHOST.SIGIP1V6	If SCTPHOST.IPVERSION is set to IPv6 , configure a local control-plane IPv6 address.
Second Local IP Address	SCTPHOST.SIGIP2V4	If SCTPHOST.IPVERSION is set to IPv4 , configure a local control-plane IPv4 address.
Second Local IPv6 Address	SCTPHOST.SIGIP2V6	If SCTPHOST.IPVERSION is set to IPv6 , configure a local control-plane IPv6 address.
Local SCTP Port No.	SCTPHOST.PN	The value recommended by 3GPP specifications is 38412 . For details, see chapter 7 "Transport layer" in 3GPP TS 38.412 V15.1.0.
Simple Mode Switch	SCTPHOST.SIMPLEMOD ESWITCH	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
SCTP Parameters Template ID	SCTPHOST.SCTPTEMPL ATEID	You are advised to use the SCTP parameter template whose ID is 0. If you do not use this template, you need to add an SCTPTEMPLATE MO and change the value of SCTPHOST.SCTPTEMPL ATEID to reference the new template.

The following table describes the parameters that must be set in a **USERPLANEHOST** MO to configure the local gNodeB information of the NG-U interface.

Parameter Name	Parameter ID	Setting Notes
User Plane Host ID	USERPLANEHOST.UPHOSTID	This parameter specifies the ID of the user-plane host.
IP Version	USERPLANEHOST.IPVERSION	Set the IP version of the local end to the same as the peer end.
Local IP Address	USERPLANEHOST.LOCIPV4	If USERPLANEHOST.IPVERSION (LTE eNodeB, 5G gNodeB) is set to IPv4 , configure a local user-plane IPv4 address.
Local IPv6 Address	USERPLANEHOST.LOCIPV6	If USERPLANEHOST.IPVERSION (LTE eNodeB, 5G gNodeB) is set to IPv6 , configure a local user-plane IPv6 address.
IPSec Auto Configure Switch	USERPLANEHOST.IPSEC SWITCH	Set this parameter to ENABLE to enable direct IPsec.
Master Flag	USERPLANEHOST.FLAG	Set this parameter when the user-plane active/standby function is required.

(Optional) The following table describes the parameters that must be set in an **SCTPPEER** MO to configure information about the peer AMF of the NG-C interface. This MO is required only during manual NG-C peer configuration.

The local gNodeB can set up an SCTP association only with the peer AMF having the same IP addresses as those configured in the **SCTPPEER** MO. In addition, the source IP addresses of SCTP packets sent by the peer AMF must be identical to those configured in the **SCTPPEER** MO on the local gNodeB.

Parameter Name	Parameter ID	Setting Notes
SCTP Peer ID	SCTPPEER.SCTPPEERID	Set this parameter based on the network plan.
IP Version	SCTPPEER.IPVERSION	Set the IP version of the local end to the same as the peer end.
First Peer IP Address	SCTPPEER.SIGIP1V4	If SCTPPEER.IPVERSION is set to IPv4 , configure a peer control-plane IPv4 address.
First Peer IPv6 Address	SCTPPEER.SIGIP1V6	If SCTPPEER.IPVERSION is set to IPv6 , configure a peer control-plane IPv6 address.
Second Peer IP Address	SCTPPEER.SIGIP2V4	If SCTPPEER.IPVERSION is set to IPv4 , configure a peer control-plane IPv4 address.
Second Peer IPv6 Address	SCTPPEER.SIGIP2V6	If SCTPPEER.IPVERSION is set to IPv6 , configure a peer control-plane IPv6 address.
Peer SCTP Port No.	SCTPPEER.PN	The value recommended by 3GPP specifications is 38412 . For details, see chapter 7 "Transport layer" in 3GPP TS 38.412 V15.1.0.

(Optional) The following table describes the parameters that must be set in a **USERPLANEPEER** MO to configure information about the peer UPF of the NG-U interface. This MO is required only during manual NG-U peer configuration.

Parameter Name	Parameter ID	Setting Notes
User Plane Peer ID	USERPLANEPEER.UPPEERID	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
IP Version	USERPLANEPEER.IPVERSION	Set the IP version of the local end to the same as the peer end.
Peer IP Address	USERPLANEPEER.PEERIPV4	If USERPLANEPEER.IPVERSION is set to IPv4 , configure a peer user-plane IPv4 address.
Peer IPv6 Address	USERPLANEPEER.PEERIPV6	If USERPLANEPEER.IPVERSION is set to IPv6 , configure a peer user-plane IPv6 address.
IPSec Auto Configure Switch	USERPLANEPEER.IPSEC SWITCH	Set this parameter to ENABLE to enable direct IPsec.

The following table describes the parameters that must be set in an **EPGROUP** MO.

Parameter Name	Parameter ID	Setting Notes
End Point Group ID	EPGROUP.EPGROUPID	Set this parameter based on the network plan.
SCTP Host List	EPGROUP.SCTPHOSTLIST	Set this parameter based on the network plan. Add an SCTP host to the endpoint group.
SCTP Peer List	EPGROUP.SCTPPEERLIST	Set this parameter based on the network plan. Add an SCTP peer to the endpoint group.
User Plane Host List	EPGROUP.USERPLANEHOSTLIST	Set this parameter based on the network plan. Add a user-plane host to the endpoint group.
User Plane Peer List	EPGROUP.USERPLANEPEERLIST	Set this parameter based on the network plan. Add a user-plane peer to the endpoint group.

Parameter Name	Parameter ID	Setting Notes
Static Check Switch	EPGROUP.STATICCHK	If the user-plane load sharing and active/standby user plane functions are required, set this parameter to ENABLE to enable GTP-U static detection on all user-plane paths in the endpoint group.

The following table describes the parameters that must be set in an **ADD SCTPHOST2EPGRP** command to add an SCTP host to an endpoint group.

Parameter Name	Parameter ID	Setting Notes
End Point Group ID	EPGROUP.EPGROUPID	Set this parameter based on the network plan.
SCTP Host ID	SCTPHOSTREF.SCTPHOSTID	Set this parameter based on the network plan.

The following table describes the parameters that must be set in an **ADD SCTPPEER2EPGRP** command to add an SCTP peer to an endpoint group.

Parameter Name	Parameter ID	Setting Notes
End Point Group ID	EPGROUP.EPGROUPID	Set this parameter based on the network plan.
SCTP Peer ID	SCTPPEERREF.SCTPPEERID	Set this parameter based on the network plan.

The following table describes the parameters that must be set in an **ADD UPHOST2EPGRP** command to add a user-plane host to an endpoint group.

Parameter Name	Parameter ID	Setting Notes
End Point Group ID	EPGROUP.EPGROUPID	Set this parameter based on the network plan.
User Plane Host ID	USERPLANEHOSTREF.UPHOSTID	Set this parameter based on the network plan.

The following table describes the parameters that must be set in an **ADD UPPEER2EPGRP** command to add a user-plane peer to an endpoint group.

Parameter Name	Parameter ID	Setting Notes
End Point Group ID	EPGROUP.EPGROUPID	Set this parameter based on the network plan.
User Plane Peer ID	USERPLANEPEERREF.UPPEERID	Set this parameter based on the network plan.

The following table describes the parameters that must be set in a **gNBCUNg** MO to configure the relationships with the **EPGROUP** MO.

Parameter Name	Parameter ID	Setting Notes
gNodeB CU NG ID	gNBCUNg.gNBCuNgId	Set this parameter based on the network plan.
Control Plane Endpoint Group ID	gNBCUNg.CpEpGroupId	Set this parameter based on the network plan. This parameter specifies the endpoint group ID of the NG control plane.
User Plane Endpoint Group ID	gNBCUNg.UpEpGroupId	Set this parameter based on the network plan. This parameter specifies the endpoint group ID of the NG user plane.

(Optional) The following table describes the parameters that must be set in a **gNBNgSonConfig** MO to configure self-setup and self-removal for the NG-C interface. This MO is required for NG-C self-configuration.

Parameter Name	Parameter ID	Setting Notes
NG-C SON Switch	gNBNgSonConfig.NgcSonSwitch	- The NGC_SON_SETUP_SW option specifies whether to enable NG-C self-setup. Set this parameter based on the network plan. - The NGC_SON_DELETE_SW option specifies whether to enable NG-C self-removal. Set this parameter based on the network plan. - The MULTI_TNLA_SELF_SETUP_SW option specifies whether to enable self-setup of multiple TNLAs over the NG interface. Set this parameter based on the network plan. - The FAULT_MULTI_TNLA_SELF_RMV_SW option specifies whether to enable self-removal of faulty TNLAs in the context of multiple TNLAs over the NG interface. Set this parameter based on the network plan.

 NOTE

If the SON function will be used to automatically configure, update, and remove NG interfaces, the **GTRANSPARA.EPOBJAUTOCFGSW** parameter must be set to the default value **ENABLE**.

(Optional) The following table describes the parameters that must be set in a **DNSSRV** MO to configure DNS server information for automatic NG-C peer configuration. This MO is required only for NG-C self-setup.

Parameter Name	Parameter ID	Setting Notes
DNS SERVER ID.	DNSSRV.DNSSRVID	Set this parameter based on the network plan.
IP Version	DNSSRV.IPVER	Set this parameter based on the network plan.
DNS Client IPv4	DNSSRV.DNSCIP4	Set this parameter based on the network plan.
DNS Server IPv4	DNSSRV.DNSSIP4	Set this parameter based on the network plan.
DNS Client IPv6	DNSSRV.DNSCIP6	Set this parameter based on the network plan.
DNS Server IPv6	DNSSRV.DNSSIP6	Set this parameter based on the network plan.
DNS Server Port No.	DNSSRV.PN	Set this parameter based on the network plan.
DSCP	DNSSRV.DSCP	Set this parameter based on the network plan.

(Optional) The following table describes the parameters that must be set in a **gNBAmfDomainName** MO to specify the domain name of the peer AMF. This MO is required only for NG-C self-setup.

Parameter Name	Parameter ID	Setting Notes
gNodeB CU NG ID	gNBAmfDomainName.gNBCuNgId	Set this parameter based on the network plan.
AMF Domain Name	gNBAmfDomainName.AmfDomainName	Set this parameter based on the network plan.

(Optional) The following table describes the parameters that must be set in a **gNBCUNg** MO to send gNodeB names to an AMF. This MO is required only if the AMF requires gNodeB names.

Parameter Name	Parameter ID	Setting Notes
Protocol CR Switch	EXTENDED_RAN_NODE_NAME_SW option of the gNBCUNg.ProtocolCrSwitch parameter	Set this parameter based on the network plan.

(Optional) The following table describes the parameters that must be set in a **GEPMODELPARA** MO to enable automatic user plane removal. This MO is required only in NG-U self-removal scenarios.

Parameter Name	Parameter ID	Setting Notes
UP Automatic Deletion Switch	GEPMODELPARA.UPAU TODELSW	Set this parameter based on the network plan.

(Optional) The following table describes the parameters that must be set in an **SRCIPRT** MO to set a source IP route. This MO is required only when **GTRANSPARA.TRANS CFGMODE** is set to **OLD** and the user-plane load sharing function is required.

Parameter Name	Parameter ID	Setting Notes
Source Route Index	SRCIPRT.SRCRTIDX	Set this parameter based on the network plan.
Slot No.	SRCIPRT.SN	Set this parameter based on the network plan.
Source IP Address	SRCIPRT.SRCIP	Set this parameter based on the network plan.
Subboard Type	SRCIPRT.SBT	You are advised to set this parameter to BASE_BOARD .
Route Type	SRCIPRT.RTTYPE	You are advised to set this parameter to NEXTHOP .
Next Hop IP	SRCIPRT.NEXTHOP	This parameter is valid when SRCIPRT.RTTYPE is set to NEXTHOP .

Parameter Name	Parameter ID	Setting Notes
Priority	SRCIPRT.PREF	This parameter can be configured if IP route backup is required. The priority is used for route identification. A smaller parameter value indicates a higher priority. Generally, activated routes have higher priorities. The system does not support route load sharing. Routes to the same destination network segment must have different priorities.

(Optional) The following table describes the parameters that must be set in an **SRCIPROUTE4** MO to configure a source IPv4 route. This MO is required only when **GTRANSPARA.TRANS CFGMODE** is set to **NEW** and the user-plane load sharing function is required.

Parameter Name	Parameter ID	Setting Notes
Source Route Index	SRCIPROUTE4.SRCRTID <i>X</i>	Set this parameter based on the network plan.
Source IP Address	SRCIPROUTE4.SRCIP	Set this parameter based on the network plan.
Route Type	SRCIPROUTE4.RTTYPE	You are advised to set this parameter to NEXTHOP .
Next Hop IP	SRCIPROUTE4.NEXTHO <i>P</i>	This parameter is valid when SRCIPROUTE4.RTTYPE is set to NEXTHOP .

Parameter Name	Parameter ID	Setting Notes
Priority	SRCIPROUTE4.PREF	This parameter can be configured if IP route backup is required. The priority is used for route identification. A smaller parameter value indicates a higher priority. Generally, activated routes have higher priorities. The system does not support route load sharing. Routes to the same destination network segment must have different priorities.

(Optional) The following table describes the parameters that must be set in an **SRCIPROUTE6** MO to configure a source IPv6 route. This MO is required only when **GTRANSPARA.TRANS CFGMODE** is set to **NEW** and the user-plane load sharing function is required.

Parameter Name	Parameter ID	Setting Notes
IPv6 Source Route Index	SRCIPROUTE6.SRCRTID <i>X</i>	Set this parameter based on the network plan.
Source IPv6 Address	SRCIPROUTE6.SRCIP	Set this parameter based on the network plan.
Route Type	SRCIPROUTE6.RTTYPE	You are advised to set this parameter to NEXTHOP .
Next-Hop IPv6 Address	SRCIPROUTE6.NEXTHO <i>P</i>	This parameter is valid when SRCIPROUTE6.RTTYPE is set to NEXTHOP .

Parameter Name	Parameter ID	Setting Notes
Preference	SRCIPROUTE6.PREF	This parameter can be configured if IP route backup is required. The priority is used for route identification. A smaller parameter value indicates a higher priority. Generally, activated routes have higher priorities. The system does not support route load sharing. Routes to the same destination network segment must have different priorities.

(Optional) The following table describes the parameters that must be set in a **GTPU** MO. This MO is required only when the user-plane load sharing and active/standby user plane functions are required.

Parameter Name	Parameter ID	Setting Notes
Static Check Switch	GTPU.STATICCHK	If the user-plane load sharing and active/standby user plane functions are required, set this parameter to ENABLE to enable GTP-U static detection for the system.

(Optional) The following table describes the parameters that must be set in a **GEPMODEL PARA** MO. This MO is required only when the user-plane load sharing and active/standby user plane functions are required.

Parameter Name	Parameter ID	Setting Notes
Static Check Mode	GEPMODEL PARA.STATICCHKMODE	If the user-plane load sharing and active/standby user plane functions are required, set this parameter to EPSTATICCHK to enable static detection for the endpoint group.

4.4.2.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

NG Self-Setup

NOTE

This section provides only MML command examples of the NG interface. For packet filtering protected NG self-setup, packet filtering also needs to be configured. For details, see *Equipment Security*. If a dedicated transmission resource group needs to be specified for the NG interface, a transmission resource group must be configured. For details, see *Transmission Resource Management*.

Common scenarios: configurations when the user-plane load sharing and active/standby user plane functions are not required

With the NG-C and NG-U peer manually configured:

```
//(Optional) Adding an SCTP template. By default, the gNodeB has an SCTP template with
SCTPTemplateID set to 0.
ADD SCTPTemplate: SCTPTemplateID=1, SWITCHBACKFLAG=ENABLE;
//Adding an NG-C host (IPv4 used as an example)
ADD SCTPHOST: SCTPHOSTID=0, IPVERSION=IPv4, SIGIPV4="5.5.3.37", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=38412, SIMPLEMODESWITCH=SIMPLE_MODE_OFF, SCTPTemplateID=1;
//Adding an NG-C peer
ADD SCTPPEER: SCTPPEERID=0, IPVERSION=IPv4, SIGIPV4="138.32.1.20", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=38412;
//Adding an NG-U host
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="5.5.3.37", IPSECSWITCH=DISABLE;
//Adding an NG-U peer
ADD USERPLANEPEER: UPPEERID=0, IPVERSION=IPv4, PEERIPV4="195.168.1.28", IPSECSWITCH=DISABLE;
//Adding an endpoint group
ADD EPGROUP: EPGROUPID=0;
//Adding the NG-C host to the endpoint group
ADD SCTPHOST2EPGRP: EPGROUPID=0, SCTPHOSTID=0;
//Adding the NG-C peer to the endpoint group
ADD SCTPPEER2EPGRP: EPGROUPID=0, SCTPPEERID=0;
//Adding the NG-U host to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=0;
//Adding the NG-U peer to the endpoint group
ADD UPPEER2EPGRP: EPGROUPID=0, UPPEERID=0;
//Adding a gNodeB CU NG object, with CpEpgroupId being EPGROUPID of the referenced NG control plane
and UpEpgroupId being EPGROUPID of the referenced NG user plane
ADD GNBCUNG: gNBCuNgId=0, CpEpgroupId=0, UpEpgroupId=0,
ProtocolCrSwitch=EXTENDED_RAN_NODE_NAME_SW-1;
```

With the NG-C peer manually configured and the NG-U peer automatically configured:

```
//(Optional) Adding an SCTP template. By default, the gNodeB has an SCTP template with
SCTPTemplateID set to 0.
ADD SCTPTemplate: SCTPTemplateID=1, SWITCHBACKFLAG=ENABLE;
//Adding an NG-C host (IPv4 used as an example)
ADD SCTPHOST: SCTPHOSTID=0, IPVERSION=IPv4, SIGIPV4="5.5.3.37", SIGIP1SECSWITCH=DISABLE,
```

```
SIGIP2SECSWITCH=DISABLE, PN=38412, SIMPLEMODESWITCH=SIMPLE_MODE_OFF, SCTPTemplateID=1;
//Adding an NG-C peer
ADD SCTPPEER: SCTPPEERID=0, IPVERSION=IPv4, SIGIP1V4="138.32.1.20", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=38412;
//Adding an NG-U host
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="5.5.3.37", IPSECSWITCH=DISABLE;
//Adding an endpoint group
ADD EPGROUP: EPGROUPID=0;
//Adding the NG-C host to the endpoint group
ADD SCTPHOST2EPGRP: EPGROUPID=0, SCTPHOSTID=0;
//Adding the NG-C peer to the endpoint group
ADD SCTPPEER2EPGRP: EPGROUPID=0, SCTPPEERID=0;
//Adding the NG-U host to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=0;
//Adding a gNodeB CU NG object, with CpEpGroupId being EPGROUPID of the referenced NG control plane
and UpEpGroupId being EPGROUPID of the referenced NG user plane
ADD GNBCUNG: gNBCuNgId=0, CpEpGroupId=0, UpEpGroupId=0,
ProtocolCrSwitch=EXTENDED_RAN_NODE_NAME_SW-1;
```

With the NG-C peer automatically configured and the NG-U peer manually configured:

```
//(Optional) Adding an SCTP template. By default, the gNodeB has an SCTP template with
SCTPTemplateID set to 0.
ADD SCTPTemplate: SCTPTemplateID=1, SWITCHBACKFLAG=ENABLE;
//Adding an NG-C host (IPv4 used as an example)
ADD SCTPHOST: SCTPHOSTID=0, IPVERSION=IPv4, SIGIP1V4="5.5.3.37", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=38412, SIMPLEMODESWITCH=SIMPLE_MODE_OFF, SCTPTemplateID=1;
//Adding a DNS server, configuring the AMF domain name, and turning on the NG-C self-setup switch
ADD DNSSRV: DNSSRVID=2, IPVER=IPv4, DNSCIp4="20.2.7.37", DNSSIP4="185.22.63.78";
ADD GNBAMFDomainName: gNBCuNgId=0, AmfDomainName="amf1";
MOD GNBNGSONCONFIG: NgcSonSwitch=NGC_SON_SETUP_SW-1;
//Adding an NG-U host
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="5.5.3.37", IPSECSWITCH=DISABLE;
//NG-U self-setup can be classified into NG-U self-setup with peers manually configured and with peers
automatically configured. Use either configuration mode.
//Adding an NG-U peer
ADD USERPLANEPEER: UPPEERID=0, IPVERSION=IPv4, PEERIPV4="195.168.1.28", IPSECSWITCH=DISABLE;
//Adding an endpoint group
ADD EPGROUP: EPGROUPID=0;
//Adding the NG-C host to the endpoint group
ADD SCTPHOST2EPGRP: EPGROUPID=0, SCTPHOSTID=0;
//Adding the NG-U host to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=0;
//Adding the NG-U peer to the endpoint group
ADD UPPEER2EPGRP: EPGROUPID=0, UPPEERID=0;
//Adding a gNodeB CU NG object, with CpEpGroupId being EPGROUPID of the referenced NG control plane
and UpEpGroupId being EPGROUPID of the referenced NG user plane
ADD GNBCUNG: gNBCuNgId=0, CpEpGroupId=0, UpEpGroupId=0,
ProtocolCrSwitch=EXTENDED_RAN_NODE_NAME_SW-1;
```

With the NG-C and NG-U peer automatically configured:

```
//(Optional) Adding an SCTP template. By default, the gNodeB has an SCTP template with
SCTPTemplateID set to 0.
ADD SCTPTemplate: SCTPTemplateID=1, SWITCHBACKFLAG=ENABLE;
//Adding an NG-C host (IPv4 used as an example)
ADD SCTPHOST: SCTPHOSTID=0, IPVERSION=IPv4, SIGIP1V4="5.5.3.37", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=38412, SIMPLEMODESWITCH=SIMPLE_MODE_OFF, SCTPTemplateID=1;
ADD DNSSRV: DNSSRVID=2, IPVER=IPv4, DNSCIp4="20.2.7.37", DNSSIP4="185.22.63.78";
ADD GNBAMFDomainName: gNBCuNgId=0, AmfDomainName="amf1";
MOD GNBNGSONCONFIG: NgcSonSwitch=NGC_SON_SETUP_SW-1;
//Adding an NG-U host
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="5.5.3.37", IPSECSWITCH=DISABLE;
//NG-U self-setup can be classified into NG-U self-setup with peers manually configured and with peers
automatically configured. Use either configuration mode.
//Adding an endpoint group
ADD EPGROUP: EPGROUPID=0;
//Adding the NG-C host to the endpoint group
ADD SCTPHOST2EPGRP: EPGROUPID=0, SCTPHOSTID=0;
```

```
//Adding the NG-U host to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=0;
//Adding a gNodeB CU NG object, with CpEpGroupId being EPGROUPID of the referenced NG control plane
and UpEpGroupId being EPGROUPID of the referenced NG user plane
ADD GNBCUNG: gNBCuNgId=0, CpEpGroupId=0, UpEpGroupId=0,
ProtocolCrSwitch=EXTENDED_RAN_NODE_NAME_SW-1;
```

User-plane load sharing scenarios (both IPv4 and IPv6 are supported): The following configurations are required for user-plane load sharing. Other configurations are the same as those in common scenarios.

Old model: Only IPv4 networking is supported.

```
//Configuring the source-based routing when GTRANSPARA.TRANSFCFGMODE is set to OLD
ADD SRCIPRT: SRCRTIDX=0, SN=7, SBT=BASE_BOARD, SRCIP="5.5.3.37", RTTYPE=NEXTHOP,
NEXTHOP="10.1.1.2", PREF=60;
ADD SRCIPRT: SRCRTIDX=1, SN=7, SBT=BASE_BOARD, SRCIP="5.5.3.38", RTTYPE=NEXTHOP,
NEXTHOP="10.1.1.3", PREF=60;
//Adding two NG-U hosts for load sharing
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="5.5.3.37", IPSECSWITCH=DISABLE;
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="5.5.3.38", IPSECSWITCH=DISABLE;
//Turning on the GTP-U static detection switches
MOD GTPU: STATICCHK=ENABLE;
SET GEPMODEL PARA: STATICCHKMODE=EPSTATICCHK;
//Adding an endpoint group with the STATICCHK parameter specified
ADD EPGROUP: EPGROUPID=0, STATICCHK=ENABLE;
//Adding the two NG-U hosts for load sharing to the same endpoint group
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=0;
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=1;
```

New model: IPv4 networking

```
//Configuring the source-based routing when GTRANSPARA.TRANSFCFGMODE is set to NEW
ADD SRCIPROUTE4: SRCRTIDX=0, SRCIP="5.5.3.37", RTTYPE=NEXTHOP, NEXTHOP="10.1.1.2", PREF=60;
ADD SRCIPROUTE4: SRCRTIDX=1, SRCIP="5.5.3.38", RTTYPE=NEXTHOP, NEXTHOP="10.1.1.3", PREF=60;
//Adding two NG-U hosts for load sharing
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="5.5.3.37", IPSECSWITCH=DISABLE;
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="5.5.3.38", IPSECSWITCH=DISABLE;
//Turning on the GTP-U static detection switches
MOD GTPU: STATICCHK=ENABLE;
SET GEPMODEL PARA: STATICCHKMODE=EPSTATICCHK;
//Adding an endpoint group with the STATICCHK parameter specified
ADD EPGROUP: EPGROUPID=0, STATICCHK=ENABLE;
//Adding the two NG-U hosts for load sharing to the same endpoint group
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=0;
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=1;
```

New model: IPv6 networking

```
//Configuring the source-based routing when GTRANSPARA.TRANSFCFGMODE is set to NEW
ADD SRCIPROUTE6: SRCRTIDX=0, SRCIP="2001::25", RTTYPE=NEXTHOP, NEXTHOP="2002::2", PREF=60;
ADD SRCIPROUTE6: SRCRTIDX=1, SRCIP="2001::26", RTTYPE=NEXTHOP, NEXTHOP="2002::3", PREF=60;
//Adding two NG-U hosts for load sharing
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv6, LOCIPV6="2001::25", IPSECSWITCH=DISABLE;
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv6, LOCIPV6="2001::26", IPSECSWITCH=DISABLE;
//Turning on the GTP-U static detection switches
MOD GTPU: STATICCHK=ENABLE;
SET GEPMODEL PARA: STATICCHKMODE=EPSTATICCHK;
//Adding an endpoint group with the STATICCHK parameter specified
ADD EPGROUP: EPGROUPID=0, STATICCHK=ENABLE;
//Adding the two NG-U hosts for load sharing to the same endpoint group
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=0;
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=1;
```

When the user-plane active/standby function is required (both IPv4 and IPv6 are supported): The following configurations are required for the user-plane active/standby function. Other configurations are the same as those in common scenarios.

IPv4 networking:

```
//Configuring the GTP-U static detection function
MOD GTPU: STATICCHK=ENABLE;
SET GEPMODELpara: STATICCHKMODE=EPSTATICCHK;
//Adding two NG-U hosts in active/standby mode with the FLAG parameter specified
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="5.5.3.37", IPSECSWITCH=DISABLE,
FLAG=MASTER;
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="5.5.3.38", IPSECSWITCH=DISABLE,
FLAG=SLAVE;
//Adding an endpoint group with the STATICCHK parameter specified
ADD EPGROUP: EPGROUPID=0, STATICCHK=ENABLE;
//Adding the two NG-U hosts in active/standby mode to the same endpoint group
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=0;
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=1;
```

IPv6 networking:

```
//Configuring the GTP-U static detection function
MOD GTPU: STATICCHK=ENABLE;
SET GEPMODELpara: STATICCHKMODE=EPSTATICCHK;
//Adding two NG-U hosts in active/standby mode with the FLAG parameter specified
ADD USERPLANEHOST: UPHOSTID=2, IPVERSION=IPv6, LOCIPV6="2001::25", IPSECSWITCH=DISABLE,
FLAG=MASTER;
ADD USERPLANEHOST: UPHOSTID=3, IPVERSION=IPv6, LOCIPV6="2001::26", IPSECSWITCH=DISABLE,
FLAG=SLAVE;
//Adding an endpoint group with the STATICCHK parameter specified
ADD EPGROUP: EPGROUPID=0, STATICCHK=ENABLE;
//Adding the two NG-U hosts in active/standby mode to the same endpoint group
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=2;
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=3;
```

NG Self-Removal

```
//Turning on the NG-C self-removal switch
MOD GNBNGSONCONFIG: NgcSonSwitch=NGC_SON_DELETE_SW-1;
//Turning on the NG-U self-removal switch
SET GEPMODELpara: UPAUTODELSW=ENABLE;
```

Self-Setup of Multiple TNLAs over the NG Interface

```
//Turning on the switch for self-setup of multiple TNLAs
MOD GNBNGSONCONFIG: NgcSonSwitch=MULTI_TNLA_SELF_SETUP_SW-1;
```

Self-Removal of Faulty TNLAs in the Context of Multiple TNLAs over the NG Interface

```
//Turning on the switch for self-removal of faulty TNLAs in the context of multiple TNLAs
MOD GNBNGSONCONFIG: NgcSonSwitch=FAULT_MULTI_TNLA_SELF_RMV_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Deactivating NG-C self-setup
MOD GNBNGSONCONFIG: NgcSonSwitch=NGC_SON_SETUP_SW-0;
//Deactivating NG-C self-removal
MOD GNBNGSONCONFIG: NgcSonSwitch=NGC_SON_DELETE_SW-0;
//Deactivating self-setup of multiple TNLAs over the NG interface
MOD GNBNGSONCONFIG: NgcSonSwitch=MULTI_TNLA_SELF_SETUP_SW-0;
//Deactivating self-removal of faulty TNLAs in the context of multiple TNLAs over the NG interface
MOD GNBNGSONCONFIG: NgcSonSwitch=FAULT_MULTI_TNLA_SELF_RMV_SW-0;
//Deactivating NG-U self-removal
SET GEPMODELpara: UPAUTODELSW=DISABLE;
```


4.4.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.3 Activation Verification

After the peer IP address of the NG interface is manually configured, perform the following activation verification procedures for the NG interface.

Step 1 Check the status of the NG interface.

- Run the **DSP GNBCUNGINTERFACE** command to query the values of **gNodeB CU NG Interface CP Bearer State** and **gNodeB CU NG Interface CP Bearer ID**.
- Run the **DSP CPBEARER** command to query the CP bearer state corresponding to the **gNodeB CU NG Interface CP Bearer ID**.

Expected result:

- If the value of **gNodeB CU NG Interface CP Bearer State** is **Normal**, the gNodeB CU NG interface is working properly.
- If the CP bearer state corresponding to the **gNodeB CU NG Interface CP Bearer ID** is normal, the SCTP link is working properly.

Step 2 Start a service.

Enable a UE to access the network, start a service on the UE, and observe whether the service runs properly.

Expected result: The service is established, indicating that NG self-setup has taken effect.

----End

After the NG-C peer is automatically configured, perform the following activation verification procedures for the NG-C self-setup, self-update, and self-removal.

Step 1 Run the **DSP DNS** command to check the DNS query results.

Expected result: The resolution succeeds.

Step 2 On the MAE-Access, choose **SON > SON Log > Synchronize NE Log**.

Step 3 On the **Synchronize NE Log** tab page, click **Manual Synchronization**. In the displayed dialog box, select NEs and click **OK**.

Step 4 On the **Query SON Log** tab page, select **NR NG Interface Self Management Log** from the **Log Category** drop-down list box in the upper left corner. In the **Event Name** area, select **Automatically Establish Signaling Link**, **Automatically Update Signaling Link**, **Automatically Delete Signalling Link**, **Automatically Establish Service Link**, and **Automatically Delete Service Link**. Then, select

target gNodeBs in the **Log Source** area to check different types of NG self-configuration SON logs.

----End

Expected result: The SON logs contain NG-C self-setup, self-update, and self-removal events.

4.4.4 Network Monitoring

If the NG interface is faulty during the gNodeB operation, check whether any of the alarms listed in [Table 4-3](#) have been reported. If an alarm has been reported, clear the alarm by referring to the alarm reference.

Table 4-3 Alarms related to the NG interface

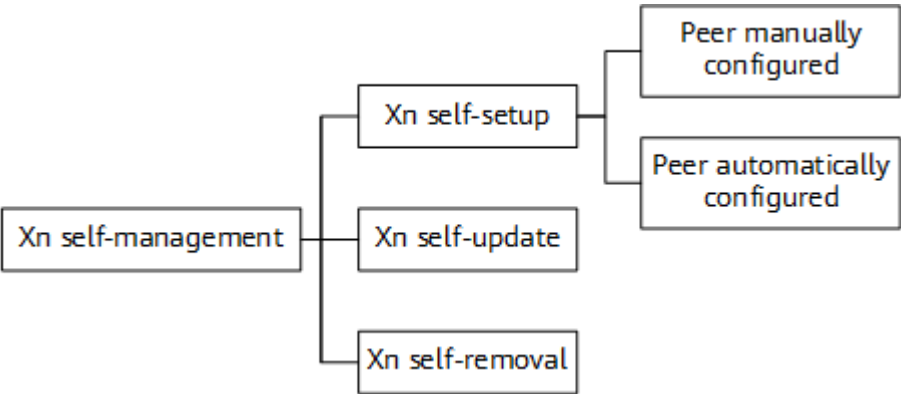
Alarm ID	Alarm Name	Severity
29815	gNodeB NG Interface Fault	Major
29816	gNodeB NG Control Plane Transmission Interruption	Critical
25954	User Plane Fault	Major
25888	SCTP Link Fault	Major
25958	DNS Resolution Failure	Major

5 Xn Self-Management

5.1 Principles

Figure 5-1 shows Xn self-management.

Figure 5-1 Xn self-management



5.1.1 Xn Self-Setup

Xn self-setup is performed in either of the following methods:

- Xn self-setup with peers manually configured
SCTPPEER and **USERPLANEPEER** MOs are configured manually. If there are multiple Xn interfaces between gNodeBs, multiple peer configurations must be manually added.
- Xn self-setup with peers automatically configured
Peers do not need to be configured manually. When the UE undergoes inter-gNodeB handovers and the source gNodeB detects that the Xn links do not exist or are faulty, the gNodeB exchanges Xn transport layer configuration information with the peer gNodeB through the AMF. In this procedure, the local gNodeB obtains the control-plane and user-plane IP addresses of the peer gNodeB.

NOTE

The **gNBCUXnOp** MO must be configured for the Xn interface in multi-operator sharing scenarios. For details, see *Multi-Operator Sharing*.

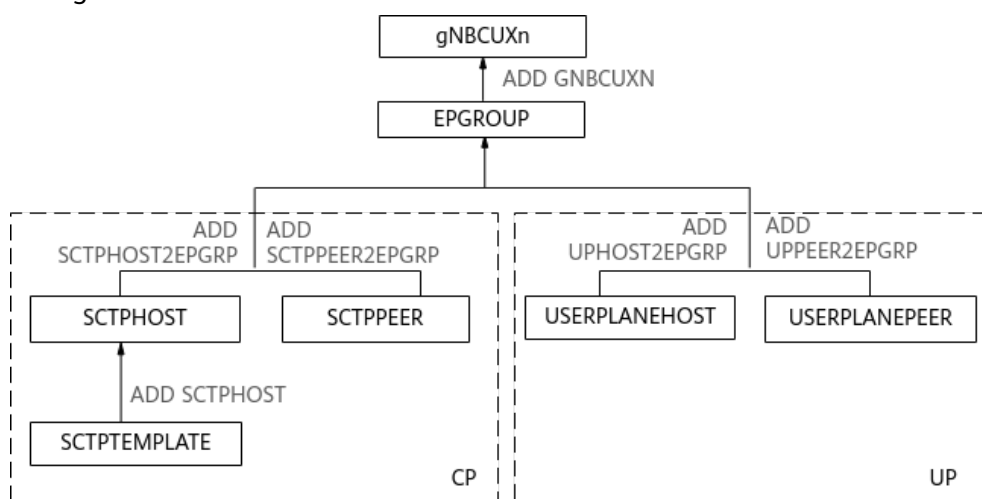
It is recommended that only one **gNBCUXn** MO be configured for one operator.

5.1.1.1 IPv4/IPv6 Single-Stack Transmission

5.1.1.1.1 Xn Self-Setup with Peers Manually Configured

An Xn interface is classified into Xn-C and Xn-U interfaces. The Xn-C interface transmits control-plane data, and the Xn-U interface transmits user-plane data. [Figure 5-2](#) shows the relationships between MOs involved in Xn self-setup with peers manually configured.

Figure 5-2 Relationships between MOs involved in self-setup with peers manually configured



ADD xxx refers to the MML command for correlating MOs.

The following describes how the MOs in [Figure 5-2](#) relate to each other.

- Control plane
 - SCTPHOST** and **SCTPPEER** MOs are used to set parameters such as the local and peer IP addresses and port numbers of a control-plane link.
 - In IPv4 networking, the Sctp link uses the values of the **SCTPHOST.SIGIP1V4** and **SCTPPEER.SIGIP1V4** parameters as the local and peer IP addresses, respectively.
 - In IPv6 networking, the Sctp link uses the values of the **SCTPHOST.SIGIP1V6** and **SCTPPEER.SIGIP1V6** parameters as the local and peer IP addresses, respectively.
 - The **SCTPHOST.PN** and **SCTPPEER.PN** parameters must be set to the same value. According to 3GPP specifications, the recommended value is **38422**. For details, see chapter 7 "Transport layer" in 3GPP TS 38.422 V15.1.0.

The gNodeB automatically sets up a multihoming Sctp link if second IP addresses are configured in the **SCTPHOST** and **SCTPPEER** MOs.

 NOTE

The **SCTPTEMPLATE** MO consists of optional parameters common for configuring SCTP links. The template is referenced by a control-plane link through the **SCTPHOST.SCTPTEMPLATEID** parameter, whose value is the same as the **SCTPTEMPLATE.SCTPTEMPLATEID** parameter.

- User plane

USERPLANEHOST and **USERPLANEPEER** MOs are used to set parameters such as the local and peer IP addresses of a user-plane path, respectively.

- In IPv4 networking, the gNodeB uses a local Xn user-plane IP address (specified by **USERPLANEHOST.LOCIPV4**) and a peer Xn user-plane IP address (specified by **USERPLANEPEER.PEERIPV4**).
- In IPv6 networking, the gNodeB uses a local Xn user-plane IPv6 address (specified by **USERPLANEHOST.LOCIPV6**) and a peer IPv6 address (specified by **USERPLANEPEER.PEERIPV6**).

- Endpoint group

In the same endpoint group, links are set up between the local control plane and the peer control plane or set up between the local user plane and the peer user plane. When adding the local and peer control and user planes to endpoint groups by using the **EPGROUP** MO, note the following:

- The local and peer control planes of an Xn interface must be added to the same endpoint group.
- The local and peer user planes of an Xn interface must be added to the same endpoint group.

- Xn interface

The **gNBCUXn** MO is used to add an Xn interface object and bind it with the endpoint group.

 NOTE

The IP version of the Xn-U links between two gNodeBs must be consistent with that of the corresponding Xn-C links.

When the Xn and X2 interfaces use the same local and peer user-plane IP addresses:

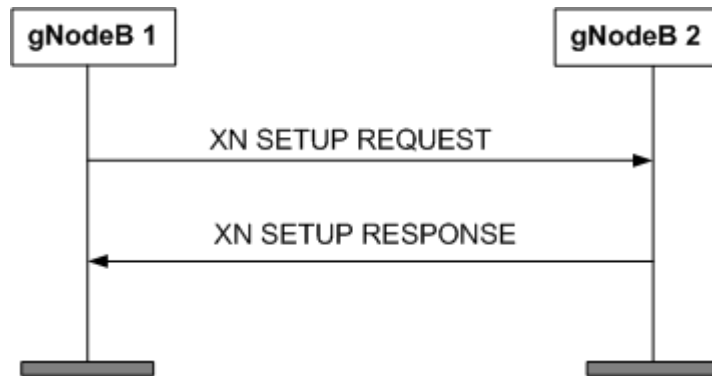
- The Xn and X2 interfaces must be configured with the same user-plane **EPGROUP** MO. Otherwise, only the first user-plane path can be set up.
- A user-plane link cannot exist in two different user-plane endpoint groups (specified by the **EPGROUP** MO) at the same time. If you need to adjust a user-plane peer (specified by the **USERPLANEPEER** MO) to the user-plane endpoint group of a different interface, ensure that the user-plane peer has been removed from the user-plane endpoint group of the original interface.

In NSA and SA hybrid networking, the IP address of the control-plane host of the Xn interface can only be identical to that of the X2 interface on the gNodeB side. If these IP addresses are identical, the local SCTP port numbers (specified by **SCTPHOST.PM**) of the Xn and X2 interfaces must be different.

If non-direct IPsec is used for the user-plane host and the Xn control plane is normal, the user-plane IP address of the peer base station is carried in a control-plane message. In such cases, the **USERPLANEPEER** MO can be automatically configured.

After the preceding configurations are complete, the gNodeB automatically initiates Xn self-setup. gNodeB 1 sends an XN SETUP REQUEST message, as shown in [Figure 5-3](#). After receiving this message, gNodeB 2 responds with an XN SETUP RESPONSE message. The Xn interface is then set up based on the process defined in section 8.4.1 "Xn Setup" in 3GPP TS 38.423 V15.3.0.

Figure 5-3 Xn self-setup with peers manually configured



5.1.1.1.2 Xn Self-Setup with Peers Automatically Configured

Self-setup with peers automatically configured is recommended because it reduces manual operations, as well as network planning and optimization costs.

AMF-based Xn Self-Setup with Peers Automatically Configured

In SA networking, when a peer is automatically configured, Xn self-setup is triggered by inter-gNodeB handovers or certain inter-gNodeB coordination-based features (such as inter-gNodeB CA).

The **XN_SON_SETUP_SW** option of the **gNBXnSonConfig.XnSonConfigSwitch** parameter determines whether to enable this function.

Figure 5-4 shows the Xn self-setup procedure.

Xn self-setup with peers automatically configured requires the following MOs to have been configured: **SCTPHOST**, **USERPLANEHOST**, **gNBCUXn**, **EPGROUP**, and **SCTEMPLATE**.

NOTE

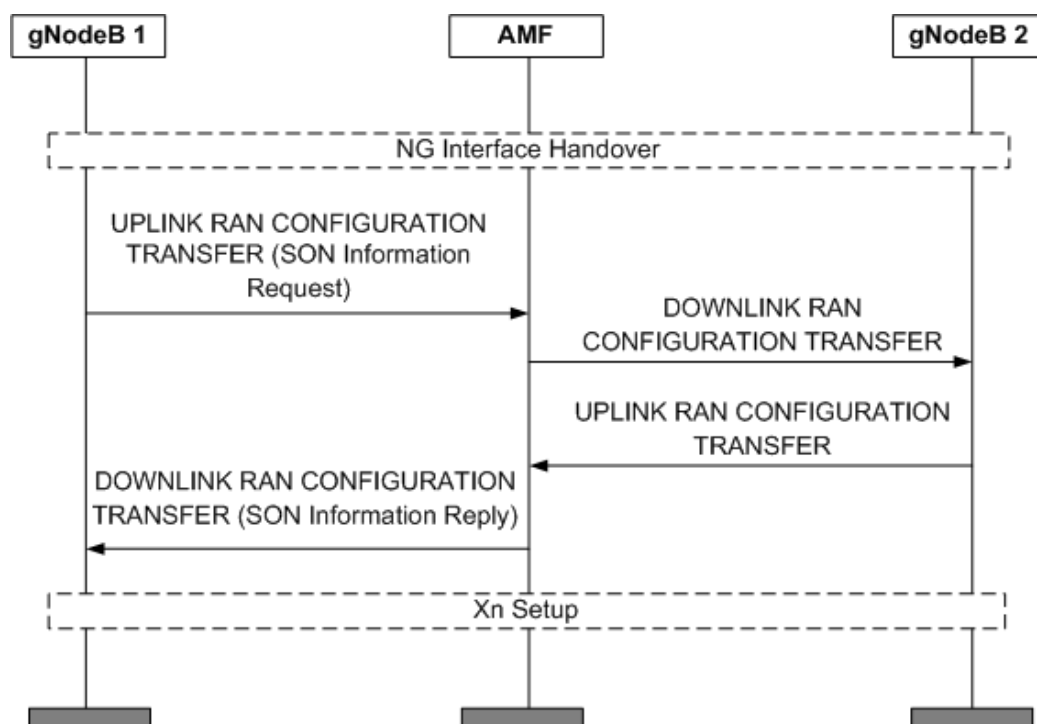
When the Xn and X2 interfaces use the same local user-plane IP address, the following rules apply:

- The Xn and X2 interfaces must be configured with the same user-plane **EPGROUP** MO. Otherwise, only the first user-plane path can be set up.
- A user-plane link cannot exist in two different user-plane endpoint groups (specified by the **EPGROUP** MO) at the same time. If you need to adjust a user-plane peer (specified by the **USERPLANEPEER** MO) to the user-plane endpoint group of a different interface, ensure that the user-plane peer has been removed from the user-plane endpoint group of the original interface.

In NSA and SA hybrid networking, the IP address of the control-plane host of the Xn interface can only be identical to that of the X2 interface on the gNodeB side. If these IP addresses are identical, the local SCTP port numbers (specified by **SCTPHOST.PM**) of the Xn and X2 interfaces must be different.

After the source base station initiates an Xn self-setup to a target base station, additional Xn self-setup to the target base station cannot be initiated within 10 minutes.

Figure 5-4 AMF-based Xn self-setup with peers automatically configured



The detailed procedure is as follows:

1. Xn self-setup, which is usually triggered by inter-gNodeB handovers or certain inter-gNodeB coordination-based features (such as inter-gNodeB CA), requires the peer gNodeB ID as the input. For details about the handover procedure, see *SA Mobility Management in Connected Mode*.
 - If the Xn link is available and operational, an Xn self-setup procedure will not be triggered.
 - If the Xn link is unavailable or faulty, an Xn self-setup procedure will be triggered.
2. The source gNodeB sends an UPLINK RAN CONFIGURATION TRANSFER message to the AMF. For details, see section 8.8.1 "Uplink RAN Configuration Transfer" in 3GPP TS 38.413 V15.2.0. The message contains the following information:
 - *Source gNB-ID* IE, which consists of the *Global gNB ID* and *Selected TAI* IEs
 - *Target gNB-ID* IE, which consists of the *Global gNB ID* and *Selected TAI* IEs
 - Control- and user-plane IP addresses of the source gNodeB
3. The AMF sends the DOWNLINK RAN CONFIGURATION TRANSFER message that carries the source gNodeB information to the target gNodeB. For details, see section 8.8.2 "Downlink RAN Configuration Transfer" in 3GPP TS 38.413 V15.2.0.
4. Upon receiving the DOWNLINK RAN CONFIGURATION TRANSFER message, the target gNodeB automatically creates related MOs to set up control-plane and user-plane transmission links.

- a. The target gNodeB automatically creates the **SCTPPEER** and **USERPLANEPEER** MOs according to the IP addresses in the message. It then automatically adds the control-plane and user-plane peers identified by the **SCTPPEER.SCTPPEERID** and **USERPLANEPEER.UPPEERID** parameters to the **EPGROUP** MO corresponding to the Xn interface.
- b. The target gNodeB automatically creates the **SCTPLNK**, **CPBEARER**, and **gNBCUXnInterface** MOs based on the following MOs:
 - Manually configured **SCTPHOST**, **USERPLANEHOST**, **gNBCUXn**, **EPGROUP**, and **SCTPTEMPLATE** MOs;
 - Automatically generated **SCTPPEER** and **USERPLANEPEER** MOs;
- c. The above message indicates the peer IP address and not the SCTP port number. In this case, the gNodeB automatically sets **SCTPPEER.PN** to the same value as **SCTPHOST.PN**. 3GPP specifications recommend that the local SCTP port number be set to 38422.
5. The target gNodeB includes its Xn-C and Xn-U IP addresses in the UPLINK RAN CONFIGURATION TRANSFER message sent to the AMF.
6. The AMF forwards the DOWNLINK RAN CONFIGURATION TRANSFER message to the source gNodeB.
7. Upon receiving the DOWNLINK RAN CONFIGURATION TRANSFER message, the source gNodeB automatically creates related MOs to set up control-plane and user-plane transmission links.
 - a. The source gNodeB automatically creates the **SCTPPEER** and **USERPLANEPEER** MOs based on the control-plane and user-plane IP addresses of the target gNodeB contained in the message sent from the AMF. It then automatically adds the control-plane and user-plane peers identified by the **SCTPPEER.SCTPPEERID** and **USERPLANEPEER.UPPEERID** parameters to the **EPGROUP** MO corresponding to the Xn interface.
 - b. The source gNodeB automatically creates the **SCTPLNK**, **CPBEARER**, and **gNBCUXnInterface** MOs based on the following MOs:
 - Manually configured **SCTPHOST**, **USERPLANEHOST**, **gNBCUXn**, **EPGROUP**, and **SCTPTEMPLATE** MOs;
 - Automatically generated **SCTPPEER** and **USERPLANEPEER** MOs;If the message sent from the AMF contains two control-plane addresses of the target gNodeB, the source gNodeB automatically configures two SCTP peer IP addresses. If the source gNodeB has two SCTP host IP addresses, it automatically establishes a dual-homing SCTP link.
 - c. The above message indicates the peer IP address and not the SCTP port number. The gNodeB automatically sets **SCTPPEER.PN** to the same value as **SCTPHOST.PN**.
8. The source gNodeB initiates an Xn setup request message and the target gNodeB responds with an Xn setup response message. The Xn interface is then set up.

During the automatic generation of the **SCTPPEER** and **USERPLANEPEER** MOs, the source and target gNodeBs automatically assign the values of the **SCTPPEER.USERLABEL** and **USERPLANEPEER.USERLABEL** parameters based on the following mechanism:

The gNodeB automatically assigns a value to the **SCTPPEER.USERLABEL** or **USERPLANEPEER.USERLABEL** parameter according to the peer gNodeB ID and the Xn interface's MCC and MNC. The format of the assigned value is "gNBXn:MCC_xxx MNC_xxx GlobalgNBId_xxx-xxx-xxxxxxx." Note that "gNBXn:MCC_xxx MNC_xxx" indicates the operator for which the **SCTPPEER** and **USERPLANEPEER** MOs are created. The gNodeB automatically assigns the **SCTPPEER.USERLABEL** value to the **SCTPLNK.DESCR** parameter.

In the preceding procedure, if the protocol versions of the source gNodeB, target gNodeB, and AMF are inconsistent, Xn self-setup fails. To address this issue, compatibility with the 3GPP specifications for Xn self-setup is therefore introduced.

The **TS38413_CR0304_SW** option of the **gNBCUNg.ProtocolCrSwitch** parameter determines whether to enable this function. If this option is selected, the UPLINK RAN CONFIGURATION TRANSFER and DOWNLINK RAN CONFIGURATION TRANSFER messages are processed according to the value of the *Xn TNL Configuration Info* IE defined in 3GPP TS 38.413 V15.6.0 or later. If this option is deselected, the UPLINK RAN CONFIGURATION TRANSFER and DOWNLINK RAN CONFIGURATION TRANSFER messages are processed according to the value of the *Xn TNL Configuration Info* IE defined in 3GPP TS 38.413 V15.5.0 or earlier. This function is disabled by default.

MAE-based Xn Self-Setup with Peers Automatically Configured

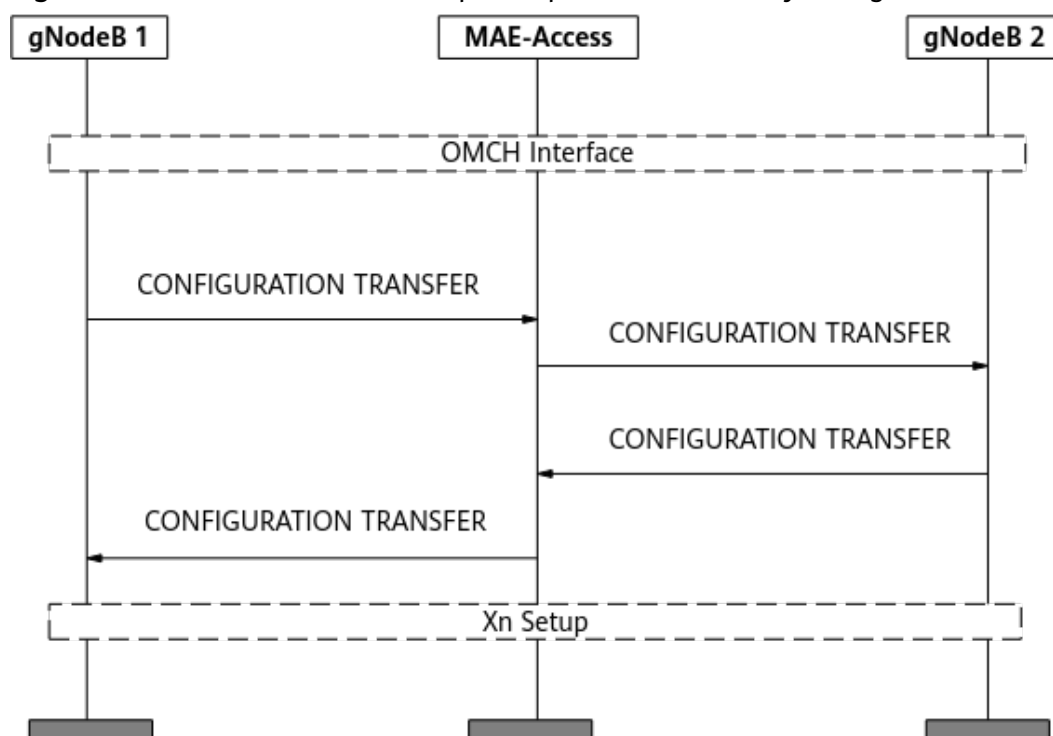
In SA and NSA hybrid networking, Xn self-setup with peers automatically configured is based on the AMF. If no NG interface is available between the gNodeB and AMF, Xn self-setup with peers automatically configured can be performed based on the MAE. Xn self-setup is usually triggered by inter-gNodeB handovers or certain inter-gNodeB coordination-based features (such as inter-gNodeB CA).

In NSA networking, Xn self-setup with peers automatically configured is based on the MAE. When a peer is automatically configured, Xn self-setup is triggered by certain inter-gNodeB coordination-based features (such as inter-gNodeB CA).

MAE-based Xn self-setup with peers automatically configured is also controlled by the **XN_SON_SETUP_SW** option of the **gNBXnSonConfig.XnSonConfigSwitch** parameter.

Figure 5-5 shows the MAE-based Xn self-setup procedure in which gNodeB 1 and gNodeB 2 must be connected to the same MAE.

Figure 5-5 MAE-based Xn self-setup with peers automatically configured



The detailed procedure is as follows:

1. Xn self-setup, which is usually triggered by inter-gNodeB handovers or certain inter-gNodeB coordination-based features (such as inter-gNodeB CA), requires the peer gNodeB ID as the input.
 - If the Xn link is available and operational, an Xn self-setup procedure will not be triggered.
 - If the Xn link is unavailable or faulty, an Xn self-setup procedure will be triggered.
2. The source gNodeB sends a CONFIGURATION TRANSFER message to the MAE-Access. The message contains the following Xn-interface-related parameters of the source gNodeB:
 - *Source gNB-ID IE*, which consists of the *Global gNB ID* and *Selected TAI* IEs
 - *Target gNB-ID IE*, which consists of the *Global gNB ID* and *Selected TAI* IEs
 - Control- and user-plane IP addresses of the source gNodeB
3. The MAE-Access forwards the CONFIGURATION TRANSFER message that carries the source gNodeB information to the target gNodeB.
4. Upon receiving the CONFIGURATION TRANSFER message, the target gNodeB automatically creates related MOs to set up control-plane and user-plane transmission links.
 - a. The target gNodeB automatically creates the **SCTPPEER** and **USERPLANEPEER** MOs according to the IP addresses in the message. It then automatically adds the control-plane and user-plane peers identified by the **SCTPPEER.SCTPPEERID** and **USERPLANEPEER.UPPEERID** parameters to the **EPGROUP** MO corresponding to the Xn interface.

- b. The target gNodeB automatically creates the **SCTPLNK**, **CPBEARER**, and **gNBCUXnInterface** MOs based on the following MOs:
 - Manually configured **SCTPHOST**, **USERPLANEHOST**, **gNBCUXn**, **EPGROUP**, and **SCTPTEMPLATE** MOs;
 - Automatically generated **SCTPPEER** and **USERPLANEPEER** MOs;
 - c. The above message indicates the peer IP address and not the SCTP port number. In this case, the gNodeB automatically sets **SCTPPEER.PN** to the same value as **SCTPHOST.PN**. 3GPP specifications recommend that the local SCTP port number be set to 38422.
 5. The target gNodeB includes its Xn-C and Xn-U IP addresses in the CONFIGURATION TRANSFER message sent to the MAE-Access.
 6. The MAE-Access forwards the CONFIGURATION TRANSFER message to the source gNodeB.
 7. Upon receiving the CONFIGURATION TRANSFER message, the source gNodeB automatically creates related MOs to set up control-plane and user-plane transmission links.
 - a. The source gNodeB automatically creates the **SCTPPEER** and **USERPLANEPEER** MOs based on the control-plane and user-plane IP addresses of the target gNodeB contained in the message sent from the MAE-Access. It then automatically adds the control-plane and user-plane peers identified by the **SCTPPEER.SCTPPEERID** and **USERPLANEPEER.UPPEERID** parameters to the **EPGROUP** MO corresponding to the Xn interface.
 - b. The source gNodeB automatically creates the **SCTPLNK**, **CPBEARER**, and **gNBCUXnInterface** MOs based on the following MOs:
 - Manually configured **SCTPHOST**, **USERPLANEHOST**, **gNBCUXn**, **EPGROUP**, and **SCTPTEMPLATE** MOs;
 - Automatically generated **SCTPPEER** and **USERPLANEPEER** MOs;If the message sent from the MAE-Access contains two control-plane addresses of the target gNodeB, the source gNodeB automatically configures two SCTP peer IP addresses. If the source gNodeB has two SCTP host IP addresses, it automatically establishes a dual-homing SCTP link.
 - c. The above message indicates the peer IP address and not the SCTP port number. The gNodeB automatically sets **SCTPPEER.PN** to the same value as **SCTPHOST.PN**.
 8. The source gNodeB initiates an Xn setup request message and the target gNodeB responds with an Xn setup response message. The Xn interface is then set up.

During the automatic generation of the **SCTPPEER** and **USERPLANEPEER** MOs, the source and target gNodeBs automatically assign the values of the **SCTPPEER.USERLABEL** and **USERPLANEPEER.USERLABEL** parameters based on the following mechanism:

The gNodeB automatically assigns a value to the **SCTPPEER.USERLABEL** or **USERPLANEPEER.USERLABEL** parameter according to the peer gNodeB ID and the Xn interface's MCC and MNC. The format of the assigned value is

"gNBXn:MCC_XXX MNC_XXX GlobalgNBId_XXX-XXX-XXXXXXX." Note that "gNBXn:MCC_XXX MNC_XXX" indicates the operator for which the **SCTPPEER** and **USERPLANEPEER** MOs are created. The gNodeB automatically assigns the **SCTPPEER.USERLABEL** value to the **SCTPLNK.DESCR** parameter.

5.1.1.2 IPv4/IPv6 Dual-Stack Transmission

The Xn interface supports IPv4/IPv6 dual-stack transmission when the local gNodeB is configured with both IPv4 and IPv6 addresses. IPv4/IPv6 dual-stack transmission over the Xn interface applies when the local gNodeB uses different IP versions to connect to different peer gNodeBs. For example, during a reconstruction from IPv4 to IPv6 transmission, some peer gNodeBs use IPv6 and some use IPv4. When the Xn interface supports IPv4/IPv6 dual-stack transmission, Xn links can be set up using either of the following methods.

Xn Self-Setup with Peers Manually Configured

During Xn self-setup with peers manually configured, if the local gNodeB has been configured with both IPv4 and IPv6 addresses, the following requirements must be met:

- The same IP version is used for the peer user and control planes of the Xn interface by setting the **SCTPPEER.IPVERSION** and **USERPLANEPEER.IPVERSION** parameters to the same value.
- Configuring both IPv4 and IPv6 addresses for a peer gNodeB should be avoided. If both the local and peer ends are configured with IPv4 and IPv6 addresses, both IPv4 and IPv6 links will be established, occupying twice the link resources, and the second Xn link is unavailable. As a result, ALM-29810 gNodeB Xn Interface Fault is generated.

Xn Self-Setup with Peers Automatically Configured

During Xn self-setup with peers automatically configured, if the local gNodeB has been configured with both IPv4 and IPv6 addresses, the local gNodeB automatically selects the IP address for the peer gNodeB based on the following rules:

- If the peer gNodeB is configured with only an IPv4 address, the IPv4 address is used for the peer.
- If the peer gNodeB is configured with only an IPv6 address, the IPv6 address is used for the peer.
- If the peer gNodeB is configured with both IPv4 and IPv6 addresses, the peer IP address is selected based on the value of the **EPGROUP.IPVERPREFERENCE** parameter.

If the local gNodeB has been configured with both IPv4 and IPv6 addresses, the following requirements must be met:

- The same IP version is used for both the user-plane and control-plane links of the Xn interface by setting the **EPGROUP.IPVERPREFERENCE** parameter to the same value for the Xn user and control planes.
- The **EPGROUP.IPVERPREFERENCE** parameter is set to the same value for the local and peer gNodeBs.

To prevent ALM-29810 gNodeB Xn Interface Fault from being generated when IPv4/IPv6 dual-stack addresses are configured for the local and peer gNodeBs, control-plane dual-stack check for the Xn interface can be enabled (by selecting the **XN_CP_DUALSTACK_CHK_SW** option of the **gNBInterfaceParam.InterfaceOmSw** parameter). In this case, only one available Xn control-plane link is retained between the local and peer gNodeBs, reducing unnecessary alarm reporting.

 NOTE

The modification of the **EPGROUP.IPVERPREFERENCE** parameter setting does not take effect for Xn links that have already been set up and are working properly, and takes effect only for links that need to be reestablished or that are newly set up.

5.1.1.3 Packet Filtering Protected Xn Self-Setup

If integrated firewall is enabled and packet filtering is configured on a gNodeB, ACL rules (defined in the **ACLRULE** MO under the **ACL** MO for IPv4, or defined in the **ACLRULE6** MO under the **ACL6** MO for IPv6, which is associated with the **PACKETFILTER** or **PACKETFILTERING** MO) must be configured for automatically established Xn-C and Xn-U links. When the **GTRANSPARA.TRANS CFGMODE** parameter is set to **OLD**, the **PACKETFILTER** MO is used. When this parameter is set to **NEW**, the **PACKETFILTERING** MO is used. ACL rules can be manually or automatically configured.

- If **EPGROUP.PACKETFILTERSWITCH** (IPv4)/**EPGROUP.PACKETFILTERSWITCH6** (IPv6) is set to **ENABLE** for an Xn object (**gNBCUXn**), the ACL rule self-configuration function for packet filtering is enabled during Xn self-setup. That is, two ACL rules are automatically created for the Xn-C and Xn-U interfaces during Xn self-setup based on the local and peer configurations of the Xn interface. The two ACL rules are then automatically added to the ACL for packet filtering.
 - You are advised not to manually modify the automatically created packet filtering rules. If you manually modify such rules, the Xn-C and Xn-U links may be disconnected. In this case, you can restore the packet filtering rules to those before modification. Alternatively, you can delete the incorrect packet filtering rules, and packet filtering rules will be automatically created in the next Xn self-setup procedure.
 - If the gNodeB determines that the number of ACL rules has exceeded the maximum allowed number, the **MOD EPGROUP** command output will be displayed after command execution, indicating that the number exceeds board capability. In this case, it is recommended that ACL rules be manually configured.
 - In Xn self-setup with peers automatically configured, if the Xn control-plane link is normal but the Xn user-plane link has not been set up or was automatically removed, the target base station cannot obtain the Xn user-plane IP address of the source base station through the data forwarding process during an inter-base-station handover. As a result, ACL rule self-configuration cannot be performed for the Xn user plane, and ACL rules for the Xn user plane must be manually configured.
- If **EPGROUP.PACKETFILTERSWITCH** (IPv4)/**EPGROUP.PACKETFILTERSWITCH6** (IPv6) is set to **DISABLE** for an Xn object (**gNBCUXn**), ACL rules for the peer base station must be manually configured on the gNodeB.

 NOTE

For details about packet filtering and ACL rule configuration, see *Equipment Security*.

5.1.2 Xn Self-Update

Xn self-update refers to Xn peer information self-update, which can be service-triggered or manually triggered.

 NOTE

When changing the IP address of the Xn interface, you are advised to disable the Xn dynamic blacklist function. For details, see [5.1.3 Xn Self-Removal](#).

5.1.2.1 Service-Triggered Self-Update of Xn Peer Information

During network operation, whether the IP address or operator information of the peer gNodeB changes depends on the following scenarios.

Changes in the IP Address of the Peer End

If the control-plane IP address of the peer end changes, the Xn interface to that peer end will become faulty. A faulty interface will trigger the Xn self-setup when the UE initiates a handover to the target gNodeB. The local end obtains the new control-plane IP address of the peer end during the self-setup. The local end searches for an **SCTPPEER** MO that contains the same operator information as indicated by **SCTPPEER.REMOTEID** based on operator information such as MNC, MCC, and gNodeB ID in the Configuration Transfer message. The local end then updates the control-plane IP address in the located **SCTPPEER** MO.

If the user-plane IP address of the peer end changes, the Xn interface to that peer end will become faulty. The local end will obtain the new user-plane IP address during the Xn self-setup or Xn-based handover to the peer gNodeB. The local end searches for a **USERPLANEPEER** MO that contains the same operator information as indicated by **USERPLANEPEER.REMOTEID** based on operator information such as MNC, MCC, and gNodeB ID in the Handover Request Acknowledge message. The local end then generates a new **USERPLANEPEER** MO based on the new user-plane IP address of the peer end. The faulty user-plane link is deleted through Xn-U self-removal. For details, see [Xn-U Self-Removal](#).

Changes in the Global gNodeB ID of the Peer End

If the peer Global gNodeB ID (composed of the MNC, MCC, and gNodeB ID) of the Xn interface changes, the peer gNodeB restarts, and the Xn interface becomes faulty. The restart will trigger an Xn setup, and the change of Global gNodeB ID of the neighboring gNodeB is considered the equivalent of adding a new gNodeB. In this case, the transmission configuration of the Xn interface will not be updated, and the **REMOTEID** parameter value will not be automatically updated either. The local end configuration can be updated to be the same as that of the peer end only after the Xn interface is re-established (re-establishment triggered when the IP address is changed or the Xn interface is removed).

Changes in the IP Address and Global gNodeB ID of the Peer End

If the peer IP address and Global gNodeB ID of the Xn interface change, the local end cannot obtain the peer information, and therefore the local end cannot

update the peer configuration. The change of Global gNodeB ID of the neighboring gNodeB is considered the equivalent of adding a new gNodeB. When a UE initiates a handover to the target gNodeB, an Xn self-setup is triggered. During this process, the local end of the Xn interface cannot find the peer end (**SCTPPEER** and **USERPLANEPEER** MOs) that matches the peer IP addresses contained in the Configuration Transfer message. Therefore, an Xn interface will be set up again based on the new peer IP addresses and Global gNodeB ID.

Changes in the SCTP Multi-Homing IP Addresses of the Peer End

SCTP multi-homing reconstruction on the peer end involves the following scenarios:

- Scenario 1: Both control-plane IP addresses of the peer endpoint are changed.
- Scenario 2: A second control-plane IP address is added to the peer endpoint, which evolves to a multi-homed endpoint.
- Scenario 3: The control-plane peer endpoint is multi-homed, and only one of its IP addresses is changed.

In scenario 1, the Xn interface between the local and peer ends becomes faulty after both control-plane IP addresses of the peer endpoint are changed. In this case, Xn self-setup is triggered only when a UE initiates a handover to the target gNodeB. If this happens, configuration of the **SCTPPEER** MO is updated. For details, see [Changes in the IP Address of the Peer End](#).

In scenarios 2 and 3, the SCTP link to the local gNodeB still works properly as one IP address of the peer SCTP multi-homed endpoint remains unchanged. The peer gNodeB sends an SCTP INIT message containing the updated peer IP address list to the local gNodeB. When the Xn interface works properly, the local gNodeB determines that the IP address list in the SCTP INIT message is inconsistent with the IP addresses in the **SCTPPEER** MO, the local and peer gNodeBs exchange proprietary messages (XNAP_PRIVATE_MSG) containing their respective configurations, to update configurations of the **SCTPPEER** MO.

5.1.2.2 Manually Triggered Self-Update of Xn Peer Information

For the Xn interface, the local end's IP addresses are referenced by the peer end's **SCTPPEER** and **USERPLANEPEER** MOs. If the local and peer gNodeBs are managed by the same MAE-Deployment, the MAE-Deployment can automatically associate and change the local IP addresses referenced by the peer end in the same virtual routing and forwarding (VRF) domain. When the local IP addresses are changed, the MAE-Deployment can automatically associate and change the IP addresses in the **SCTPPEER** and **USERPLANEPEER** MOs.

5.1.3 Xn Self-Removal

Xn self-removal includes Xn-C self-removal and Xn-U self-removal, which are implemented as follows:

- Xn self-removal: When the Xn control plane is removed, the Xn user plane is also removed. For details, see [Xn Self-Removal](#).
- Xn-U self-removal: Only the Xn user plane is removed and the Xn control plane is not removed. For details, see [Xn-U Self-Removal](#).

Xn Self-Removal

Adjustment and other operations during network construction result in redundant Xn interfaces on gNodeBs. The maximum number of Xn interfaces that a gNodeB supports is limited. New Xn interfaces cannot be configured if the total number of configured Xn interfaces of a gNodeB has reached the maximum allowed number. The Xn self-removal function automatically recognizes and removes redundant Xn interfaces that are generated during the self-setup, reducing manual maintenance costs. This function is controlled by the **XN_SON_DEL_SW** option of the **gNBXnSonConfig.XnSonConfigSwitch** parameter. Before enabling this function, you are advised to select the **XN_SON_SETUP_SW** option of the **gNBXnSonConfig.XnSonConfigSwitch** parameter to ensure that removed Xn interfaces can be automatically set up again when necessary.

When the **XN_SON_DEL_SW** option of the **gNBXnSonConfig.XnSonConfigSwitch** parameter is selected, the gNodeB supports Xn self-removal in the following scenarios:

- Xn self-removal and Xn dynamic blacklist based on the initial setup fault
An Xn interface is automatically removed if all of the following conditions are met:
 - The format of the **SCTPPEER.REMOTEID** parameter is "gNBXn:MCC_xxx MNC_xxx GlobalgNBId_xxx-xxx-xxxxxxx". The value of **SCTPPEER.REMOTEID** can be queried using the **LST SCTPPEER** command.
-  **NOTE**
- If the **SCTPPEER.CTRLMODE** parameter is set to **AUTO_MODE**, the Xn interface and the **SCTPLNK** and **SCTPPEER** MOs are removed.
- If the **SCTPPEER.CTRLMODE** parameter is set to **MANUAL_MODE**, the Xn interface and the **SCTPLNK** MO are removed, but the **SCTPPEER** MO is not removed.
- The Xn interface is initially set up and the lower-layer signaling link is faulty (in the initial setup phase, the gNodeB takes 2 to 3 minutes to determine whether the lower-layer SCTP link is faulty).
 - When the Xn interface is faulty for 10 minutes, the Xn self-removal from the local end is triggered. However, if the length of the timer for Xn self-removal based on Xn link faults is shorter than 10 minutes, Xn self-removal based on Xn link faults is preferentially triggered.

At the initial stage of network construction, lower-layer link faults frequently occur because network planning is not optimum and the transport network is unstable. To avoid unnecessary self-setup, users can manually remove the faulty Xn interfaces and manually configure the Xn blacklist. However, as manual operations are complex and prone to errors, the Xn dynamic blacklist function has been introduced. After Xn interfaces are removed by Xn self-removal based on the initial setup fault, the local gNodeB automatically creates an Xn dynamic blacklist based on the local and peer base station information about the removed Xn interfaces. This simplifies the configuration and reduces risks introduced by manual configurations. This function is controlled by the **XN_SON_DYN_BLACKLIST_SW** option of the **gNBXnSonConfig.XnSonConfigSwitch** parameter.

When the local and peer gNodeBs of the Xn interface are in the Xn dynamic blacklist, Xn interface self-setup is prohibited before the Xn dynamic blacklist

aging timer (420 minutes) expires. That is, the local gNodeB does not initiate a self-setup request for the corresponding Xn interface and does not respond to the Xn self-setup request initiated by the peer gNodeB. When the timer expires, the local gNodeB removes the peer gNodeB from the local gNodeB's dynamic blacklist.

When the Xn dynamic blacklist function is enabled, if the Xn self-update fails due to transmission faults after the local or peer IP address of the Xn interface is changed, the peer base station to which the Xn interface belongs is added to the Xn dynamic blacklist. As a result, services depending on the Xn link are affected. Therefore, it is recommended that the Xn dynamic blacklist function be enabled only in the early stage of network deployment when the transport network is unstable, then disabled once the transport network becomes stable.

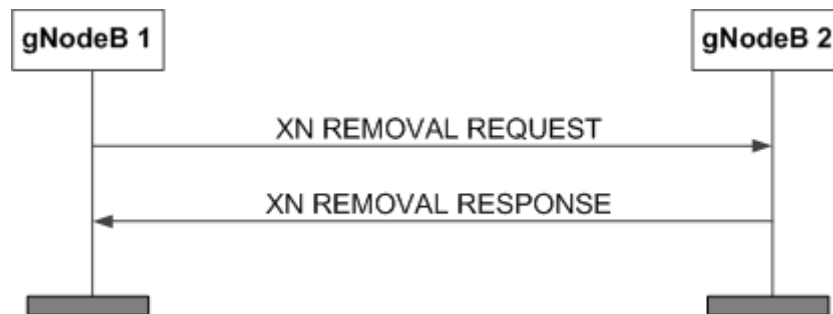
 NOTE

- The specification for the Xn dynamic blacklists is the same as those for Xn interfaces. If the number of entries in an Xn dynamic blacklist has reached the maximum and a new entry needs to be added, the new entry will replace the one with the shortest remaining aging time.
 - The **CLR GNBXNDYNBLACKLIST** command can be executed to delete Xn dynamic blacklists before the Xn dynamic blacklist aging timer expires.
 - The **DSP GNBXNDYNBLACKLIST** command can be executed to query Xn dynamic blacklists.
 - When the Xn dynamic blacklist switch is turned on, the peer gNodeB is removed even if it is in the Xn whitelist. This is because the peer gNodeB information cannot be obtained when an initial Xn self-setup procedure fails.
- Xn self-removal based on link faults
Xn self-removal at the local gNodeB will be triggered when the Xn interface malfunctions and remains that way for a specified period (one hour by default).
The **gNBXnSonConfig.XnSonDeleteTimerForXnFault** parameter specifies the length of the timer for Xn self-removal based on Xn link faults. Set this parameter based on the network plan. It is recommended that the timer length be set to a value smaller than 30 minutes only when Xn interfaces need to be quickly removed. Such scenarios include migration, capacity expansion, and network reconstruction.
If the **gNBXnSonConfig.XnSonDeleteTimerForXnFault** parameter is set to an excessively small value, Xn interfaces will be mistakenly removed when the network is unstable or the peer base station of the Xn interface is restarted. Therefore, it is recommended that the parameter value be greater than 5 minutes even when Xn interfaces need to be quickly removed.
 - Xn self-removal based on interface usage
The gNodeB periodically measures the Xn interface usage to remove Xn links with low usage. The usage is calculated based on the inter-gNodeB information (such as the number of inter-gNodeB handovers, number of inter-gNodeB coordination UEs, and other inter-gNodeB messages exchanged through Xn links) periodically measured between neighboring gNodeBs. The measurement period is specified by the **gNBXnSonConfig.XnSonDeleteTimerForXnUsage** parameter. The Xn links that are not occupied within the measurement period will be removed.
The default value of **gNBXnSonConfig.XnSonDeleteTimerForXnUsage** is 7 days. That is, an Xn link will be removed if it is not used within 7 days. A

smaller value of this parameter results in a higher probability of meeting the conditions for Xn link removal. As a result, Xn links are quickly removed and then automatically set up, causing frequent Xn link addition and removal. Therefore, set this parameter based on the network plan. It is recommended that the parameter value be greater than one day even when Xn interfaces need to be quickly removed.

Xn self-removal based on interface usage is performed in accordance with the procedure defined in 8.4.6 "Xn Removal" in 3GPP TS 38.423 V15.3.0, as shown in [Figure 5-6](#).

Figure 5-6 Xn self-removal based on interface usage



The detailed procedure is as follows:

- gNodeB 1 sends gNodeB 2 an XN REMOVAL REQUEST message, instructing gNodeB 2 to remove its Xn interface. gNodeB 2 sends an XN REMOVAL RESPONSE message and removes the Xn interface.
- After receiving the XN REMOVAL RESPONSE message, gNodeB 1 removes its Xn interface.

NOTE

When the user planes of the X2 and Xn interfaces use the same IP address, the peer of the Xn-U interface is not removed when Xn self-removal is triggered to ensure that services on the X2 interface are not affected.

Xn-U Self-Removal

The gNodeB can automatically remove the **USERPLANEPEER** MOs with **USERPLANEPEER.CTRLMODE** set to **AUTO_MODE** and the corresponding Xn-U transmission links. However, the gNodeB does not automatically remove the **USERPLANEPEER** MOs with **USERPLANEPEER.CTRLMODE** set to **MANUAL_MODE** and the corresponding Xn-U transmission links. This function is controlled by the **GEPMODELPARA.UPAUTODELSW** parameter.

The gNodeB determines whether to automatically remove Xn-U transmission links as follows:

- Self-removal based on link faults
If the GTP-U detection result indicates that all the transmission links corresponding to **USERPLANEPEER** MOs are faulty, the gNodeB periodically (every 60 minutes) attempts to remove those transmission links (up to 50 links at one time) that remain faulty for over one hour and the corresponding **USERPLANEPEER** MOs.

- Self-removal based on the aging mechanism
 - In case of an excess over the specifications of a base station
The gNodeB marks a transmission link as aged if the link is not used by any service.
 - The gNodeB performs self-removal based on the aging mechanism once every hour and checks whether the number of established transmission links reaches or exceeds 90% of the maximum number of transmission links supported. If it does, the gNodeB removes aged transmission links (up to 50 links at one time) and the corresponding **USERPLANEPEER** MOs. If it does not, no aged transmission links are removed.
 - When UEs need to establish new transmission links for initiating services, the gNodeB checks whether the number of established transmission links reaches or exceeds 90% of the maximum number of transmission links supported. If it does, the gNodeB removes aged transmission links (up to 50 links at one time) and the corresponding **USERPLANEPEER** MOs, and establishes new transmission links. If it does not, no aged transmission links are removed.
 - If the number of established transmission links has reached the maximum number supported and there is no aged link when UEs need to establish new transmission links for initiating services, the gNodeB rejects new UEs' access requests.
 - In case of an excess over the specifications of a single endpoint group
"Self-removal in case of an excess over the specifications of a single endpoint group" takes effect for the gNodeB when the **SINGLE_EPUP_AUTO_DEL_SW** option of the **TRANSALGEXTPARA.TNLBEARERMNGALGSW** parameter is selected. The base station removes aged links when the number of peers in a single endpoint group exceeds 90% of the maximum number allowed. The condition for marking a link as aged and the self-removal mechanism are the same as those for "self-removal in case of an excess over the specifications of a base station".

5.1.4 Xn Blacklist and Xn Whitelist

Operators can define Xn blacklists and whitelists. If the peer gNodeB is in the Xn blacklist of a local gNodeB, the Xn interface between the two gNodeBs cannot be automatically set up. If the peer gNodeB is in the Xn whitelist of a local gNodeB, the Xn interface between the two gNodeBs cannot be automatically removed. Users can run the **ADD GNBXNBLACKWHITELIST** command to add gNodeBs to an Xn blacklist or whitelist. The Xn blacklist and whitelist combined can contain a maximum of 64 entries.

If an Xn interface is no longer required after the peer gNodeB that has established the Xn link is added to the Xn blacklist, you are advised to manually delete the SCTP peer of the Xn interface.

 NOTE

To remove the SCTP peer corresponding to the peer gNodeB, perform the following steps:

1. Run the **DSP GNBCUXNINTERFACE** command to obtain the peer IP address of the control plane of the gNodeB CU Xn interface corresponding to the peer gNodeB ID.
2. Run the **LST SCTPPEER** command to obtain the SCTP peer ID corresponding to the peer IP address.
3. Run the **RMV SCTPPEER2EPGRP** command to remove the SCTP peer from the endpoint group.
4. Run the **RMV SCTPPEER** command to remove the SCTP peer.

5.1.5 Xn-C Interface Specifications

Xn-C is the interface between gNodeBs for implementing control plane functions. The specifications for a gNodeB are as follows:

- When a UMPTe series board is used, a maximum of 384 Xn-C interfaces are supported.
- When a UMPTg/UMPTHa,/UMPTHb series board is used, a maximum of 768 Xn-C interfaces are supported.
- When a UMPTga series board is used, a maximum of 512 Xn-C interfaces are supported.

5.1.6 Application Restrictions

The gNodeB supports the self-setup of two **USERPLANEPEER** MOs at the same time. The self-setup request of additional **USERPLANEPEER** MOs will be rejected, which affects handover services.

It is recommended that one **gNBCUXn** MO be configured for each operator. If multiple **gNBCUXn** MOs are configured for an operator, only the **gNBCUXn** MO with the smallest **gNBCUXn.gNBCuXnId** parameter value takes effect.

5.2 Network Analysis

5.2.1 Benefits

Xn self-management simplifies configuration operations, reduces the OPEX of operators, and improves gNodeB resource usage.

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

This function has no impact on the network.

5.3 Requirements

5.3.1 Licenses

N/A

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Automatic user plane removal	GEPMODEL PARA.UPAUT ODELSW	<i>NG and Xn Self-Management</i>	Ensure that the GEPMODEL PARA.UPAUTODELSW parameter is set to ENABLE before enabling "self-removal in case of an excess over the specifications of a single endpoint group" (by selecting the SINGLE_EPUP_AUTO_DEL_SW option of the TRANSALGEXTPARA.TNLBEARER MNGALGSW parameter).
Low-frequency TDD High-frequency TDD	Self-removal in case of an excess over the specifications of a single endpoint group	SINGLE_EPUP_AUTO_DEL_SW option of the TRANSALGEXTPARA.TNLBEARER MNGALGSW parameter	<i>NG and Xn Self-Management</i>	Ensure that "self-removal in case of an excess over the specifications of a single endpoint group" is disabled before disabling automatic user plane removal (by setting the GEPMODEL PARA.UPAUTODELSW parameter to DISABLE).

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Xn self-removal	XN_SON_DE L_SW option of the gNBXnSonC onfig.XnSon ConfigSwitc h parameter	<i>NG and Xn Self- Management</i>	Control-plane dual-stack check for the Xn interface can be enabled (by selecting the XN_CP_DUALSTA CK_CHK_SW option of the gNBInterfacePar- am.InterfaceOmS w parameter) only when Xn self-removal is enabled.

5.3.2.2 Mutually Exclusive Functions

None

5.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Referenc e	Description
Low-frequency TDD	Operator-specific NCI configuration	gNBSharing Mode.Multipl eNCISwitch	<i>Multi- Operator Sharing</i>	An operator-specific gNodeB ID must be configured if the operator-specific NCI configuration function is enabled on the peer gNodeB to be added to the Xn blacklist or whitelist.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Endpoint group IDs for the NG interface	gNBCUNg.CpEpGroupId and gNBCUNg.UpEpGroupId	<i>NG and Xn Self-Management</i>	- The value of the gNBCUXn.CpEpGroupId parameter configured for the Xn interface must be different from that of the gNBCUNg.CpEpGroupId parameter configured for the NG interface. - The value of the gNBCUXn.UpEpGroupId parameter configured for the Xn interface must be different from that of the gNBCUNg.UpEpGroupId parameter configured for the NG interface.
Low-frequency TDD High-frequency TDD	User plane endpoint group ID for the S1 interface	gNBCUS1.UpEpGroupId	<i>X2 and S1 Self-Management in NSA Networking</i>	The value of the gNBCUXn.UpEpGroupId parameter configured for the Xn interface must be different from that of the gNBCUS1.UpEpGroupId parameter configured for the S1 interface.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Endpoint group IDs for the X2 interface	gNBCUX2.CpEpGroupId and gNBCUX2.UpEpGroupId	<i>X2 and S1 Self-Management in NSA Networking</i>	• The value of the gNBCUXn.CpEpGroupId parameter configured for the Xn interface must be different from that of the gNBCUX2.CpEpGroupId parameter configured for the X2 interface. • If the value of the gNBCUXn.UpEpGroupId parameter configured for the Xn interface is different from that of the gNBCUX2.UpEpGroupId parameter configured for the X2 interface, the value of the USERPLANEHOST.UPHOSTID parameter in the EPGROUP MO referenced by the Xn interface must be different from that of the USERPLANEHOST.UPHOSTID parameter in the EPGROUP MO referenced by the X2 interface.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	User plane endpoint group ID for the eXn interface	gNBDUExn.UpEpGroupId	<i>eXn Self-Management</i>	- If the value of the gNBCUXn.UpEpGroupId parameter configured for the Xn interface is different from that of the gNBDUExn.UpEpGroupId parameter configured for the eXn interface, the value of the USERPLANEHOST.UPHOSTID parameter in the EPGROUP MO referenced by the Xn interface must be different from that of the USERPLANEHOST.UPHOSTID parameter in the EPGROUP MO referenced by the eXn interface. - If the value of the gNBCUXn.UpEpGroupId parameter configured for the Xn interface is the same as that of the gNBDUExn.UpEpGroupId parameter configured for the eXn interface, the number of USERPLANEHOST.UPHOSTID parameters referenced in the EPGROUP MO cannot exceed 2.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards support this function. To learn which main control boards are NR-capable, see the related BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

5.3.4 Networking

For the QoS requirements of Xn interface for transmission networking, see *Transmission Resource Management*.

5.3.5 Others

The implementation of this function requires that the core network or MAE-Access support Xn self-setup.

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

The following table describes the parameters that must be set in an SCTEMPLATE MO.

Parameter Name	Parameter ID	Setting Notes
SCTP Parameters Template ID	SCTEMPLATE.SCTPTEMPLATEID	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Switch Back Flag	SCTPTEMPLATE.SWITCHBACKFLAG	Set this parameter based on the network plan.

The following table describes the parameters that must be set in an **SCTPHOST** MO.

Parameter Name	Parameter ID	Setting Notes
SCTP Host ID	SCTPHOST.SCTPHOSTID	Set this parameter based on the network plan.
IP Version	SCTPHOST.IPVERSION	Set the IP version of the local end to the same as the peer end. Set the IP version of the user plane to the same as the control plane.
First Local IP Address	SCTPHOST.SIGIP1V4	If SCTPHOST.IPVERSION is set to IPv4 , configure a local control-plane IPv4 address.
First Local IPv6 Address	SCTPHOST.SIGIP1V6	If SCTPHOST.IPVERSION is set to IPv6 , configure a local control-plane IPv6 address.
Second Local IP Address	SCTPHOST.SIGIP2V4	If SCTPHOST.IPVERSION is set to IPv4 , configure a local control-plane IPv4 address.
Second Local IPv6 Address	SCTPHOST.SIGIP2V6	If SCTPHOST.IPVERSION is set to IPv6 , configure a local control-plane IPv6 address.
Local SCTP Port No.	SCTPHOST.PN	The value recommended by 3GPP specifications is 38422 . For details, see chapter 7 "Transport layer" in 3GPP TS 38.422 V15.1.0.
Simple Mode Switch	SCTPHOST.SIMPLEMODESWITCH	Set this parameter based on the network plan.
SCTP Parameters Template ID	SCTPHOST.SCTPTEMPLATEID	Set this parameter based on the network plan.

(Optional) The following table describes the parameters that must be set in an **SCTPPEER** MO. This MO is required only for manual peer configuration.

Parameter Name	Parameter ID	Setting Notes
SCTP Peer ID	SCTPPEER.SCTPPEERID	Set this parameter based on the network plan.
IP Version	SCTPPEER.IPVERSION	Set the IP version of the local end to the same as the peer end. Set the IP version of the user plane to the same as the control plane.
First Peer IP Address	SCTPPEER.SIGIP1V4	If SCTPPEER.IPVERSION is set to IPv4 , configure a peer control-plane IPv4 address.
First Peer IPv6 Address	SCTPPEER.SIGIP1V6	If SCTPPEER.IPVERSION is set to IPv6 , configure a peer control-plane IPv6 address.
Second Peer IP Address	SCTPPEER.SIGIP2V4	If SCTPPEER.IPVERSION is set to IPv4 , configure a peer control-plane IPv4 address.
Second Peer IPv6 Address	SCTPPEER.SIGIP2V6	If SCTPPEER.IPVERSION is set to IPv6 , configure a peer control-plane IPv6 address.
Peer SCTP Port No.	SCTPPEER.PN	The value recommended by 3GPP specifications is 38422 . For details, see chapter 7 "Transport layer" in 3GPP TS 38.422 V15.1.0.

The following table describes the parameters that must be set in a **USERPLANEHOST MO**.

Parameter Name	Parameter ID	Setting Notes
User Plane Host ID	USERPLANEHOST.UPHOSTID	Set this parameter based on the network plan.
IP Version	USERPLANEHOST.IPVERSION	Set the IP version of the local end to the same as the peer end. Set the IP version of the user plane to the same as the control plane.
Local IP Address	USERPLANEHOST.LOCIPV4	If USERPLANEHOST.IPVERSION is set to IPv4 , configure a local user-plane IPv4 address.

Parameter Name	Parameter ID	Setting Notes
Local IPv6 Address	USERPLANEHOST.LOCIPV6	If USERPLANEHOST.IPVERSION is set to IPv6 , configure a local user-plane IPv6 address.
IPSec Auto Configure Switch	USERPLANEHOST.IPSEC SWITCH	Set this parameter to ENABLE to enable direct IPsec.

(Optional) The following table describes the parameters that must be set in a **USERPLANEPEER** MO. This MO is required only for manual peer configuration.

Parameter Name	Parameter ID	Setting Notes
User Plane Peer ID	USERPLANEPEER.UPPEERID	Set this parameter based on the network plan.
IP Version	USERPLANEPEER.IPVERSION	Set the IP version of the local end to the same as the peer end. Set the IP version of the user plane to the same as the control plane.
Peer IP Address	USERPLANEPEER.PEERIPV4	If USERPLANEPEER.IPVERSION is set to IPv4 , configure a peer user-plane IPv4 address.
Peer IPv6 Address	USERPLANEPEER.PEERIPV6	If USERPLANEPEER.IPVERSION is set to IPv6 , configure a peer user-plane IPv6 address.
IPSec Auto Configure Switch	USERPLANEPEER.IPSECSWITCH	Set this parameter to ENABLE to enable direct IPsec.

The following table describes the parameters that must be set in an **EPGROUP** MO. It is recommended that the **USERPLANEHOST** and **USERPLANEPEER** MOs or the **SCTPHOST** and **SCTPPEER** MOs for an operator be configured in the same **EPGROUP** MO.

Parameter Name	Parameter ID	Setting Notes
End Point Group ID	EPGROUP.EPGROUPID	Set this parameter based on the network plan.
SCTP Host List	EPGROUP.SCTPHOSTLIST	Set this parameter based on the network plan. Add an SCTP host to the endpoint group.

Parameter Name	Parameter ID	Setting Notes
SCTP Peer List	EPGROUP.SCTPPEERLIST	Set this parameter based on the network plan. Add an SCTP peer to the endpoint group.
User Plane Host List	EPGROUP.USERPLANEHOSTLIST	Set this parameter based on the network plan. Add a user-plane host to the endpoint group.
User Plane Peer List	EPGROUP.USERPLANEPEERLIST	Set this parameter based on the network plan. Add a user-plane peer to the endpoint group.
Packet Filter ACL Rule Auto-Setup-Deletion SW	EPGROUP.PACKETFILTERSWITCH	Set this parameter based on the network plan.
IPv6 Packet Filter ACL Rule Auto-Setup-Deletion SW	EPGROUP.PACKETFILTERSWITCH6	Set this parameter based on the network plan.
IP Protocol Version Preference	EPGROUP.IPVERPREFERENCE	Set this parameter based on the network plan.

The following table describes the parameters that must be set in a **gNBXnSonConfig** MO.

Parameter Name	Parameter ID	Option	Setting Notes
Xn SON Configuration Switch	gNBXnSonConfig.XnSonConfigSwitch	• XN_SON_SETUP_SW • XN_SON_DELETE_SW • XN_SON_DYNAMIC_BLACKLIST_SW	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
Xn SON Delete Timer for Xn Fault	gNBXnSonConfig.XnSonDeleteTimerForXnFault	None	This parameter takes effect when the XN_SON_DEL_SW option of the gNBXnSonConfig.XnSonConfigSwitch parameter is selected. For details about the configuration requirements, see 5.1.3 Xn Self-Removal .
Xn SON Delete Timer for Xn Usage	gNBXnSonConfig.XnSonDeleteTimerForXnUsage	None	This parameter takes effect when the XN_SON_DEL_SW option of the gNBXnSonConfig.XnSonConfigSwitch parameter is selected. For details about the configuration requirements, see 5.1.3 Xn Self-Removal .

 NOTE

If the SON function is planned to automatically configure, update, and remove Xn interfaces, the **GTRANSPARA.EPOBJAUTOCFGSW** parameter must be set to the default value **ENABLE**.

The following table describes the parameters that must be set to configure the reference relationship between the **gNBCUXn** and **EPGROUP** MOs.

Parameter Name	Parameter ID	Setting Notes
gNodeB CU Xn ID	gNBCUXn.gNBCuXnId	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Control Plane Endpoint Group ID	gNBCUXn.CpEpGroupId	Set this parameter based on the network plan.
User Plane Endpoint Group ID	gNBCUXn.UpEpGroupId	Set this parameter based on the network plan.

(Optional) The following table describes the parameters that must be set in a **gNBCUNg** MO to configure the 3GPP specification release which the messages transmitted over the NG interface during Xn self-setup comply with.

Parameter Name	Parameter ID	Setting Notes
Protocol CR Switch	gNBCUNg.ProtocolCrSwitch	Set this parameter based on the network plan. Select the TS38413_CR0304_SW option if message processing over the NG interface needs to comply with 3GPP TS 38.413 CR 0304.
Control Plane Endpoint Group ID	gNBCUNg.CpEpGroupId	Set this parameter based on the network plan.
User Plane Endpoint Group ID	gNBCUNg.UpEpGroupId	Set this parameter based on the network plan.

The following table describes the parameters that must be set in a **gNBXnBlackWhiteList** MO.

Parameter Name	Parameter ID	Setting Notes
Mobile Country Code	gNBXnBlackWhiteList.Mcc	Set this parameter based on the network plan.
Mobile Network Code	gNBXnBlackWhiteList.Mnc	Set this parameter based on the network plan.
gNodeB ID	gNBXnBlackWhiteList.gNBId	Set this parameter based on the network plan.
Xn List Type	gNBXnBlackWhiteList.XnListType	Set this parameter based on the network plan.

The following table describes the parameters that must be set to remove a gNodeB Xn dynamic blacklist.

Parameter Name	Parameter ID	Setting Notes
Clear Mode	<i>ClearMode</i>	Set this parameter based on the network plan.

(Optional) The following table describes the parameters that must be set in a **GNBINTERFACEPARAM** MO.

Parameter Name	Parameter ID	Option	Setting Notes
Interface Operation and Maintenance Switch	gNBInterfaceParam.InterfaceOmSw	XN_CP_DUALSTACK_CHK_SW	Set this parameter based on the network plan.

5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Xn Self-Setup

NOTE

This section provides only MML command examples of the Xn interface. For packet filtering protected Xn self-setup, packet filtering also needs to be configured. For details, see *Equipment Security*. If a dedicated transmission resource group needs to be specified for the Xn interface, a transmission resource group must be configured. For details, see *Transmission Resource Management*.

IPv4 or IPv6 single-stack scenario:

Xn self-setup with peers manually configured

```
//Configuring common parameters
//(Optional) Adding an SCTP template. By default, the gNodeB has an SCTP template with
SCTPTemplateID set to 0.
ADD SCTPTemplate: SCTPTemplateID=1, SWITCHBACKFLAG=ENABLE;
//Adding an endpoint group for the Xn interface (IPv4 used as an example)
```

```
ADD EPGROUP: EPGROUPID=1, PACKETFILTERSWITCH=DISABLE;  
//Adding an Xn control-plane host (IPv4 used as an example)  
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv4, SIGIPV4="5.5.3.38", SIGIP1SECSWITCH=DISABLE,  
SIGIP2SECSWITCH=DISABLE, PN=38422, SIMPLEMODESWITCH=SIMPLE_MODE_OFF, SCTPTEMPLATEID=1;  
//Adding an Xn control-plane peer  
ADD SCTPPEER: SCTPPEERID=1, IPVERSION=IPv4, SIGIPV4="5.5.3.21", PN=38422;  
//Adding an Xn user-plane host  
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="5.5.3.38", IPSECSWITCH=DISABLE;  
//Adding an Xn user-plane peer  
ADD USERPLANEPEER: UPPEERID=1, IPVERSION=IPv4, PEERIPV4="5.5.3.21", IPSECSWITCH=DISABLE;  
//Adding the Xn user-plane host to the endpoint group  
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;  
//Adding the Xn user-plane peer to the endpoint group  
ADD UPPEER2EPGRP: EPGROUPID=1, UPPEERID=1;  
//Adding the Xn control-plane host to the endpoint group  
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=1;  
//Adding the Xn control-plane peer to the endpoint group  
ADD SCTPPEER2EPGRP: EPGROUPID=1, SCTPPEERID=1;  
//Turning on the Xn self-setup switch  
MOD GNBXNJSONCONFIG: XnSonConfigSwitch=XN_SON_SETUP_SW-1;  
//Adding a gNodeB CU Xn object  
ADD GNBCUXN: gNBCuXnId=1, CpEpGroupId=1, UpEpGroupId=1;  
//((Optional) Adding a gNodeB CU NG object, and enabling messages over the NG interface to be processed  
in accordance with 3GPP TS 38.413 V15.6.0 or later during Xn self-setup  
ADD GNBCUNG: gNBCuNgId=0, CpEpGroupId=0, UpEpGroupId=0, ProtocolCrSwitch=TS38413_CR0304_SW-1;
```

Xn self-setup with peers automatically configured

```
//Configuring common parameters  
//((Optional) Adding an SCTP template. By default, the gNodeB has an SCTP template with  
SCTPTEMPLATEID set to 0.  
ADD SCTPTEMPLATE: SCTPTEMPLATEID=1, SWITCHBACKFLAG=ENABLE;  
//Adding an endpoint group for the Xn interface (IPv4 used as an example)  
ADD EPGROUP: EPGROUPID=1, PACKETFILTERSWITCH=DISABLE;  
//Adding an Xn control-plane host (IPv4 used as an example)  
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv4, SIGIPV4="5.5.3.38", SIGIP1SECSWITCH=DISABLE,  
SIGIP2SECSWITCH=DISABLE, PN=38422, SIMPLEMODESWITCH=SIMPLE_MODE_OFF, SCTPTEMPLATEID=1;  
//Adding an Xn user-plane host  
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="5.5.3.38", IPSECSWITCH=DISABLE;  
//Adding the Xn user-plane host to the endpoint group  
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;  
//Adding the Xn control-plane host to the endpoint group  
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=1;  
//Turning on the Xn self-setup switch  
MOD GNBXNJSONCONFIG: XnSonConfigSwitch=XN_SON_SETUP_SW-1;  
//Adding a gNodeB CU Xn object  
ADD GNBCUXN: gNBCuXnId=1, CpEpGroupId=1, UpEpGroupId=1;  
//((Optional) Adding a gNodeB CU NG object, and enabling messages over the NG interface to be processed  
in accordance with 3GPP TS 38.413 V15.6.0 or later during Xn self-setup  
ADD GNBCUNG: gNBCuNgId=0, CpEpGroupId=0, UpEpGroupId=0, ProtocolCrSwitch=TS38413_CR0304_SW-1;
```

IPv4 or IPv6 dual-stack scenario:

```
//Configuring common parameters  
//((Optional) Adding an SCTP template. By default, the gNodeB has an SCTP template with  
SCTPTEMPLATEID set to 0.  
ADD SCTPTEMPLATE: SCTPTEMPLATEID=1, SWITCHBACKFLAG=ENABLE;  
//Adding an endpoint group for the Xn interface  
ADD EPGROUP: EPGROUPID=1, IPVERPREFERENCE=IPv6;  
//Adding Xn control-plane hosts  
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv4, SIGIPV4="10.131.64.35", SIGIP1SECSWITCH=DISABLE,  
SIGIP2SECSWITCH=DISABLE, PN=38422, SIMPLEMODESWITCH=SIMPLE_MODE_OFF, SCTPTEMPLATEID=1;  
ADD SCTPHOST: SCTPHOSTID=2, IPVERSION=IPv6, SIGIPV6="2001::25", SIGIP1SECSWITCH=DISABLE,  
SIGIP2SECSWITCH=DISABLE, PN=38422, SIMPLEMODESWITCH=SIMPLE_MODE_OFF, SCTPTEMPLATEID=1;  
//Adding Xn user-plane hosts  
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="10.131.64.35", IPSECSWITCH=DISABLE;  
ADD USERPLANEHOST: UPHOSTID=2, IPVERSION=IPv6, LOCIPV6="2001::25", IPSECSWITCH=DISABLE;  
//Adding the Xn user-plane hosts to the endpoint group  
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;  
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=2;
```

```
//Adding the Xn control-plane hosts to the endpoint group
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=1;
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=2;
//Adding a gNodeB CU Xn object
ADD GNBCUXN: gNBCuXnId=1, CpEpGroupId=1, UpEpGroupId=1;
//Turning on the Xn self-setup switch
MOD GNBXNJSONCONFIG: XnSonConfigSwitch=XN_SON_SETUP_SW-1;
//((Optional) Adding a gNodeB CU NG object, and enabling messages over the NG interface to be processed
in accordance with 3GPP TS 38.413 V15.6.0 or later during Xn self-setup
ADD GNBCUNG: gNBCuNgId=0, CpEpGroupId=0, UpEpGroupId=0, ProtocolCrSwitch=TS38413_CR0304_SW-1;
//((Optional) Enabling control-plane dual-stack check for the Xn interface to reduce unnecessary alarm
reporting when both IPv4 and IPv6 addresses are configured for the local and peer gNodeBs in scenarios
with peers automatically configured over the Xn interface
MOD GNBINTERFACEPARAM: InterfaceOmSw=XN_CP_DUALSTACK_CHK_SW-1;
```

Xn Self-Removal

```
//Enabling the Xn self-removal function (with XnSonDeleteTimerForXnUsage and
XnSonDeleteTimerForXnFault set based on the network plan)
MOD GNBXNJSONCONFIG: XnSonConfigSwitch=XN_SON_DEL_SW-1, XnSonDeleteTimerForXnUsage=10080,
XnSonDeleteTimerForXnFault=60;
//Turning on the Xn dynamic blacklist switch
MOD GNBXNJSONCONFIG: XnSonConfigSwitch=XN_SON_DYN_BLACKLIST_SW-1;
//Removing gNodeBs from a gNodeB Xn dynamic blacklist
CLR GNBXNDYNBLACKLIST: ClearMode=CLR_ALL;
```

Xn Blacklist and Xn Whitelist

```
//Adding a gNodeB Xn blacklist and whitelist
ADD GNBXNBLACKWHITELIST: Mcc="262", Mnc="02", gNBId=1, XnListType=XN_BLACKLIST_TYPE;
ADD GNBXNBLACKWHITELIST: Mcc="262", Mnc="02", gNBId=2, XnListType=XN_WHITELIST_TYPE;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Deactivating Xn self-setup
MOD GNBXNJSONCONFIG: XnSonConfigSwitch=XN_SON_SETUP_SW-0;
//Deactivating Xn self-removal
MOD GNBXNJSONCONFIG: XnSonConfigSwitch=XN_SON_DEL_SW-0;
//Deactivating Xn dynamic blacklisting
MOD GNBXNJSONCONFIG: XnSonConfigSwitch=XN_SON_DYN_BLACKLIST_SW-0;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

(Automatic Xn peer configuration) The activation verification procedures for Xn self-setup, self-update, and self-removal are as follows:

- Step 1** On the MAE-Access, choose **SON > SON Log > Synchronize NE Log**.
- Step 2** On the **Synchronize NE Log** tab page, click **Manual Synchronization**. In the displayed dialog box, select NEs and click **OK**.

Step 3 On the **Query SON Log** tab page, select **NR Xn Interface Self Management Log** from the **Log Category** drop-down list box. In the **Event Name** area, select **Automatically Establish Xn Control Plane Link, Automatically Update Xn Control Plane Link, Automatically Delete Xn Control Plane Link, Automatically Establish Xn User Plane Link, Automatically Delete Xn User Plane Link, and Automatically Update Xn User Plane Link**. Then, select target gNodeBs in the **Log Source** area to check different types of Xn self-configuration operation logs.

----End

Expected result: The operation logs contain Xn self-setup, self-update, and self-removal events.

5.4.3 Network Monitoring

If the Xn interface is faulty, check whether any of the alarms listed in [Table 5-1](#) have been reported. If an alarm has been reported, clear the alarm by referring to the alarm reference.

Table 5-1 Alarms related to the Xn interface

Alarm ID	Alarm Name	Severity
29810	gNodeB Xn Interface Fault	Major
25954	User Plane Fault	Major
25888	SCTP Link Fault	Major

6 Direct IPsec for Xn Interface

6.1 Principles

This function creates direct IPsec tunnels between two gNodeBs to reduce transmission delay over the Xn interface when connected through an SeGW, as routing through the SeGW can introduce unwanted delay, depending on its location and the network topology.

Figure 6-1 and **Figure 6-2** show the typical networking of direct IPsec for the Xn interface between gNodeBs.

Figure 6-1 Typical networking of direct IPsec type 1

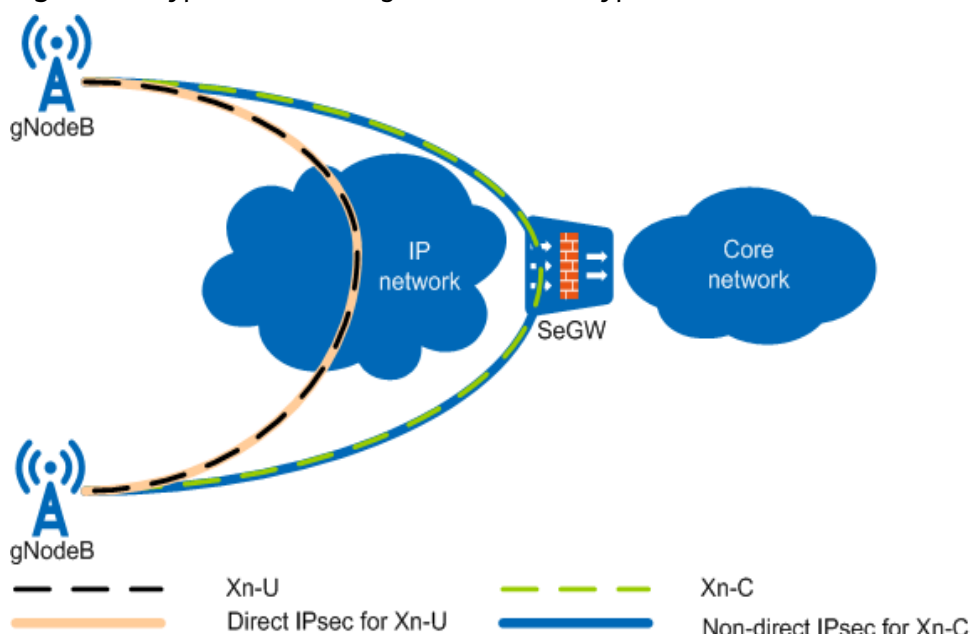
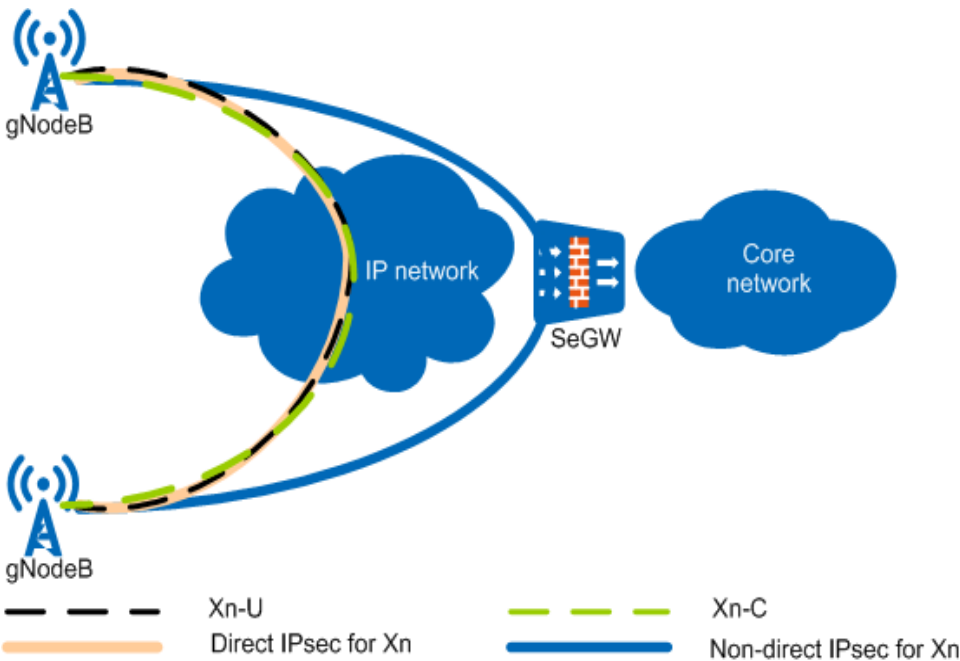


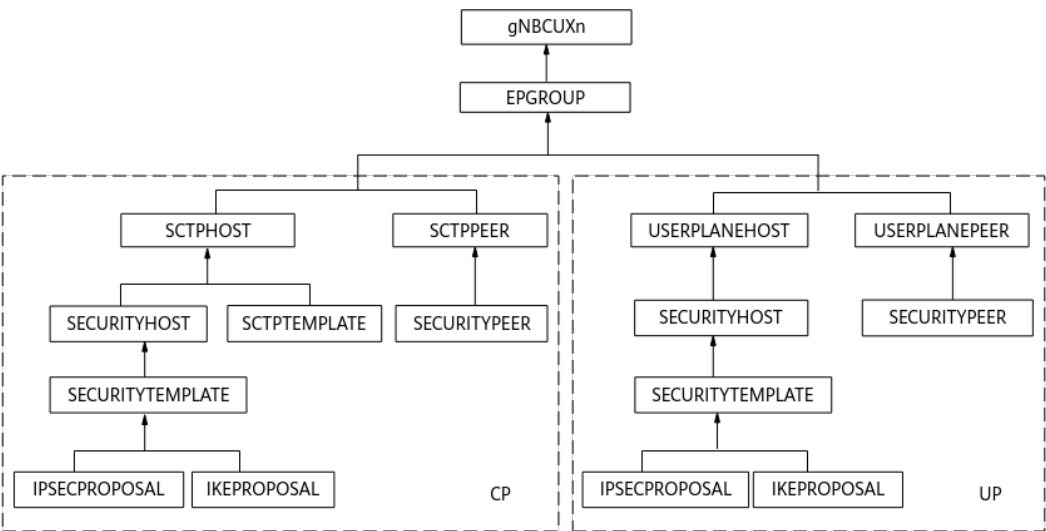
Figure 6-2 Typical networking of direct IPsec type 2



6.1.1 Direct IPsec Setup

Xn control plane peer and user plane peer (corresponding to **SCTPPEER** and **USERPLANEPEER** MOs) can be automatically or manually configured to achieve direct IPsec setup between gNodeBs. If the **SCTPPEER** or **USERPLANEPEER** MO is manually configured, the **SECURITYPEER** MO must be configured as well. (The **SECURITYPEER** MO cannot be manually configured in IPv6 networking.) [Figure 6-3](#) shows the relationships between MOs.

Figure 6-3 Relationships between MOs



Direct IPsec is classified into direct IPv4 IPsec and direct IPv6 IPsec.

- Only one security host IPv4 address can be configured for direct IPv4 IPsec self-setup.

- Only one security host IPv6 address can be configured for direct IPv6 IPsec self-setup.
- In IPv4/IPv6 dual-stack scenarios, it is recommended that one security host IPv4 address and one security host IPv6 address be configured for direct IPsec self-setup.

If the local and peer base stations are configured with IPv4 and IPv6 addresses and enabled with direct IPv4 IPsec self-setup and direct IPv6 IPsec self-setup, the peer base station selects the address type based on the value of the **EPGROUP.IPVERPREFERENCE** parameter.

6.1.1.1 Direct IPv4 IPsec Self-Setup

IPv4 IPsec supports Ethernet-interface-based IPsec and tunnel-interface-based IPsec configuration modes, which are specified by the **IKECFG.IPSECCFGMODE** parameter. In different IPv4 IPsec configuration modes, direct IPv4 IPsec self-setup for the Xn interface can be performed in either of the following methods:

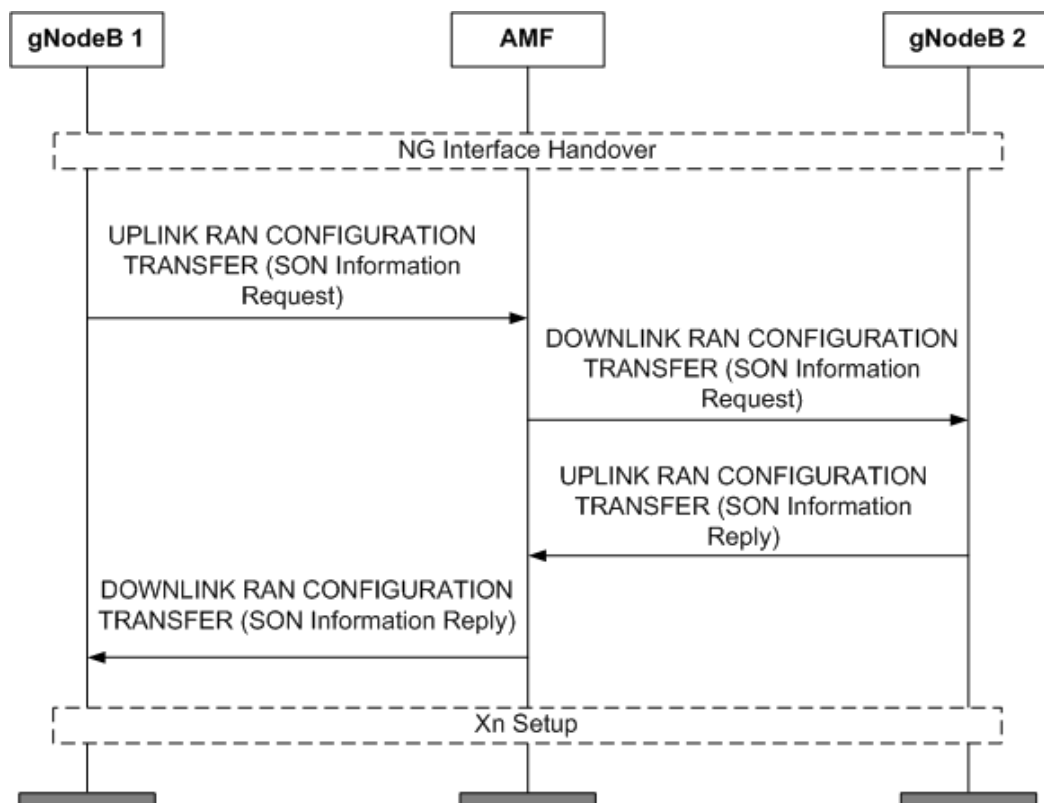
- Method 1: Direct IPv4 IPsec self-setup only for the Xn user plane
- Method 2: Direct IPv4 IPsec self-setup for the Xn control plane and user plane

NOTE

- Inter-gNodeB coordination-based services in non-ideal transmission scenarios have transmission latency requirements only for the Xn user plane. As such, when the IPv4 IPsec configuration mode is Ethernet-interface-based IPsec, the IPv4 IPsec specification (maximum number of allowed IPv4 IPsec tunnels) of the gNodeB does not allow direct IPsec for both the control and user planes of all allowed Xn interfaces. Therefore, method 1 is recommended.
- When the IPv4 IPsec configuration mode is tunnel-interface-based IPsec and the Xn control plane and user plane share an IP address, only method 2 is supported.
- It is recommended that the same IPsec configuration mode (Ethernet-interface-based IPsec or tunnel-interface-based IPsec) be used for the local and peer gNodeBs in direct IPv4 IPsec self-setup.

The direct IPv4 IPsec self-setup procedure in both methods is almost the same. As shown in [Figure 6-4](#), through the AMF, the local gNodeB 1 and the peer gNodeB 2 send the UPLINK RAN CONFIGURATION TRANSFER and DOWNLINK RAN CONFIGURATION TRANSFER messages between each other. These messages carry the SON Information Request and SON Information Reply messages to implement the automatic configuration of the Xn interface between gNodeB 1 and gNodeB 2. The main difference between method 1 and method 2 lies in the content carried in SON Information Request and SON Information Reply.

Figure 6-4 Procedure for direct IPv4 IPsec self-setup for an Xn interface



The self-setup procedure in both methods is described as follows:

NOTE

- The **USERPLANEHOST.IPSECSWITCH** parameter must be set to **ENABLE** for both gNodeBs. Otherwise, the direct IPv4 IPsec setup will not be enabled during the Xn-U setup.
- The **SCTPHOST.SIGIP1SECSWITCH** parameter must be set to **ENABLE** for both gNodeBs. Otherwise, the direct IPv4 IPsec setup will not be enabled during the Xn-C setup.

A peer IKE IP address (specified by **SECURITYPEER.PEERIKEIPV4**) can be used to set up only one IPv4 IPsec tunnel.

- When the IPsec configuration mode is tunnel-interface-based IPsec (with **IKECFG.IPSECCFGMODE** set to **TUNNELITF_BASED**) and the control-plane and user-plane hosts of the Xn interface share an IP address, IPsec self-setup must be simultaneously enabled or disabled for both the user plane and control plane.
- Method 1: (Recommended) Direct IPv4 IPsec self-setup only for the Xn user plane
The **USERPLANEHOST.IPSECSWITCH** parameter must be set to **ENABLE** and the **SCTPHOST.SIGIP1SECSWITCH** parameter must be set to **DISABLE**. Otherwise, direct IPv4 IPsec self-setup cannot be successfully performed for the user plane.
 - a. gNodeB 1 sends a SON Information Request message to gNodeB 2. This message carries the transport network layer addresses of the gNodeB 1 Xn interface, including the Xn-C address, Xn-U address, and Xn-U IPv4 IPsec address.

- b. gNodeB 2 automatically configures the Xn-C interface, Xn-U interface, and Xn-U IPv4 IPsec based on the received information.
- c. gNodeB 2 replies to gNodeB 1 with a SON Information Reply message. This message carries the transport network layer addresses of the gNodeB 2 Xn interface, including the Xn-C address, Xn-U address, and Xn-U IPv4 IPsec address.
- d. gNodeB 1 automatically configures the Xn-C interface, Xn-U interface, and Xn-U IPv4 IPsec based on the returned information.

 NOTE

- gNodeB 1 sets up Xn-C and Xn-U transmission links based on the following MOs: (1) manually configured **gNBCUXn**, **SCTPHOST**, **USERPLANEHOST**, and **EPGROUP** MOs; (2) automatically generated **SCTPPEER** and **USERPLANEPEER** MOs.
- gNodeB 1 sets up a direct IPv4 IPsec tunnel for the Xn-U interface based on the following information: (1) gNodeB 2's user-plane IPv4 IPsec address, control-plane IP address, and user-plane IP address contained in the signaling messages; (2) automatically generated MOs **SECURITYPEER**, **ACL**, **ACLRULE**, **IPSECPOLICY**, **IKEPEER**, and **IPSECBIND** (old model)/**IPSECBINDITF** (new model). When **GTRANSPARA.TRANSCFGMODE** is set to **OLD**, the old model is used. When this parameter is set to **NEW**, the new model is used.

When direct IPv4 IPsec is set up only for the Xn-U interface, the current IPv4 IPsec specification allows direct IPsec tunnels to be set up for all Xn interfaces for a gNodeB.

- Method 2: Direct IPv4 IPsec self-setup for the Xn control plane and user plane
The **SCTPHOST.SIGIP1SECSWITCH** and **USERPLANEHOST.IPSECSWITCH** parameters must be set to **ENABLE**. Otherwise, direct IPv4 IPsec self-setup cannot be successfully performed for the control plane or user plane.
 - a. gNodeB 1 sends a SON Information Request message to gNodeB 2. This message carries the transport network layer addresses of the gNodeB 1 Xn interface, including the Xn-C address, Xn-C IPv4 IPsec address, Xn-U address, and Xn-U IPv4 IPsec address.
 - b. gNodeB 2 automatically configures the Xn-C interface, Xn-C IPv4 IPsec, Xn-U interface, and Xn-U IPv4 IPsec based on the received information.
 - c. gNodeB 2 replies to gNodeB 1 with a SON Information Reply message. This message carries the transport network layer addresses of the gNodeB 2 Xn interface, including the Xn-C address, Xn-C IPv4 IPsec address, Xn-U address, and Xn-U IPv4 IPsec address.
 - d. gNodeB 1 automatically configures the Xn-C interface, Xn-C IPv4 IPsec, Xn-U interface, and Xn-U IPv4 IPsec based on the returned information.

 NOTE

- gNodeB 1 sets up Xn-C and Xn-U transmission links based on the following MOs: (1) manually configured **gNBCUXn**, **SCTPHOST**, **USERPLANEHOST**, and **EPGROUP** MOs; (2) automatically generated **SCTPPEER** and **USERPLANEPEER** MOs.
- gNodeB 1 sets up direct IPv4 IPsec tunnels for the Xn-U and Xn-C interfaces based on the following information: (1) gNodeB 2's control-plane IPv4 IPsec address, user-plane IPv4 IPsec address, control-plane IP address, and user-plane IP address contained in the signaling messages; (2) automatically generated MOs **SECURITYPEER**, **ACL**, **ACLRULE**, **IPSECPOLICY**, **IKEPEER**, and **IPSECBIND** (old model)/**IPSECBINDITF** (new model).

When the IPv4 IPsec specification is limited, the self-setup of new Xn interfaces fails and the ALM-29810 gNodeB Xn Interface Fault is reported. For details about the IPv4 IPsec specification, see MOs **ACL**, **ACLRULE**, **IPSECPOLICY**, **IKEPEER**, and **IPSECBIND** (old model)/**IPSECBINDITF** (new model) in *3900 & 5900 Series Base Station MO and Parameter Reference* in *3900 & 5900 Series Base Station Product Documentation*. The possible fault scenarios are as follows:

- When direct IPv4 IPsec self-setup is required for Xn-C and Xn-U interfaces, separate direct IPv4 IPsec tunnels are also set up. However, the current IPv4 IPsec specification does not allow direct IPsec tunnels to be set up for all Xn interfaces for a gNodeB.
- When service specifications exceed the IPsec specifications or those configured in the **SECURITYPEER** MO:
 - With **GEPMODELPARA.DIRECTIPSECAUTOSETUPMODE** set to **STRICT_MODE**, even if a non-direct IPsec-based Xn interface has been manually set up, it will be removed after direct IPv4 IPsec is configured due to limited specifications of the direct IPv4 IPsec.
 - With **GEPMODELPARA.DIRECTIPSECAUTOSETUPMODE** set to **LOOSE_MODE**, an Xn interface is automatically set up even if direct IPv4 IPsec setup fails.

When an Xn interface matches multiple IPsec tunnels, the **GTRANSPARA.DIRECTIPSECPRIMATCHSW** parameter can be used to adjust the IPv4 IPsec policy matching priority. That is, the parameter determines whether the Xn interface preferentially matches a direct IPv4 IPsec tunnel or an SeGW-based IPv4 IPsec tunnel.

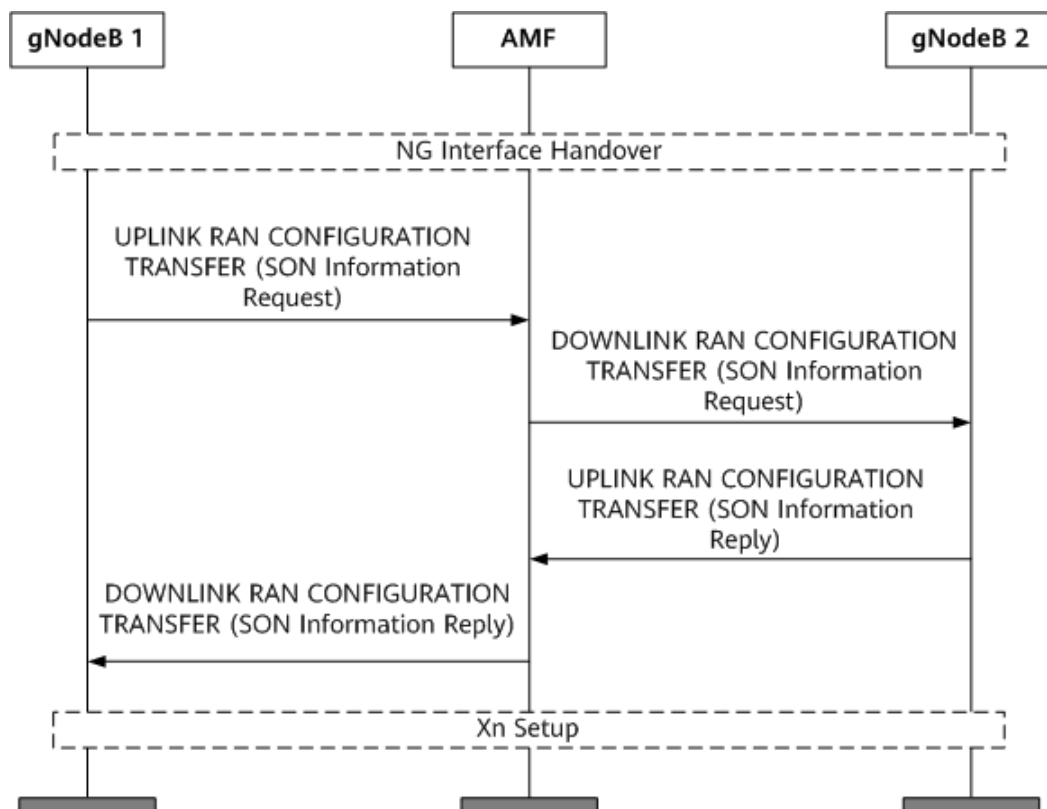
- When this parameter is set to **DISABLE**, the IPv4 IPsec policy with a smaller sequence number (specified by **IPSECPOLICY.SPSM**) is preferentially matched.
- When this parameter is set to **ENABLE**, the IPv4 IPsec policy of a self-setup direct IPsec tunnel is preferentially matched. If this direct IPsec tunnel is faulty, the IPv4 IPsec policy of a normal IPsec tunnel is preferentially used.

If the IPsec Redundancy Among Multiple SeGWs function is enabled on the gNodeB, it is recommended that the IPv4 IPsec tunnels passing through the active and standby SeGWs and the direct IPv4 IPsec tunnels be deployed on the same port. If the IPsec Redundancy Among Multiple SeGWs function is deployed on multiple ports, the Xn interface may only match direct IPv4 IPsec tunnels during an SeGW switchover. In this case, if the direct IPv4 IPsec tunnels are faulty, the SeGW-based IPv4 IPsec tunnels will not be matched for the Xn interface. As a result, the Xn interface is faulty. For details about IPsec Redundancy Among Multiple SeGWs, see *IPsec*.

6.1.1.2 Direct IPv6 IPsec Self-Setup

In direct IPv6 IPsec self-setup for the Xn interface, only direct IPv6 IPsec self-setup for the Xn control plane and user plane is supported. This method is similar to direct IPv4 IPsec self-setup for the Xn control plane and user plane. See [Figure 6-5](#). The main difference between the two procedures lies in the content carried in the SON Information Request and SON Information Reply messages.

Figure 6-5 Procedure for direct IPv6 IPsec self-setup for an Xn interface



The procedure for direct IPv6 IPsec self-setup for the Xn control plane and user plane is as follows:

NOTE

The **USERPLANEHOST.IPSECSWITCH** and **SCTPHOST.SIGIP1SECSWITCH** parameters must be set to **ENABLE** for the two gNodeBs with direct IPv6 IPsec self-setup for the Xn control plane and user plane.

When the user plane and control plane of the Xn interface share an IPv6 service address, both the Xn user plane and control plane are protected by direct IPv6 IPsec if either **USERPLANEHOST.IPSECSWITCH** or **SCTPHOST.SIGIP1SECSWITCH** is set to **ENABLE**.

1. gNodeB 1 sends a SON Information Request message to gNodeB 2. This message carries the transport network layer addresses of the gNodeB 1 Xn interface, including the Xn-C address, Xn-C IPv6 IPsec address, Xn-U address, and Xn-U IPv6 IPsec address.
2. gNodeB 2 automatically configures the Xn-C interface, Xn-C IPv6 IPsec, Xn-U interface, and Xn-U IPv6 IPsec based on the received information.
3. gNodeB 2 replies to gNodeB 1 with a SON Information Reply message. This message carries the transport network layer addresses of the gNodeB 2 Xn interface, including the Xn-C address, Xn-C IPv6 IPsec address, Xn-U address, and Xn-U IPv6 IPsec address.
4. gNodeB 1 automatically configures the Xn-C interface, Xn-C IPv6 IPsec, Xn-U interface, and Xn-U IPv6 IPsec based on the returned information.

 NOTE

- gNodeB 1 sets up Xn-C and Xn-U transmission links based on the following MOs: (1) manually configured **gNBCUXn**, **SCTPHOST**, **USERPLANEHOST**, and **EPGROUP** MOs; (2) automatically generated **SCTPPEER** and **USERPLANEPEER** MOs.
- gNodeB 1 sets up direct IPv6 IPsec tunnels for the Xn-C and Xn-U interfaces based on the following information: (1) gNodeB 2's control-plane IPv6 IPsec address, user-plane IPv6 IPsec address, control-plane IP address, and user-plane IP address in the signaling messages; (2) automatically generated MOs **SECURITYPEER**, **ACL**, **ACLRULE**, **IPSECPOLICY**, **IKEPEER**, **IPSECBINDITF**, **TUNNELITF**, and **INTERFACE** as well as the internal object **POLICYBASEDROUTING6**.
- After direct IPv6 IPsec is enabled, the control- and user-plane IP addresses of the Xn interface cannot be directly accessed through external networks. That is, the Xn interface cannot be negotiated to serve as a plaintext tunnel.
- When direct IPv6 IPsec self-setup is enabled for base stations, the routing domains of the control plane and user plane of the Xn interface must be the same. That is, the values of the **SCTPHOST.VRFID** and **USERPLANEHOST.VRFID** parameters must be the same among all base stations for which direct IPv6 IPsec self-setup is enabled.

When service specifications exceed the IPsec specifications or those configured in the **SECURITYPEER** MO:

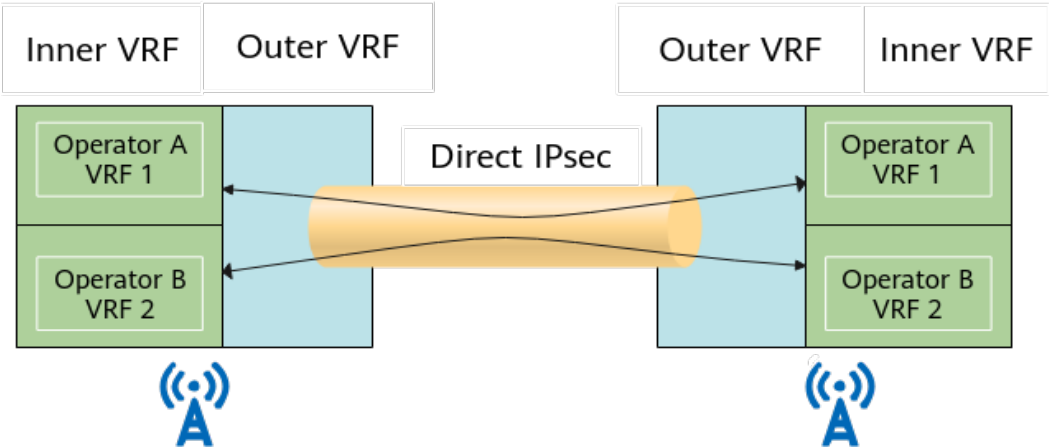
- With **GEPMODELpara.DIRECTIPSECAUTOSETUPMODE** set to **STRICT_MODE**, even if a non-direct IPsec-based Xn interface has been manually set up, it will be removed after direct IPv6 IPsec is configured due to limited specifications of the direct IPv6 IPsec.
- With **GEPMODELpara.DIRECTIPSECAUTOSETUPMODE** set to **LOOSE_MODE**, an Xn interface is automatically set up even if direct IPv6 IPsec setup fails.

When an Xn interface is configured with both a direct IPv6 IPsec tunnel and an SeGW-based IPv6 IPsec tunnel, the direct IPv6 IPsec tunnel is preferentially used. This is because the **POLICYBASEDROUTING6** MO has a higher priority than the **IPROUTE6** and **SRCIPROUTE6** MOs.

6.1.1.3 Direct IPsec Sharing Among Multiple Operators

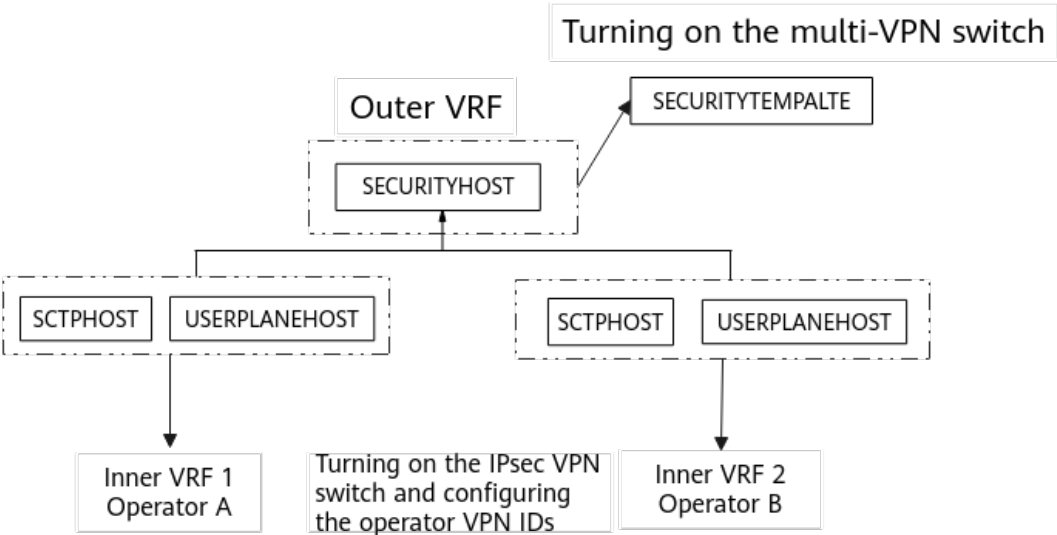
In multi-operator sharing scenarios, a direct IPsec tunnel can be shared over the Xn interface using the direct IPsec demultiplexing technology to provide direct IPsec services for multiple operators based on their VPN IDs. This saves resources and improves the usage rate of direct IPsec. Different inner VRFs are configured for different operators. IPsec can use VRFs to isolate outer packets from inner service packets. This function enables deployment of multiple operators on different inner VRFs of a base station. In this way, multiple operators can use a direct IPsec tunnel to share Xn services. In addition, Xn services no longer need to pass through the SeGW. [Figure 6-6](#) shows the principle using two operators as an example.

Figure 6-6 Direct IPsec sharing among multiple operators



In multi-operator sharing scenarios, after the multi-VPN switch is turned on (by using the **SECURITYTEMPLATE.MULTIVPNSWITCH** parameter), the IPsec VPN switch is turned on (by using the **VRF.IPSECVPNSW** parameter) and the corresponding operator VPN IDs are configured (by using the **VRF.IPSECVPNID** parameter) to implement direct IPsec sharing among multiple operators. The VPN IDs (specified by the **VRF.IPSECVPNID** parameter) must be unique for different operators and the VPN ID of each operator must be consistent across all of its base stations. Otherwise, direct IPsec sharing among multiple operators cannot work properly. Figure 6-7 shows the configuration model. After the multi-VPN switch is turned on, multiple operators (up to six) can reference the same direct IPsec tunnel. The multi-VPN switch referenced by the **SECURITYTEMPLATE** MO and the IPsec VPN switch referenced by the **SCTPHOST** and **USERPLANEHOST** MOs must be both turned on or off.

Figure 6-7 Configuration model of direct IPsec sharing among multiple operators



6.1.2 Static Blacklist & Whitelist

Static Blacklist

If a security peer is included in the static blacklist of a local end for direct IPsec tunnels, the local end will not establish a direct IPsec tunnel to the peer end.

If a local gNodeB can establish an Xn interface with a peer gNodeB through the SeGW, the peer gNodeB can be added to the local gNodeB's static blacklist for direct IPsec tunnels.

The **ADD DIRECTIPSECBLACKLIST** command can be used to configure the static direct IPsec blacklist function for a gNodeB.

You are advised to configure the static direct IPsec blacklist before enabling the direct IPsec function. If direct IPsec tunnels have been set up between the local and peer base stations before the blacklist is configured, you need to manually remove the **SECURITYPEER** MO before configuring a static direct IPsec blacklist.

Static Whitelist

If a network segment is included in the static whitelist of a local end for direct IPsec tunnels, the local end will only establish direct IPsec tunnels to the peer ends on the network segments within the whitelist.

If the gNodeB only needs to establish direct IPsec tunnels to the peer ends on some network segments, the segments can be added to the static whitelist for direct IPsec tunnels.

The **ADD DIRECTIPSECWHITELIST** command can be used to configure the static direct IPsec whitelist function.

NOTE

The static blacklist and static whitelist functions cannot both be configured.

The static whitelist and direct IPsec self-removal functions cannot both be configured.

6.1.3 Self-Update

gNodeBs support self-update of direct IPsec. This function is controlled by the **GEPMODEL PARA.DIRIPSECUPDATESW** parameter. Assume that an Xn interface is functioning properly after self-update of direct IPsec is enabled. An Xn self-setup procedure is triggered if the Xn control plane is still functioning properly after any of the following reconfigurations is performed. Then, the local base station sends information about direct IPsec to the peer base station. A direct IPsec self-update is performed if direct IPsec self-update is enabled on the peer base station. The reconfigurations are as follows:

- The value of the **USERPLANEHOST.IPSECSWITCH** parameter corresponding to the **EPGROUP** MO of the Xn interface is changed from **DISABLE** to **ENABLE**, and the corresponding **SECURITYHOST.SEGWSWITCH** parameter value is **DISABLE**.
- (Applied only to direct IPv4 IPsec self-setup) The endpoint group of the Xn interface is modified with a new **SECURITYHOST** MO referenced, and the corresponding **SECURITYHOST.SEGWSWITCH** parameter value is **DISABLE**.

- The value of the **USERPLANEHOST.IPSEC SWITCH** parameter corresponding to the endpoint group of the Xn interface is **ENABLE**, and the value of the corresponding **SECURITYHOST.SEGW SWITCH** parameter is changed from **ENABLE** to **DISABLE**.
- The value of the **USERPLANEHOST.IPSEC SWITCH** parameter corresponding to the endpoint group of the Xn interface is **ENABLE**, the value of the corresponding **SECURITYHOST.SEGW SWITCH** parameter is **DISABLE**, and the **SECURITYHOST.LOCIKE IPV4** parameter value is changed.
- The value of the **USERPLANEHOST.IPSEC SWITCH** parameter corresponding to the endpoint group of the Xn interface is **ENABLE**, the value of the corresponding **SECURITYHOST.SEGW SWITCH** parameter is **DISABLE**, and the **SECURITYHOST.LOCIKE IPV6** parameter value is changed.
- The **ADD GNBCUXN** command is executed to add an Xn interface, or the **MOD GNBCUXN** command is executed to modify the **GNBCUXN.UpEpGroupId** parameter corresponding to an Xn interface.
- The value of the **SCTPHOST.SIGIP1SEC SWITCH** parameter corresponding to the **EPGROUP** MO of the Xn interface is changed from **DISABLE** to **ENABLE**, and the corresponding **SECURITYHOST.SEGW SWITCH** parameter value is **DISABLE**.
- The value of the **SCTPHOST.SIGIP1SEC SWITCH** parameter corresponding to the endpoint group of the Xn interface is **ENABLE**, and the value of the corresponding **SECURITYHOST.SEGW SWITCH** parameter is changed from **ENABLE** to **DISABLE**.
- The value of the **SCTPHOST.SIGIP1SEC SWITCH** parameter corresponding to the endpoint group of the Xn interface is **ENABLE**, the value of the corresponding **SECURITYHOST.SEGW SWITCH** parameter is **DISABLE**, and the **SECURITYHOST.LOCIKE IPV4** parameter value is changed.
- The value of the **SCTPHOST.SIGIP1SEC SWITCH** parameter corresponding to the endpoint group of the Xn interface is **ENABLE**, the value of the corresponding **SECURITYHOST.SEGW SWITCH** parameter is **DISABLE**, and the **SECURITYHOST.LOCIKE IPV6** parameter value is changed.

There is a 3-minute configuration protection period for the self-update of direct IPsec. If any of the preceding configurations is changed, the configuration protection is triggered. After the configuration protection period ends, if the configuration is changed, a self-update of direct IPsec is triggered.

After an Xn interface between the base station and each neighboring base station is automatically set up, a 100s protection timer is automatically created. In direct IPsec scenarios, if a user-plane IP address is changed within the protection timer length, the Xn self-update procedure fails. The Xn self-update procedure can be restarted only after the protection timer expires.

The base station automatically checks every 20 hours for interfaces whose direct IPsec switch in the **SCTPHOST/USERPLANEHOST** MO is turned on but no direct IPsec tunnels are established for them:

- If the automatic IPsec update switch is turned on, the base station automatically triggers a direct IPsec tunnel setup. When IPv4 plaintext transmission is changed to direct IPsec transmission, the Xn interface is interrupted if the IPsec tunnel fails to be set up.
- If the automatic IPsec update switch is turned off, the base station does not automatically trigger a direct IPsec setup.

 NOTE

If the Xn link is normal but no direct IPsec tunnel is set up, you can set the **GEPMODELpara.DIRIPSECUPDATESW** parameter to **ENABLE** to enable the self-setup of direct IPsec triggered by Xn self-update.

If the Xn control plane is normal and direct IPsec is configured on the local end, direct IPsec self-update is initiated on the user plane in the case of a base station reset.

6.1.4 Self-Removal

Self-removal of direct IPsec applies when direct IPsec is enabled on a gNodeB and its connected gNodeBs, but direct IPsec tunnels are unreachable to the connected gNodeBs due to transmission network planning. If this scenario does not occur, this function is not recommended. This function is controlled by the **GEPMODELpara.DIRIPSECAUTODELSW** parameter.

When detecting a direct IPsec fault caused by an IKE SA negotiation failure: The gNodeB does not report an alarm but removes the faulty link if the **GEPMODELpara.DIRIPSECAUTODELTIMER** parameter is set to 0; The gNodeB reports ALM-25891 IKE Negotiation Failure if the **GEPMODELpara.DIRIPSECAUTODELTIMER** parameter is set to a non-zero value. This facilitates transmission fault locating. The gNodeB waits for a period of time specified by the **GEPMODELpara.DIRIPSECAUTODELTIMER** parameter. If the fault persists after the timer expires, the gNodeB removes the direct IPsec tunnel and ALM-25891 IKE Negotiation Failure is cleared. It is recommended that the **GEPMODELpara.DIRIPSECAUTODELTIMER** parameter value be set to one week or longer. If the configured period is excessively short, it is possible that the gNodeB has triggered direct IPsec self-removal and related alarms have been cleared before users troubleshoot transmission faults based on the alarms, or that ALM-25891 IKE Negotiation Failure is reported again after it is cleared due to the check mechanism.

If gNodeB configuration is prohibited, or there is a large number of to-be-removed direct IPsec tunnels, the fault recovery time will be longer than the time specified by the **GEPMODELpara.DIRIPSECAUTODELTIMER** parameter. This gap is dependent on the gNodeB configuration prohibition duration or the number of to-be-removed direct IPsec tunnels.

6.1.5 Application Restrictions

Direct IPsec does not apply to Xn-C interfaces adopting SCTP dual-homing.

Direct IPsec does not apply to any Xn-U interface that uses multiple IP addresses of the same protocol version.

Direct IPv6 IPsec requires that the value of the **EPGROUP.IPV6VRFIDX** parameter differs from that of the **SECURITYHOST.VRFIDX** parameter.

When direct IPsec is configured for the user plane of the Xn interface, it is recommended that the NG and Xn interfaces reference different **USERPLANEHOST** MOs. This is because the **USERPLANEHOST** MO referenced by the NG interface requires that the IPsec self-setup switch be set to off but the **USERPLANEHOST** MO referenced by the Xn interface requires that the IPsec self-setup switch be set to on.

When direct IPsec is configured only for the user plane of the Xn interface and the configuration of the IPsec self-setup switch in the **USERPLANEHOST** MO changes,

the configuration changes take effect only when Xn self-setup is performed. When direct IPsec is configured for both the user and control planes of the Xn interface and the configurations of the IPsec self-setup switches in the **USERPLANEHOST** and **SCTPPEER** MOs change, the configuration changes take effect only when Xn self-setup is performed.

Direct IPsec self-setup does not apply to IPv4-over-IPv6 IPsec or IPv6-over-IPv4 IPsec. IPv4-over-IPv6 IPsec indicates IPv4 for the inner IP header and IPv6 for the outer IP header. IPv6-over-IPv4 IPsec indicates IPv6 for the inner IP header and IPv4 for the outer IP header. For details about IPv4-over-IPv6 IPsec and IPv6-over-IPv4 IPsec, see *IPsec*.

When the **GEPMODELPARA.DIRECTIPSECAUTOSETUPMODE** parameter is set to **LOOSE_MODE**, direct IPsec can be enabled only by automatic configuration, not manual configuration, of the **SCTPPEER** and **USERPLANEPEER** MOs. If the loose mode is used, an IPsec tunnel from the base station to the SeGW must be configured to carry the Xn interface. Otherwise, the Xn interface may become faulty or there are security risks in sending Xn packets in plaintext.

In direct IPsec self-setup in loose mode, non-direct IPsec tunnels must be set up using both the **SCTPPEER** and **USERPLANEPEER** MOs when only one IPsec tunnel is allowed.

Modifying the **GEPMODELPARA.DIRECTIPSECAUTOSETUPMODE** parameter configuration does not affect the established Xn links.

When direct IPsec is deployed, transmission between base stations must be normal and stable. Frequent transmission status changes will cause frequent IKE and IPsec negotiations between base stations. As a result, the CPU usage of the main control board (**VS.BBUBoard.CPULoad.Mean** or **VS.BBUBoard.CPULoad.Max**) increases.

6.2 Network Analysis

6.2.1 Benefits

On a network secured using IPsec, the SeGW networking results in significant delay over the Xn interface; the direct IPsec networking reduces the length of the transmission path, which reduces the delay over the Xn interface.

6.2.2 Impacts

The impacts between this function and other functions are described in [6.3.2.3 Function Impacts](#).

IPsec ensures transmission security by encapsulating and encrypting IP packets. This reduces the transmission efficiency of service packets on the bearer network. For details, see *IPsec*.

6.3 Requirements

6.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-010080	IPsec	NR0S0IPSEC00	per gNodeB

6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Transmission configuration mode	GTRANSPARA.TRANS CFG MODE	<i>NG and Xn Self-Management</i>	The multi-VPN switch (specified by the SECURITYTEMPLATE.MULTIVPNSWITCH parameter) can be turned on only when the transmission mode is configured in new mode (by setting the GTRANSPARA.TRANS CFGMODE parameter to NEW).

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Self-removal of direct IPsec	GEPMODELPARA.DIRIPSEC.AUTODELSW	<i>NG and Xn Self-Management</i>	The multi-VPN switch (specified by the SECURITYTEMPLATE.MULTIVPNSWITCH parameter) can be turned on only when the switch for self-removal of direct IPsec is turned on.
Low-frequency TDD High-frequency TDD	Security protocol	IPSECPROPOSAL.TRANMODE	<i>IPsec</i>	The multi-VPN switch (specified by the SECURITYTEMPLATE.MULTIVPNSWITCH parameter) can be turned on only when the IPSECPROPOSAL.TRANMODE parameter is set to ESP .

6.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	IPsec configuration mode	IKECFG.IPSEC_CFGMODE	<i>NG and Xn Self-Management</i>	In IPv4 scenarios, if IKECFG.IPSEC_CFGMODE is set to ETHINTERFACE_BASED , the multi-VPN switch (SECURITYTEMPLATE.MULTIVPN_SWITCH) of the SECURITYTEMPLATE MO referenced by the SECURITYHOST MO for direct IPsec (with SECURITYHOST.SEGW_SWITCH set to DISABLE) cannot be turned on.

6.3.2.3 Function Impacts

None

6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

Direct IPsec sharing among multiple operators over the Xn interface: Only the UMPTe, UMPTg, UMPTga, UMPTha, and UMPThb series main control boards and all NR-capable baseband processing units can be used. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

Other functions: All NR-capable main control boards can be used. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

There are no requirements for RF modules.

6.3.4 Others

None

6.4 Operation and Maintenance

6.4.1 When to Use

If Xn interfaces utilize IPsec tunnels passing through the SeGW and service delay requirements are not met, it is recommended that the transmission delays of direct transmission paths of the Xn interface be measured before direct IPsec is enabled:

- If the measured delay is less than the transmission delay of indirect transmission paths of the Xn interface and meets the delay requirements for to-be-enabled coordination-based features, the transport network performance allows for enabling direct IPsec.
- If the measured delay is greater than or equal to the transmission delay of indirect transmission paths of the Xn interface or does not meet the delay requirements for to-be-enabled coordination-based features, the transport network performance does not allow for enabling direct IPsec. In this case, it is recommended that the transport network be optimized until the requirements for enabling direct IPsec are met.

NOTE

For details about the delay requirements of coordination-based features, see the corresponding documentation of coordination-based features. For example, for inter-gNodeB CA, see *CA Experience Improvement*. When a feature has requirements for the transmission delay over an interface (for example, the eXn interface) and the interface shares transmission with the Xn interface (the local and peer user-plane IP addresses of the two interfaces are the same), the delay requirements are equivalent to those for the Xn interface.

6.4.2 Precautions

When direct IPsec is configured on the Xn interface where the two gNodeBs at two ends of the Xn interface are located in different network segments, respective routes between the two gNodeBs must be configured.

6.4.3 Data Configuration

6.4.3.1 Data Preparation

Direct IPsec for Xn interface is based on Xn self-setup between gNodeBs. For details about the configurations for Xn self-setup, see [5.4.1.1 Data Preparation](#). This section describes only IPsec configurations.

The following table describes the parameters that must be set in a **GTRANSPARA** MO to configure the direct IPsec priority matching switch.

Parameter Name	Parameter ID	Setting Notes
Direct IPsec Priorities Match Switch	GTRANSPARA.DIRECTIPSECPRIMATCHSW	Set this parameter to ENABLE to enable direct IPsec. This parameter applies only to direct IPv4 IPsec.

The following table describes the parameters that must be set in an **IKECFG** MO to configure the IPsec configuration mode.

Parameter Name	Parameter ID	Setting Notes
IPsec Config Mode	IKECFG.IPSECCFGMODE	The value TUNNELITF_BASED (tunnel-interface-based IPsec) is recommended.

The following table describes the parameters that must be set in an **IKEPROPOSAL** MO.

Parameter Name	Parameter ID	Setting Notes
Proposal ID	IKEPROPOSAL.PROPID	Set this parameter based on the network plan.
Authentication Method	IKEPROPOSAL.AUTHMETHOD	Set this parameter to IKE_CERT_SIG when digital certificate authentication is used. Set this parameter to the same value for all gNodeBs in the entire network.
IKE Reauthentication Lifetime	IKEPROPOSAL.REAUTHLT	Set this parameter based on the network plan. It is recommended that the configured value be greater than 3600 .

Parameter Name	Parameter ID	Setting Notes
ISAKMP SA Duration	IKEPROPOSAL.DURATION	Set this parameter based on the network plan. It is recommended that the configured value be greater than 3600 .
IKE Reauthentication Switch	IKEPROPOSAL.REAUTH SW	Set this parameter based on the network plan.
Authentication Algorithm	IKEPROPOSAL.AUTHAL G	Set this parameter based on the network plan.
Encryption Algorithm	IKEPROPOSAL.ENCALG	Set this parameter based on the network plan.

The following table describes the parameters that must be set in an **IPSECPROPOSAL** MO.

Parameter Name	Parameter ID	Setting Notes
IPSec Proposal Name	IPSECPROPOSAL.PROPN AME	Set this parameter based on the network plan.
Encapsulation Mode	IPSECPROPOSAL.ENCAP MODE	Set this parameter based on the network plan.
Transform	IPSECPROPOSAL.TRAN MODE	Set this parameter based on the network plan.
ESP Encryption Algorithm	IPSECPROPOSAL.ESPEN CALG	Set this parameter based on the network plan.
ESP Authentication Algorithm	IPSECPROPOSAL.ESPAU THALG	Set this parameter based on the network plan.

The following table describes the parameters that must be set in a **SECURITYTEMPLATE** MO. This MO applies only to IPsec-enabled scenarios. When the following parameters are to be modified, the base station needs to adjust the negotiation status of all security links automatically created using the **SECURITYTEMPLATE** MO. The adjustment takes a long time. Therefore, you are advised to run the **MOD SECURITYTEMPLATE** command to modify parameters, wait for about 30 seconds, and then run other commands.

Parameter Name	Parameter ID	Setting Notes
Security Parameters Template ID	SECURITYTEMPLATE.SECURITYTEMPLATEID	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
SeGW Switch	SECURITYTEMPLATE.<i>SE GWSWITCH</i>	Set this parameter to DISABLE when you enable direct IPsec, so that gNodeBs adopt direct IPsec negotiation.
IKE Proposal ID	SECURITYTEMPLATE.<i>PR OPID</i>	Set this parameter based on the network plan.
IPSec Proposal Name	SECURITYTEMPLATE.<i>IPS ECPROPNAME</i>	Set this parameter based on the network plan.
IKE Version	SECURITYTEMPLATE.<i>IKE VERSION</i>	Set this parameter based on the network plan.
Pre-shared Key	SECURITYTEMPLATE.<i>PK EY</i>	Set this parameter based on the network plan.
DPD Mode	SECURITYTEMPLATE.<i>DP D</i>	Set this parameter based on the network plan.
Certificate Source	SECURITYTEMPLATE.<i>CE RTSOURCE</i>	Set this parameter based on the network plan.
Certificate File Name	SECURITYTEMPLATE.<i>CE RTNAME</i>	Set this parameter based on the network plan. This parameter is valid when SECURITYTEMPLATE.<i>CE RTSOURCE</i> is set to CERTMK .
IKEv2 Fragmentation Switch	SECURITYTEMPLATE.<i>IKE V2FRGSW</i>	Set this parameter based on the network plan. The default value is OFF . For details about IKEv2 packet fragmentation, see <i>IPsec</i> .
IKEv2 Fragmentation MTU	SECURITYTEMPLATE.<i>IKE V2FRGMTU</i>	Set this parameter based on the network plan. For details about IKEv2 packet fragmentation, see <i>IPsec</i> . This parameter is valid when SECURITYTEMPLATE.<i>IKE V2FRGSW</i> is set to ON .
Trust Group ID	SECURITYTEMPLATE.<i>TR USTGROUPID</i>	Set this parameter based on the network plan. The default value is 65535 .

Parameter Name	Parameter ID	Setting Notes
OCSwitch	SECURITYTEMPLATE.OCSW	Set this parameter based on the network plan. The default value is OFF . For details about IKEv2 OCSW, see <i>IPsec</i> .
CRL Using Policy	SECURITYTEMPLATE.CRLPOLICY	Set this parameter based on the network plan. The default value is GLOBAL . For details about CRL using policies, see <i>IPsec</i> .
Narrow Down Switch	SECURITYTEMPLATE.NARROWDOWNSW	Set this parameter based on the network plan. Set this parameter to ON if the IPv4 IPsec configuration mode is tunnel-interface-based IPsec.
Multiple VPN Switch	SECURITYTEMPLATE.MULTIVPNSWITCH	Set this parameter to ON when direct IPsec sharing among multiple operators over the Xn interface is enabled.

The following table describes the parameters that must be set in a **SECURITYHOST** MO.

Parameter Name	Parameter ID	Setting Notes
Security Host ID	SECURITYHOST.SECURITYHOSTID	Set this parameter based on the network plan.
SeGW Switch	SECURITYHOST.SEGWSWITCH	Set this parameter to DISABLE when you enable direct IPsec, so that gNodeBs adopt direct IPsec negotiation.
IP Version	SECURITYHOST.IPVERSION	Set this parameter based on the network plan.
Local IKE IP Address	SECURITYHOST.LOCIKEIPV4	Set this parameter based on the network plan.
Local IKE IPv6 Address	SECURITYHOST.LOCIKEIPV6	Set this parameter based on the network plan.
Security Parameters Template ID	SECURITYHOST.SECURITYTEMPLATEID	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
VRF Index	SECURITYHOST.VRFIDX	Set this parameter based on the network plan. - In IPv6, the value of this parameter must be different from the values of the SCTPHOST.VRFIDX and USERPLANEHOST.VRFIDX parameters. - In IPv4, if the IKECFG.IPSECCFGMODE parameter is set to ETHINTERFACE_BASED, the value of this parameter must be the same as the values of the SCTPHOST.VRFIDX and USERPLANEHOST.VRFIDX parameters. If the IKECFG.IPSECCFGMODE parameter is set to TUNNELITF_BASED, the value of this parameter must be different from the values of the SCTPHOST.VRFIDX and USERPLANEHOST.VRFIDX parameters.

Configure the parameters in an **SCTPTEMPLATE** MO that is used to set mandatory parameters for Sctp link setup over the Xn interface. For details about key parameter settings, see [5.4.1.1 Data Preparation](#).

The following table describes the parameters that must be set in an **SCTPHOST** MO to configure the local gNodeB information of the Xn-C interface.

Parameter Name	Parameter ID	Setting Notes
SCTP Host ID	SCTPHOST.SCTPHOSTID	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
VRF Index	SCTPHOST.VRFID$\mathcal{X}$	Set this parameter based on the network plan. - In IPv6, the value of this parameter must be the same as that of the USERPLANEHOST.VRFID$\mathcal{X}$ parameter but different from that of the SECURITYHOST.VRFID$\mathcal{X}$ parameter. - In IPv4, if the IKECFG.IPSECCFGMO$\mathcal{D}$ parameter is set to ETHINTERFACE_BASE$\mathcal{D}$, the value of this parameter must be the same as the values of the USERPLANEHOST.VRFID$\mathcal{X}$ and SECURITYHOST.VRFID$\mathcal{X}$ parameters. If the IKECFG.IPSECCFGMO$\mathcal{D}$ parameter is set to TUNNELITF_BASE$\mathcal{D}$, the value of this parameter must be the same as that of the USERPLANEHOST.VRFID$\mathcal{X}$ parameter but different from that of the SECURITYHOST.VRFID$\mathcal{X}$ parameter.
IP Version	SCTPHOST.IPVERSION	Set the IP version of the local end to the same as the peer end.
First Local IP Address	SCTPHOST.SIGIP1V4	If SCTPHOST.IPVERSION is set to IPv4 , configure a local control-plane IPv4 address.
First Local IPv6 Address	SCTPHOST.SIGIP1V6	If SCTPHOST.IPVERSION is set to IPv6 , configure a local control-plane IPv6 address.

Parameter Name	Parameter ID	Setting Notes
Second Local IP Address	SCTPHOST.SIGIP2V4	If SCTPHOST.IPVERSION is set to IPv4 , configure a local control-plane IPv4 address.
Second Local IPv6 Address	SCTPHOST.SIGIP2V6	If SCTPHOST.IPVERSION is set to IPv6 , configure a local control-plane IPv6 address.
IPSec Auto Configure Switch for First Local IP	SCTPHOST.SIGIP1SECSWITCH	Set this parameter to ENABLE to enable direct IPsec.
Security Host ID for First Local IP	SCTPHOST.SIGIP1SECHOSTID	Set this parameter based on the network plan.
Local SCTP Port No.	SCTPHOST.PN	Set this parameter based on the network plan.
User Label	SCTPHOST.USERLABEL	Set this parameter based on the network plan.

The following table describes the parameters that must be set in a **USERPLANEHOST** MO to configure the local gNodeB information of the Xn-U interface.

Parameter Name	Parameter ID	Setting Notes
User Plane Host ID	USERPLANEHOST.UPHOSTID	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
VRF Index	USERPLANEHOST.VRFID_X	Set this parameter based on the network plan. - In IPv6, the value of this parameter must be the same as that of the SCTPHOST.VRFID_X parameter but different from that of the SECURITYHOST.VRFID_X parameter. - In IPv4, if the IKECFG.IPSECCFGMO_{DE} parameter is set to ETHINTERFACE_BASED, the value of this parameter must be the same as the values of the SCTPHOST.VRFID_X and SECURITYHOST.VRFID_X parameters. If the IKECFG.IPSECCFGMO_{DE} parameter is set to TUNNELITF_BASED, the value of this parameter must be the same as that of the SCTPHOST.VRFID_X parameter but different from that of the SECURITYHOST.VRFID_X parameter.
IP Version	USERPLANEHOST.IPVERSION	Set the IP version of the local end to the same as the peer end.
Local IP Address	USERPLANEHOST.LOCIPV4	If USERPLANEHOST.IPVERSION is set to IPv4 , configure a local user-plane IPv4 address.

Parameter Name	Parameter ID	Setting Notes
Local IPv6 Address	USERPLANEHOST.LOCIPV6	If USERPLANEHOST.IPVERSION is set to IPv6 , configure a local user-plane IPv6 address.
IPSec Auto Configure Switch	USERPLANEHOST.IPSEC SWITCH	Set this parameter to ENABLE to enable direct IPsec.
Security Host ID	USERPLANEHOST.SECHOSTID	Set this parameter based on the network plan.
User Label	USERPLANEHOST.USER LABEL	Set this parameter based on the network plan.

Configure the parameters in an **EPGROUP** MO that is used to set the endpoint resource groups for an Xn interface. For details about key parameter settings, see [5.4.1.1 Data Preparation](#).

It is recommended that the **USERPLANEHOST**, **USERPLANEPEER**, **SCTPHOST**, and **SCTPPEER** MOs that are used by Xn interfaces of an operator be configured in the same **EPGROUP** MO. For details about key parameter settings, see [5.4.1.1 Data Preparation](#).

Configure the reference relationship between the **gNBCUXn** and **EPGROUP** MOs. For details about settings, see [5.4.1.1 Data Preparation](#).

The following table describes the parameters that must be set in the **NRCELLALGOSWITCH** MO to enable ANR.

Parameter Name	Parameter ID	Setting Notes
ANR Switch	NRCellAlgoSwitch.AnrSwitch	Select the NR_NR_ANR_SW and NR_EUTRAN_ANR_SW options.

(Optional) The following table describes the parameters that must be set in a **GEPMODEL PARA** MO to configure self-removal of direct IPsec for a gNodeB.

Parameter Name	Parameter ID	Setting Notes
Direct IPsec Automatic Deletion Switch	GEPMODEL PARA.DIRIPSECAUTODELSW	Set this parameter based on the network plan.
Direct IPsec Automatic Deletion Timer	GEPMODEL PARA.DIRIPSECAUTODELTIMER	Set this parameter based on the network plan.

(Optional) The following table describes the parameters that must be set in a **GEPMODELPARA** MO to configure self-update of direct IPsec for a gNodeB.

Parameter Name	Parameter ID	Setting Notes
Direct IPsec Update Switch	GEPMODELPARA.DIRIPSECUPDATESW	Set this parameter based on the network plan.

(Optional) The following table describes the parameters that must be set in a **GEPMODELPARA** MO to configure self-setup of direct IPsec for a gNodeB.

Parameter Name	Parameter ID	Setting Notes
Direct IPsec Automatic Setup Mode	GEPMODELPARA.DIRECTIPSECAUTOSETUP-MODE	Set this parameter based on the network plan.

The following table describes the parameters that must be set in a **DIRECTIPSECBLACKLIST** MO.

Parameter Name	Parameter ID	Setting Notes
ID of Blacklist for Direct IPsec Tunnel	DIRECTIPSECBLACKLIST.DIRECTIPSECBLACKLISTID	Set this parameter based on the network plan.
IP Version	DIRECTIPSECBLACKLIST.IPVERSION	Set this parameter based on the network plan.
Security Peer IPv4 Address	DIRECTIPSECBLACKLIST.SECURITYPEERIPV4	Set this parameter based on the network plan. This parameter is valid only when the DIRECTIPSECBLACKLIST.IPVERSION parameter is set to IPV4 .
Security Peer IPv6 Address	DIRECTIPSECBLACKLIST.SECURITYPEERIPV6	Set this parameter based on the network plan. This parameter is valid only when the DIRECTIPSECBLACKLIST.IPVERSION parameter is set to IPV6 .

The following table describes the parameters that must be set in a **DIRECTIPSECWHITELIST** MO.

Parameter Name	Parameter ID	Setting Notes
ID of Whitelist for Direct IPsec Tunnel	DIRECTIPSECWHITELIST.DIRECTIPSECWHITELISTID	Set this parameter based on the network plan.
IP Version	DIRECTIPSECWHITELIST.IPVERSION	Set this parameter based on the network plan.
Security Peer IPv4 Address	DIRECTIPSECWHITELIST.SECURITYPEERIPV4	Set this parameter based on the network plan. This parameter is valid only when the DIRECTIPSECWHITELIST.IPVERSION parameter is set to IPV4 .
Mask	DIRECTIPSECWHITELIST.MASK	Set this parameter based on the network plan.
Security Peer IPv6 Address	DIRECTIPSECWHITELIST.SECURITYPEERIPV6	Set this parameter based on the network plan. This parameter is valid only when the DIRECTIPSECWHITELIST.IPVERSION parameter is set to IPV6 .
IPv6 Address Prefix Length	DIRECTIPSECWHITELIST.IPV6PFXLEN	Set this parameter based on the network plan. This parameter is valid only when the DIRECTIPSECWHITELIST.IPVERSION parameter is set to IPV6 .

(Optional) Configure the **VRF** MO to configure direct IPsec sharing among multiple operators over the Xn interface.

Parameter Name	Parameter ID	Setting Notes
User Label	VRF.USERLABEL	Set this parameter based on the network plan.
IPSec VPN Switch	VRF.IPSECVPNBW	Set this parameter to ENABLE when direct IPsec sharing among multiple operators over the Xn interface is enabled.

Parameter Name	Parameter ID	Setting Notes
IPSec VPN ID	VRF.IPSECVPNID	Set this parameter based on the network plan. Operators must negotiate with each other and ensure that the parameter setting of each operator is consistent across all of its base stations.

6.4.3.2 Using MML Commands

Before using MML commands, refer to [6.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Activating direct IPsec for Xn interface:

NOTE

If a dedicated transmission resource group needs to be specified for the Xn interface, a transmission resource group must be configured. For details, see *Transmission Resource Management*.

The following uses the IPv4 configuration as an example.

Scenario 1: The IPv4 IPsec configuration mode is Ethernet-interface-based IPsec.

```
//Turning on the Xn self-setup switch
MOD GNBXNJSONCONFIG: XnSonConfigSwitch=XN_SON_SETUP_SW-1;
//Setting the direct IPsec matching prioritization with DIRECTIPSECPRIMATCHSW set to ENABLE. This
parameter applies only to direct IPv4 IPsec.
SET GTRANSPARA: DIRECTIPSECPRIMATCHSW=ENABLE;
//Setting IPSECCFGMODE to ETHINTERFACE_BASED
SET IKECFG: IPSECCFGMODE=ETHINTERFACE_BASED;
//Adding an IKE security proposal
ADD IKEPROPOSAL: PROPID=10, ENCALG=AES128, AUTHALG=SHA256, DURATION=86400,
REAUTHSW=ON, REAUTHLT=604800;
//Adding an IPsec proposal
ADD IPSECPROPOSAL: PROPNAME="prop1", ENCAPMODE=TUNNEL, TRANMODE=ESP,
ESPAUTHALG=SHA256, ESPENCALG=AES128;
//Adding a security parameter template for direct IPv4 IPsec with SeGW Switch set to DISABLE
ADD SECURITYTEMPLATE: SECURITYTEMPLATEID=0, SEGWSWITCH=DISABLE, PROPID=10,
IPSECPROPNAME="prop1", IKEVERSION=IKE_V2, PKEY="*****", DPD=PERIODIC, LTCFG=LOCAL,
CERTSOURCE=APPCERT, TRUSTGROUPID=65535, OCSPSW=OFF, CRLPOLICY=GLOBAL;
//Adding a security host with SeGW Switch set to DISABLE
ADD SECURITYHOST: SECURITYHOSTID=0, SEGWSWITCH=DISABLE, IPVERSION=IPv4,
```

```
LOCIKEIPV4="7.7.4.146", SECURITYTEMPLATEID=0;
//Adding an endpoint group for the Xn interface
ADD EPGROUP: EPGROUPID=1, IPV6VRFIDX=1;
//(Optional) Adding an SCTP template. By default, the gNodeB has an SCTP template with
SCTPTemplateID set to 0.
ADD SCTPTemplate: SCTPTemplateID=1, SWITCHBACKFLAG=ENABLE;
//Adding an Xn control-plane host
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv4, SIGIPV4="7.7.42.93", SIGIP1SECSWITCH=DISABLE,
SIGIP2V4="0.0.0.0", SIGIP2SECSWITCH=DISABLE, PN=38422, SIMPLEMODESWITCH=SIMPLE_MODE_OFF,
SCTPTemplateID=1;
//Adding an Xn user-plane host
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="7.7.42.93", IPSECSWITCH=ENABLE,
SECHOSTID=0;
//Adding the Xn user-plane host to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
//Adding the Xn control-plane host to the endpoint group
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=1;
//Turning on the ANR switch
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=NR_NR_ANR_SW-1&NR_EUTRAN_ANR_SW-1;
//Adding an Xn MO
ADD GNBCUXN: gNBCuXnId=0, CpEpGroupId=1, UpEpGroupId=1;
```

Scenario 2: The IPv4 IPsec configuration mode is tunnel-interface-based IPsec.

```
//Turning on the Xn self-setup switch
MOD GNBXNSONCONFIG: XnSonConfigSwitch=XN_SON_SETUP_SW-1;
//Setting the direct IPsec matching prioritization with DIRECTIPSECPRIMATCHSW set to ENABLE. This
parameter applies only to direct IPv4 IPsec.
SET GTRANSPARA: DIRECTIPSECPRIMATCHSW=ENABLE;
//Setting IPSECCFGMODE to TUNNELITF_BASED
SET IKECFG: IPSECCFGMODE=TUNNELITF_BASED;
//Adding an IKE security proposal
ADD IKEPROPOSAL: PROPID=10, ENCALG=AES128, AUTHALG=SHA256, DURATION=86400,
REAUTHSW=ON, REAUTHLT=604800;
//Adding an IPsec proposal
ADD IPSECPROPOSAL: PROPNAME="prop1", ENCAPMODE=TUNNEL, TRANMODE=ESP,
ESPAUTHALG=SHA256, ESPENCALG=AES128;
//Adding a security parameter template for direct IPv4 IPsec with SeGW Switch set to DISABLE and
NARROWDOWNSW set to ON
ADD SECURITYTEMPLATE: SECURITYTEMPLATEID=0, SEGWSWITCH=DISABLE, PROPID=10,
IPSECPROPNAME="prop1", IKEVERSION=IKE_V2, PKEY="*****", DPD=PERIODIC, CERTSOURCE=APPCERT,
LTCFG=LOCAL, NARROWDOWNSW=ON, IKEV2FRGSW=OFF;
//Adding a security host with SeGW Switch set to DISABLE
ADD SECURITYHOST: SECURITYHOSTID=0, SEGWSWITCH=DISABLE, IPVERSION=IPv4, VRFIDX=2,
LOCIKEIPV4="7.7.4.146", SECURITYTEMPLATEID=0;
//Adding an endpoint group for the Xn interface
ADD EPGROUP: EPGROUPID=1, VRFIDX=1, APPTYPE=NULL;
//(Optional) Adding an SCTP template. By default, the gNodeB has an SCTP template with
SCTPTemplateID set to 0.
ADD SCTPTemplate: SCTPTemplateID=1, SWITCHBACKFLAG=ENABLE;
//Adding an Xn control-plane host
ADD SCTPHOST: SCTPHOSTID=1, VRFIDX=1, IPVERSION=IPv4, SIGIPV4="7.7.42.93",
SIGIP1SECSWITCH=ENABLE, SIGIP2SECHOSTID=0, SIGIP2SECSWITCH=DISABLE, PN=38422,
SIMPLEMODESWITCH=SIMPLE_MODE_OFF, SCTPTemplateID=1, DTLSPOLICYID=NULL;
//Adding an Xn user-plane host
ADD USERPLANEHOST: UPHOSTID=1, VRFIDX=1, IPVERSION=IPv4, LOCIPV4="7.7.42.93",
IPSECSWITCH=ENABLE, SECHOSTID=0;
//Adding the Xn user-plane host to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
//Adding the Xn control-plane host to the endpoint group
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=1;
//Turning on the ANR switch
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=NR_NR_ANR_SW-1&NR_EUTRAN_ANR_SW-1;
//Adding an Xn MO
ADD GNBCUXN: gNBCuXnId=0, CpEpGroupId=1, UpEpGroupId=1;
```

The following uses the IPv6 configuration as an example.

```
//Turning on the Xn self-setup switch
MOD GNBXNSONCONFIG: XnSonConfigSwitch=XN_SON_SETUP_SW-1;
```

```
//Adding an IKE security proposal
ADD IKEPROPOSAL: PROPID=10, ENCALG=AES128, AUTHALG=SHA256, DURATION=86400,
REAUTHSW=ON, REAUTHLT=604800;
//Adding an IPsec proposal
ADD IPSECPROPOSAL: PROPNAME="prop1", ENCAPMODE=TUNNEL, TRANMODE=ESP,
ESPAUTHALG=SHA256, ESPENCALG=AES128;
//Adding a security parameter template for direct IPv6 IPsec with SeGW Switch set to DISABLE
ADD SECURITYTEMPLATE: SECURITYTEMPLATEID=1, SEGWSWITCH=DISABLE, PROPID=10,
IPSECPROPNAME="prop1", IKEVERSION=IKE_V2, PKEY="*****", DPD=PERIODIC, LTCFG=LOCAL,
CERTSOURCE=APPCERT, TRUSTGROUPID=65535, OCSPSW=OFF, CRLPOLICY=GLOBAL;
//Adding a security host with SeGW Switch set to DISABLE
ADD SECURITYHOST: SECURITYHOSTID=1, SEGWSWITCH=DISABLE, IPVERSION=IPv6,
LOCIKEIPV6="2000:0:0:0:0:0:1", SECURITYTEMPLATEID=1, VRFIDX=2;
//Adding an endpoint group for the Xn interface
ADD EPGROUP: EPGROUPID=1, IPV6VRFIDX=1;
//Optional Adding an SCTP template. By default, the gNodeB has an SCTP template with
SCTPTemplateID set to 0.
ADD SCTPTemplate: SCTPTemplateID=1, SWITCHBACKFLAG=ENABLE;
//Adding an Xn control-plane host
ADD SCTPHOST: SCTPHOSTID=1, VRFIDX=1, IPVERSION=IPv6, SIGIPV6="2000:0:0:0:0:0:1:1",
SIGIP1SECSWITCH=ENABLE, SIGIP1SECHOSTID=1, SIGIP2V6="0::0", SIGIP2SECSWITCH=DISABLE, PN=38422,
SIMPLEMODESWITCH=SIMPLE_MODE_OFF, SCTPTemplateID=1;
//Adding an Xn user-plane host
ADD USERPLANEHOST: UPHOSTID=2, VRFIDX=1, IPVERSION=IPv6, LOCIPV6="2000:0:0:0:0:0:1:1",
IPSECSWITCH=ENABLE, SECHOSTID=1;
//Adding the Xn user-plane host to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
//Adding the Xn control-plane host to the endpoint group
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=1;
//Turning on the ANR switch
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch = NR_NR_ANR_SW-1&NR_EUTRAN_ANR_SW-1;
//Adding an Xn MO
ADD GNBCUXN: gNBCuXnId=0, CpEpGroupId=1, UpEpGroupId=1;
```

Activating self-update of direct IPsec for Xn interface:

```
SET GEPMODELpara: DIRIPSECUPDATESW=ENABLE;
```

Activating self-removal of direct IPsec for Xn interface:

```
//Activating self-removal of direct IPsec
SET GEPMODELpara: DIRIPSECAUTODELSW=ENABLE, DIRIPSECAUTODELTIMER=10080;
```

Configuring self-setup of direct IPsec for Xn interface:

```
//Setting the direct IPsec self-setup mode to strict
SET GEPMODELpara: DIRECTIPSECAUTOSETUPMODE=STRICT_MODE;
```

Configuring a direct IPsec blacklist for Xn interface:

```
//Adding a direct IPsec blacklist entry (IPv4 used as an example)
ADD DIRECTIPSECBLACKLIST: DIRECTIPSECBLACKLISTID=0, IPVERSION=IPV4,
SECURITYPEERIPV4="7.7.4.154";
```

Configuring a direct IPsec whitelist for Xn interface:

```
//Adding a direct IPsec whitelist entry (IPv4 used as an example)
ADD DIRECTIPSECWHITELIST: DIRECTIPSECWHITELISTID=0, IPVERSION=IPV4,
SECURITYPEERIPV4="192.168.0.0", MASK="255.255.255.0";
```

Configuring direct IPsec sharing among multiple operators over the Xn interface (using two operators as an example):

- Configuring for a newly deployed base station with IPv4 used as an example


```
//Configuring inner VRFs for two operators
ADD VRF: VRFIDX=1, USERLABEL="OP1", IPSECVPNV6=ENABLE, IPSECVPNID=1;
ADD VRF: VRFIDX=2, USERLABEL="OP2", IPSECVPNV6=ENABLE, IPSECVPNID=2;
//Creating a security parameter template and enabling the multi-VPN function
ADD IKEPROPOSAL: PROPID=1, ENCALG=AES128, AUTHMETH=IKE_CERT_SIG, REAUTHSW=ON;
ADD IPSECPROPOSAL: PROPNAME="1", TRANMODE=ESP, ESPENCALG=AES128;
SET GEPMODELpara: DIRIPSECAUTODELSW=ENABLE, DIRIPSECAUTODELTIMER=10;
```

```
ADD SECURITYTEMPLATE: SECURITYTEMPLATEID=1, SEGWSWITCH=DISABLE, PROPID=1,
IPSECPROPNAME="1", IKEVERSION=IKE_V2, DPD=PERIODIC, CERTSOURCE=APPCERT, LTCFG=LOCAL,
NARROWDOWNSW=ON, IKEV2FRGSW=OFF, MULTIVPNSWITCH=ON;
//Configuring the security host information and referencing the preceding security parameter
template (VRF 0 is the outer VRF, and the IPsec VPN switch does not need to be turned on.)
ADD SECURITYHOST: SECURITYHOSTID=1, SEGWSWITCH=DISABLE, IPVERSION=IPv4, VRFIDX=0,
LOCIKEIPV4="10.1.1.1", SECURITYTEMPLATEID=1;
//Adding control-plane hosts for the two operators
ADD SCTPHOST: SCTPHOSTID=1, VRFIDX=1, IPVERSION=IPv4, SIGIP1V4="1.1.1.1",
SIGIP1SECSWITCH=ENABLE, SIGIP1SECHOSTID=1, SIGIP2SECSWITCH=DISABLE, PN=36412,
SIMPLEMODESWITCH=SIMPLE_MODE_OFF, SCTPTEMPLATEID=0, DTLSPOLICYID=NULL,
USERLABEL="OP1";
ADD SCTPHOST: SCTPHOSTID=2, VRFIDX=2, IPVERSION=IPv4, SIGIP1V4="2.2.2.2",
SIGIP1SECSWITCH=ENABLE, SIGIP1SECHOSTID=1, SIGIP2SECSWITCH=DISABLE, PN=36412,
SIMPLEMODESWITCH=SIMPLE_MODE_OFF, SCTPTEMPLATEID=0, DTLSPOLICYID=NULL,
USERLABEL="OP2";
//Adding user-plane hosts for the two operators
ADD USERPLANEHOST: UPHOSTID=1, VRFIDX=1, IPVERSION=IPv4, LOCIPV4="1.1.1.1",
IPSECSWITCH=ENABLE, SECHOSTID=1, USERLABEL="OP1";
ADD USERPLANEHOST: UPHOSTID=2, VRFIDX=2, IPVERSION=IPv4, LOCIPV4="2.2.2.2",
IPSECSWITCH=ENABLE, SECHOSTID=1, USERLABEL="OP2";
```

- **Configuring for an additional operator B with IPv4 used as an example**

```
//Modifying the control-plane host for operator A and disabling direct IPsec
MOD SCTPHOST: SCTPHOSTID=1, VRFIDX=1, IPVERSION=IPv4, SIGIP1V4="1.1.1.1",
SIGIP1SECSWITCH=DISABLE, SIGIP2SECSWITCH=DISABLE, PN=36412,
SIMPLEMODESWITCH=SIMPLE_MODE_OFF, SCTPTEMPLATEID=0, DTLSPOLICYID=NULL,
USERLABEL="OP1";
//Modifying the user-plane host for operator A and disabling direct IPsec
MOD USERPLANEHOST: UPHOSTID=1, VRFIDX=1, IPVERSION=IPv4, LOCIPV4="1.1.1.1",
IPSECSWITCH=DISABLE, USERLABEL="OP1";
//Creating a security parameter template and enabling the multi-VPN function
ADD IKEPROPOSAL: PROPID=1, ENCALG=AES128, AUTHMETH=IKE_CERT_SIG, REAUTHSW=ON;
ADD IPSECPROPOSAL: PROPNAME="1", TRANMODE=ESP, ESPENCALG=AES128;
SET GEPMODEL PARA: DIRIPSECAUTODELSW=ENABLE, DIRIPSECAUTODELTIMER=10;
ADD SECURITYTEMPLATE: SECURITYTEMPLATEID=1, SEGWSWITCH=DISABLE, PROPID=1,
IPSECPROPNAME="1", IKEVERSION=IKE_V2, DPD=PERIODIC, CERTSOURCE=APPCERT, LTCFG=LOCAL,
NARROWDOWNSW=ON, IKEV2FRGSW=OFF, MULTIVPNSWITCH=ON;
//Configuring the security host information and referencing the preceding security parameter
template (VRF 0 is the outer VRF, and the IPsec VPN switch does not need to be turned on.)
ADD SECURITYHOST: SECURITYHOSTID=1, SEGWSWITCH=DISABLE, IPVERSION=IPv4, VRFIDX=0,
LOCIKEIPV4="10.1.1.1", SECURITYTEMPLATEID=1;
//Configuring inner VRFs for two operators
MOD VRF: VRFIDX=1, USERLABEL="OP1", IPSECVPNV4=ENABLE, IPSECVPNID=1;
ADD VRF: VRFIDX=2, USERLABEL="OP2", IPSECVPNV4=ENABLE, IPSECVPNID=2;
//Modifying the control-plane host for operator A and enabling direct IPsec again
MOD SCTPHOST: SCTPHOSTID=1, VRFIDX=1, IPVERSION=IPv4, SIGIP1V4="1.1.1.1",
SIGIP1SECSWITCH=ENABLE, SIGIP1SECHOSTID=1, SIGIP2SECSWITCH=DISABLE, PN=36412,
SIMPLEMODESWITCH=SIMPLE_MODE_OFF, SCTPTEMPLATEID=0, DTLSPOLICYID=NULL,
USERLABEL="OP1";
//Modifying the user-plane host for operator A and enabling direct IPsec again
MOD USERPLANEHOST: UPHOSTID=1, VRFIDX=1, IPVERSION=IPv4, LOCIPV4="1.1.1.1",
IPSECSWITCH=ENABLE, SECHOSTID=1, USERLABEL="OP1";
//Adding a control-plane host for operator B
ADD SCTPHOST: SCTPHOSTID=2, VRFIDX=2, IPVERSION=IPv4, SIGIP1V4="2.2.2.2",
SIGIP1SECSWITCH=ENABLE, SIGIP1SECHOSTID=1, SIGIP2SECSWITCH=DISABLE, PN=36412,
SIMPLEMODESWITCH=SIMPLE_MODE_OFF, SCTPTEMPLATEID=0, DTLSPOLICYID=NULL,
USERLABEL="OP2";
//Adding a user-plane host for operator B
ADD USERPLANEHOST: UPHOSTID=2, VRFIDX=2, IPVERSION=IPv4, LOCIPV4="2.2.2.2",
IPSECSWITCH=ENABLE, SECHOSTID=1, USERLABEL="OP2";
```

Optimization Command Examples

Optimization command examples are not involved.

Deactivation Command Examples

```
//Setting IPsec Auto Configure Switch for First Local IP and IPsec Auto Configure Switch for Second Local IP  
in the SCTPHOST MO to DISABLE  
MOD SCTPHOST: SCTPHOSTID=1, SIGIP1SECSWITCH=DISABLE, SIGIP2SECSWITCH=DISABLE;  
//Setting IPsec Auto Configure Switch in the USERPLANEHOST MO to DISABLE  
MOD USERPLANEHOST: UPHOSTID=1, IPSECSWITCH=DISABLE;  
//Disabling direct IPsec sharing among multiple operators over the Xn interface (using two operators as an  
example):  
MOD USERPLANEHOST: UPHOSTID=1, IPSECSWITCH=DISABLE;  
MOD USERPLANEHOST: UPHOSTID=2, IPSECSWITCH=DISABLE;  
MOD SECURITYTEMPLATE: SECURITYTEMPLATEID=1, SEGWSWITCH=DISABLE, IKEVERSION=IKE_V2,  
MULTIVPNSWITCH=OFF;  
MOD VRF: VRFID=1, USERLABEL="OP1", IPSECVPSW=DISABLE;  
MOD VRF: VRFID=2, USERLABEL="OP2", IPSECVPSW=DISABLE;
```

6.4.3.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.4 Activation Verification

Direct IPsec for the Xn-U Interface

Step 1 Run the **DSP IKESA** command to check the IKE SA negotiation results.

Expected result: The IDs of ACL rules generated during direct IPsec self-setup for the Xn interface range from 80000 to 84999. If the command output includes an IKE SA with the rule ID in this range, direct IPsec negotiation between the local and peer base stations has succeeded, and the IKE SA between the base station pair is normal.

Step 2 Run the **LST IPSECPOLICY** command to obtain **Policy Group Name** and **IPsec Sequence No.** corresponding to the ACL ID queried in the previous step.

Step 3 Run the **DSP IPSECSA** command to display the results of IPsec SA negotiation based on **IPsec Policy Group Name** and **IPsec Sequence No.**

Expected result: IPsec SA information is displayed, which indicates that direct IPsec negotiation between the local and peer base stations has succeeded, and the IPsec SA between the base stations is normal.

NOTE

This procedure uses direct IPsec SA setup for the Xn-U interface as an example. If direct IPsec is set up for the control plane between the local and the peer gNodeBs, IPsec SA information about the control plane is also displayed.

----End

Self-Update of Direct IPsec

Start an Xn interface trace task three minutes after self-update of direct IPsec is enabled. If an XNAP_PRIVATE_MSG message exists, this function has been successfully activated.

Self-Removal of Direct IPsec

If ALM-25891 IKE Negotiation Failure is reported, the alarm is cleared automatically when the **Direct IPsec Automatic Deletion Timer** expires (if this timer length is not 0). Then, run the **LST IKEPEER** command to view the configurations of the IKE peer.

Expected result: The information about the IKE peer corresponding to the peer IP address is unavailable. This indicates that this function has taken effect.

6.4.5 Network Monitoring

None

7 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

8 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

9 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

10 Reference Documents

- 3GPP TS 38.412: "NG-RAN; NG signalling transport"
- 3GPP TS 38.422: "NG-RAN; Xn signalling transport"
- 3GPP TS 38.413: "NG-RAN; NG Application Protocol (NGAP)"
- 3GPP TS 38.423: "NG-RAN; Xn Application Protocol (XnAP)"
- 3GPP TS 29.303: "Technical Specification Group Core Network and Terminals; Domain Name System Procedures"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *Transmission Resource Management*
- *SA Mobility Management in Connected Mode*
- *IPsec*
- *Multi-Operator Sharing*
- *IPv4 Transmission*
- *IPv6 Transmission*
- *Equipment Security*
- *X2 and S1 Self-Management in NSA Networking*
- *eXn Self-Management*
- *CA Experience Improvement*
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*
- *3900 & 5900 Series Base Station MO and Parameter Reference in 3900 & 5900 Series Base Station Product Documentation*